Tsinghua Lectures on Ordinary Differential Equations

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CHAPTER 1

Introduction

1. **Differential equations and their classifications**
2. **The fundamental problems**
3. **The contents of the course**

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CHAPTER 2

Well-posedness theory

1. Three classes of solvable first order equations

To understand the general theory of ODE, we start with some special cases. There are mainly three classes first order ODEs that can be solved explicitly.

1.1. First order linear equations. The first class is linear equation. Let us start with an example:

$$x' + x = 0.$$

We want to solve for a function $x(t)$ satisfying this equation.

There is an observation: if we multiple the equation by e^t, then we get by Leibniz rule

$$e^t(x' + x) = e^t x' + e^t x = (e^t x)' = 0.$$

This shows that $e^t x = C$, where C is an arbitrary constant, so $x = Ce^{-t}$. One can verify that any such function solves the original equation.

This idea can be used to solve more general equations, i.e all the first order linear equations for the form

$$x' + P(t)x = Q(t), \tag{1.1}$$

where P, Q are continuous function on a given interval $(a, b) \subset \mathbb{R}$.

To form an expression that can be applied the Leibniz rule, we multiply to the equation the function $\mu(t) = e^{\int_a^t P(s),ds}$, then we have

$$\mu(t)Q(t) = \mu(t)x' + \mu(t)P(t)x = (\mu(t)x)'.$$

We integrate the equation to obtain

$$\int_a^t \mu(s)Q(s)\,ds + C = \mu(t)x(t),$$

$$\begin{aligned} x(t) &= C\mu(t)^{-1} + \int_a^t \mu(t)^{-1}\mu(s)Q(s)\,ds \\ &= Ce^{-\int_a^t P(s)\,ds} + \int_a^t e^{-\int_s^t P(s)\,ds}Q(s)\,ds. \end{aligned} \tag{1.2}$$

The is the expression for the general solutions of equation 1.1. The method of obtaining the solution is called the method of integral factor and μ is called an integral factor. This resulting formula is very important. It is a special case of a general formula called DuHamel's principle appearing frequently in analysis.

Thus we have proved the following statement.

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THEOREM 1.1. *Let $P, Q : [a, b] \to \mathbb{R}$ be continuous. Then the IVP* $\begin{cases} x' + P(t)x = Q(t) \\ x(0) = x_0 \end{cases}$ *has a solution*

$$x(t) = x_0 e^{-\int_a^t P(s)\,ds} + \int_a^t e^{-\int_s^t P(s)\,ds} Q(s)\,ds.$$

There is a second method of solving the first order linear equations, that is called variation of parameters.

Let us introduce the notation $L = \frac{d}{dt} + P(t)$ to rewrite (1.1) into the form $Lx = Q(t)$. It is clear that L is a linear operator on the space of differentiable functions. i.e. given $x(t), y(t)$ differentiable and any two real numbers a and b, we have $L(ax + by) = aLx + bLy$.

If $Q \equiv 0$, then the resulting equation $Lx = 0$ is called homogeneous equation and if $Q \neq 0$, then $Lx = Q(t)$ is called inhomogeneous.

For linear equations, there is a general principle called superposition principle, which is fundamental in quantum mechanics

(1) If x_1 and x_2 solve the homogeneous equation $Lx = 0$, then so does $ax_1 + bx_2$ for all $a, b \in \mathbb{R}$. So the solution set $\mathcal{C}$ of $Lx = 0$ is a linear space.
(2) Let y be a particular solution to the inhomogeneous equation $Lx = Q(t)$, then the solution set of $Lx = Q(t)$ is $y + \mathcal{C}$.

PROOF. Part (1) is straightforward, so we only prove part (2). First, we show any $x = y + z$ where $z \in \mathcal{C}$ is a solution to $Lx = Q(t)$. We have $L(x) = L(y + z) = Ly + Lz = Ly = Q(t)$. Next, we show that any solution of $Lx = Q(t)$ is of the form $y + z$ with $z \in C$. Given two solutions x_1 and x_2, so we have $Lx_1 = Q(t)$ and $Lx_2 = Q(t)$. Taking the difference we get

$$Lx_1 - Lx_2 = L(x_1 - x_2) = 0.$$

This says that $x_1 - x_2 \in \mathcal{C}$. Since we have known a particular solution y. Taking $x_2 = y$, we get that $x_1 \in y + C$ for any solution x_1. □

The superposition principle gives us a recipe to solve linear equations. We first find all the solutions to the homogeneous equation and then find one particular equation for the inhomogeneous equation.

Let us now go back to our equation 1.1 pretending that we only know how to solve the homogeneous equations. The homogeneous equation is $Lx = x' + P(t)x = 0$. We can use the integrating factor method to find its general solution as

$$x(t) = Ce^{-\int_a^t P(s)ds}.$$

Next, for the inhomogeneous equation, we need to find a particular solution. The idea is to treat C as a function of t instead of only a constant. We then substitute $x(t) = C(t)e^{-\int_a^t P(s)ds}$ to the equation $Lx = Q(t)$ to obtain a homogeneous equation for C that we can solve. We get

$$Q(t) = x' + P(t)x = C'e^{-\int_a^t P(s)ds} + C(t)e^{-\int_a^t P(s)ds}(-P(t)) + P(t)C(t)e^{-\int_a^t P(s)ds} = C'e^{-\int_a^t P(s)ds}.$$

So we get

$$C'(t) = e^{\int_a^t P(s)ds} Q(t), \quad C(t) = \int e^{\int_a^t P(s)ds} Q(t)dt.$$

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So we obtain a particular solution

$$x(t) = C(t)e^{-\int_a^t P(s)ds} = e^{-\int_a^t P(s)ds}\int_a^t e^{\int_a^s P(r)dr}Q(s)ds = \int_a^t e^{-\int_s^t P(r)dr}Q(s)ds.$$

Applying the superposition principle, we arrive at the same DuHamel principle as before.

1.2. Exact equations. The second type is called exact equations.

This is a special class of equations of the form

$$M(x,y)dx + N(x,y)dy = 0 \tag{1.3}$$

satisfying the condition

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}. \tag{1.4}$$

Such a differential equation is called an exact equation.

If we write $\omega(x,y) = M(x,y)dx+N(x,y)dy$, then the condition **??** implies that $d\omega = 0$, i.e. ω is a closed-one form. We hope to find a function $\phi(x,y)$ such that $d\phi(x,y) = M(x,y)dx+N(x,y)dy$. Then the solution to the equation is $\phi(x,y) = C$. If such a ϕ exists, then $\omega = d\phi$ is called an exact one-form. The existence of such an exact one-form is determined by the topology of the domain.

We state the following theorem, in which case, the domain of ω is a square that is contractible to a point. In de Rham cohomology theory, it is known that such a closed one-form is always exact.

THEOREM 1.2. *Let $M(x,y)$ and $N(x,y)$ be continuous functions on the domain $R := [a,b]\times[c,d] \subset \mathbb{R}^2$ with continuous partial derivatives $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$. Then there exists $\phi:\ R \to \mathbb{R}$ such that $d\phi = \omega$ if and only if* (1.4) *holds.*

PROOF. We first show $\Longrightarrow$. If we have $\phi:\ R \to \mathbb{R}$ with $d\phi = \omega = M(x,y)dx + N(x,y)dy$, then we have $\frac{\partial \phi}{\partial x} = M$ and $\frac{\partial \phi}{\partial y} = N$. By assumption that partial derivatives $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ exist and are continuous, we have that the mixed partial derivatives are equal $\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$. This gives (1.4).

We next prove $\Longleftarrow$. It is enough to construct ϕ from the condition (1.4). Pick any $(x_0, y_0) \in R$.

We naturally expect that $\frac{\partial \phi}{\partial x} = M(x,y)$. Integrating this equation with respect to x, we obtain

$$\phi(x,y) = \int_{x_0}^x M(s,y)ds + h(y).$$

Here we have $h(y)$ since we integrate with respect to x in which case we treat y as a constant, therefore the arbitrary constant in the indefinite integral should be a function of y in this case. So to determine ϕ, it is enough to determine h.

On the other hand, we expect $\frac{\partial \phi}{\partial y} = N(x,y)$. That is

$$\begin{aligned} N(x,y) &= \int_{x_0}^x \frac{\partial}{\partial y} M(s,y)ds + h'(y) \\ &= \int_{x_0}^x \frac{\partial}{\partial x} N(s,y)ds + h'(y) \\ &= N(x,y) - N(x_0,y) + h'(y). \end{aligned} \tag{1.5}$$

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This gives $h'(y) = N(x_0, y)$ hence we obtain $h(y) = \int_{y_0}^{y} N(x_0, t)dt$.

We thus obtain $\phi(x, y) = \int_{x_0}^{x} M(s, y)ds + \int_{y_0}^{y} N(x_0, t)dt$. □

This class of equation appears in many contexts in mathematics, such as differential forms, complex analysis, 2D fluid mechanics, symplectic geometry, electromagnetism etc. We give an example in complex analysis. Let $f(z) : U(\subset \mathbb{C}) \to \mathbb{C}$ be a complex function that is differentiable at the point $z_0 \in U$. That is to say, there is a complex number w such that

$$\lim_{h\to 0} \frac{1}{|h|}|f(z_0 + h) - f(z_0) - wh| = 0.$$

If we write $z = x + iy$ and separate the real part and complex part of $f(z)$ to write $f(z) = u(x, y) + iv(x, y)$. Then the differentiability is equivalent to the following Cauchy Riemann equation $\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases}$. In general, if u is known, then v can be recovered, and vice versa. The functions u and v are harmonic functions, i.e. $\Delta u = \Delta v = 0$. For example, Let $f(z) = z^2$, then $f(z) = (x+iy)^2 = (x^2-y^2)+i2xy$.

1.3. Separable equations. Separable equations are of the following form

$$M(x)dx = N(y)dy.$$

This is a special case of exact equations, so its solution can be found as

$$\int M(x)dx + \int N(y)dy = C.$$

1.4. Riccati equation. There is a special class of equations called Riccati equations of the following form

$$x' = p(t)x^2 + q(t)x + r(t).$$

In general such kind of equations are not solvable. We state a theorem showing that some special Riccati equations are indeed solvable.

THEOREM 1.3. *Ricatti equation of the form*

$$x' = ax^2 + bt^m$$

when $m = 0, -2, \frac{-4k}{2k+1}, \frac{-4k}{2k-1},\ k = 1, 2, \ldots,$ *can be solved explicitly by integration.*

We skip the proof but refer the readers to [**DL**] Theorem 2.3.

For Riccati equations, there is an important transformation that is useful. We can covert a Riccati equation to a second order linear equation. Given a second order linear equation $x'' = u(t)x$, we introduce $y = x'/x = (\ln x)'$. Then we get a Riccati equation for y

$$y' = (x'/x)' = x''/x - (x'/x)^2 = u(t) - y^2.$$

Most of Riccati equations are not solvable by integration. The situation is similar to the historical famous problem of nonsolvability of algebraic equations of degree greater than 4, which was settled by Abel and Galois. Galois's work was ignored for some time until Liouville realized its value. It was also Liouville who invented a so called differentiable Galois theory, which was used to prove that most of ODEs are not solvable by integrations.

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The Riccati equation $\begin{cases} x' &= x^2+1 \\ x(0) &= x_0 \end{cases}$ has explicit solutions $x(t)=\tan(t+t_0)$ where t_0 satisfies $\tan t_0 = x_0$. Consider, for instance, $x_0=0, t_0=0$. Recall that

$$\lim_{x\to(\pi/2)-0}\tan x=\infty, \qquad \lim_{x\to(-\pi/2)+0}\tan x=-\infty,$$

we see that the solution to the IVP is defined only on the open interval $(-\pi/2,\pi/2)$.

Note that this equation behaves drastically different from the linear equation $x'=ax$, whose solultion is $x(t)=x_0e^{at}$ which is defined on the whole real line.

A solution defined on the whole half line $[0,\infty)$ is called a global solution. Recall that some physical laws are written in terms of ODEs, if a solution is only defined on a finite time interval, for instance, a solution of the Newtonian N-body problem, then it means that it is possible that our solar system may exist for only a finite time.

2. Existence and uniqueness theory

As a differential equation is a physical law. It is expected that when the current information is known, then we can solve the differential equation to predict the future and trace the past. So we expect that the solution with the given initial condition exists and is unique.

As we discussed in the previous section, the explicit integral solution of an ODE may not exist. However, this does not mean that an ODE has no solution.

Given a differential equation, the first question that we would like a positive answer is the well-posedness, which means the following three aspects of contents

(1) the existence of solution.
(2) the uniqueness of solution,
(3) the continuous dependence of solution on the initial conditions.

Only when a differential equation is well-posed, it can be a candidate of a good physical law. In general, people are not interested in those ill-posed differential equations.

In this section, we prove the following classical theorem.

A function $f:\ [a,b]\to\mathbb{R}$ is called uniformly Lipschitz, if there exists $L>0$ such that for each pair $x,y\in[a,b]$, we have

$$|f(x)-f(y)|\le L|x-y|.$$

THEOREM 2.1 (Cauchy's theorem). *Let $x'=f(x,t)$ be a differential equation with $f:\ [a,b]\times[c,d]$ continuous in t and Lipschitz in x. Then for each given $x_0\in(a,b),\ t_0\in(c,d)$, there exists $\delta>0$ and a unique solution $x:\ [-\delta+t_0,\delta+t_0]\to\mathbb{R}$ solving the initial value problem*

$$\begin{cases} x' &= f(x,t) \\ x(t_0) &= x_0. \end{cases}$$

When the Lipschitzness is not satisfied, the uniqueness may be lost. Here is an example. Consider the IVP

$$\begin{cases} x' &= x^{1/3} \\ x(0) &= 0. \end{cases}$$

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By integrating this ODE explicitly, we find that there are three solutions of the ODE

$$x_1 = 0,\ x_2(t) = \left(\frac{2}{3}t\right)^{3/2},\ x_3(t) = -\left(\frac{2}{3}t\right)^{3/2}.$$

The Lipschitzness is violated at a single point $x = 0$.

Let us now move on to the proof of the theorem. The proof is the standard Picard iteration, which appears in most of the wellposedness proofs.

THE FIRST PROOF OF CAUCHY THEOREM, PICARD ITERATION. Instead of solving the ODE, we solve the integral equation

$$x(t) = x_0 + \int_{t_0}^{t} f(x(s), s)\, ds.$$

We start with any initial continuous curve $x^0 :\ I := [t_0 - \delta, t_0 + \delta] \to \mathbb{R}$ with $x^0(0) = x_0$, where δ is a small number to be determined, and construct a sequence of functions $\{x^n(t) \mid t \in I\}$ with $x^n(0) = x_0$, as follows

$$x^{n+1}(t) = x_0 + \int_{t_0}^{t} f(x^n(s), s)\, ds. \tag{2.1}$$

For convenience, we choose the starting curve x^0 to be a constant curve $x^0(t) = x_0$.

We will show that this sequence converges uniformly to a limiting function $x^* :\ I \to \mathbb{R}$. Taking limit $n \to \infty$ in the above equation, in particular, on the RHS, we can take limit under the integral by the uniform convergence, we obtain

$$x_*(t) = x_0 + \int_{t_0}^{t} f(x_*(s), s)\, ds.$$

First, there is a constant M such that $|f| \le M$ on the domain $[a, b] \times [c, d]$. From (2.1), we have

$$|x_1(t) - x_0| \le M|t - t_0| \le M\delta. \tag{2.2}$$

This shows that if δ is chosen small enough, then the curve $\{x_1(t)\}_{t\in I}$ also lies in $[a, b] \times [c, d]$. Inductively, we get $|x_n(t) - x_0| \le M|t - t_0| \le M\delta$, hence each curve $\{x_n(t)\}_{t\in I}$ lies in the domain $[a, b] \times [c, d]$.

We next show that the sequence $\{x_n(t)\}_{n\in\mathbb{N}}$ converges uniformly. For this purpose, we claim that

$$|x_{n+1}(t) - x_n(t)| \le \frac{M}{L}\frac{(L|t - t_0|)^{n+1}}{(n + 1)!}.$$

The inequality holds in the case of $n = 0$ by inequality (2.2). Suppose the case n holds then we proceed to verify the case of $n+1$. Taking difference of the equations

$$x^{n+1}(t) = x_0 + \int_{t_0}^{t} f(x^n(s), s)\, ds, \quad x^{n+2}(t) = x_0 + \int_{t_0}^{t} f(x^{n+1}(s), s)\, ds$$

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we get

$$
\begin{aligned}
|x^{n+1}(t) - x^{n+2}(t)| &\le \left| \int_{t_0}^{t} f(x^n(s), s) - f(x^{n+1}(s), s)\, ds \right| \\
&\le \int_{t_0}^{t} L \left| x^n(s) - x^{n+1}(s) \right| ds \\
&\le \int_{t_0}^{t} L \frac{M}{L} \frac{(L|t - t_0|)^{n+1}}{(n+1)!}\, ds \\
&\le \frac{M}{L} \frac{(L|t - t_0|)^{n+2}}{(n+2)!}.
\end{aligned} \tag{2.3}
$$

This completes the induction hence proves the claim.

By the claim, we have

$$
|x_{n+1}(t) - x_n(t)| \le \frac{M}{L} \frac{(L|t - t_0|)^{n+1}}{(n+1)!} \le \frac{M}{L} \frac{(L\delta)^{n+1}}{(n+1)!}.
$$

Note that the estimate on the RHS has no dependence on t and $n!$ diverges to infinity much faster than exponential functions, so we conclude that the sequence $\{x_n(t)\}_{n\in\mathbb{N}}$ is uniformly convergent.

We next show that the solution is unique. Suppose there are two solutions x and y both solving the IVP on the interval I. When writing in terms of integral equations, we have

$$
x(t) = x_0 + \int_{t_0}^{t} f(x(s), s)\, ds, \quad y(t) = x_0 + \int_{t_0}^{t} f(y(s), s)\, ds.
$$

Taking difference we get

$$
|x(t) - y(t)| \le \int_{t_0}^{t} |f(x(s), s) - f(y(s), s)|\, ds \le L \int_{t_0}^{t} |x(s) - y(s)|\, ds.
$$

We next introduce $Z(t) = \int_{t_0}^{t} |x(s) - y(s)|\, ds$, then we get $Z'(t) = |x(t) - y(t)|$, and we obtain the differential inequality

$$
Z'(t) \le LZ(t),\ Z(t_0) = 0.
$$

The differential inequality can be solved using the method of integrating factor method. We have

$$
e^{-Lt} Z'(t) - e^{-Lt} LZ(t) = (e^{-Lt} Z(t))' \le 0.
$$

Integrating this inequality on the interval $[t_0, t_0 + \delta]$, we get

$$
e^{-Lt} Z(t) - e^{-Lt_0} Z(t_0) \le 0, \quad Z(t) \le 0.
$$

However, since we have $Z(t) \ge 0$ for $t \in [t_0, t_0 + \delta]$, we conclude $Z(t) \equiv 0$, for $t \in [t_0, t_0 + \delta]$. On the interval $[t_0 - \delta, t_0]$, we get

$$
e^{-Lt_0} Z(t_0) - e^{-Lt} Z(t) \le 0, \quad Z(t) \ge 0.
$$

On the interval $t \in [t_0 - \delta, t_0]$, we get $Z(t) \le 0$ by the definition of $Z(t)$, so we again conclude $Z(t) \equiv 0$ for $t \in [t_0 - \delta, t_0]$.

Therefore, we obtain $x(t) \equiv y(t)$ on the interval I.

□

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REMARK 2.2. *The proof essentially says that the solution on a short interval can be bounded by* $e^{Lt} = \sum_{n=0}^{\infty} \frac{(Lt)^n}{n!}$.

Now we invoke the fixed point theorem, which is the main theme in all the well-posedness proofs.

Recall that a metric d on a space X is a function $d: X \times X \to \mathbb{R}$ satisfying the axioms

(1) positive definiteness, $d(x, y) \geq 0$ and $d(x, y) = 0$ iff $x = y$;
(2) symmetry, $d(x, y) = d(y, x)$;
(3) triangle inequality, $d(x, y) \leq d(x, z) + d(y, z), \quad \forall x, y, z, \in X$.

A metric space (X, d) is called complete, if every Cauchy sequence has limiting point in the space X. An example of incomplete space is the $\mathbb{R}^2$ with 0 removed and the usual Euclidean distance.

We next state the important contraction mapping theorem:

THEOREM 2.3 (Banach fixed point theorem). *Let (X, d) be a complete metric space and $T: X \to X$ be a contracting mapping, i.e. T is a map and there exists $\lambda \in (0, 1)$ such that $d(Tx, Ty) \leq \lambda d(x, y)$ for all $x, y \in X$. Then T has a unique fixed point. i.e. there is a unique $x_* \in E$ such that $Tx_* = x_*$.*

PROOF. We pick any initial point x_0 and produce a sequence $\{x_n\}$ via $x_{n+1} = Tx_n$. By assumption we have

$$d(x_{n+2}, x_{n+1}) \leq \lambda d(x_{n+1}, x_n) \leq \lambda^{n+1} d(x_1, x_0).$$

We first show that this sequence is Cauchy. For all $\varepsilon > 0$, we pick N such that $\frac{\lambda^{N+2}}{1-\lambda} d(x_1, x_0) < \varepsilon$. So we have for all $m > n > N$

$$d(x_m, x_n) \leq \sum_{k=1}^{m-n} d(x_{n+k}, x_{n+k-1}) \leq \sum_{k=1}^{m-n} \lambda^{n+k+1} d(x_1, x_0) \leq \frac{\lambda^{n+2}}{1-\lambda} d(x_1, x_0) < \varepsilon.$$

By the completeness, the sequence $\{x_n\}$ has a limiting point x_*. Taking limit in the equation $Tx_n = x_{n+1}$ by the continuity of T (implied by the definition of contracting mapping), we have

$$Tx_* = x_*.$$

Therefore we obtain a fixed point.

We next prove the uniqueness of the fixed point. Suppose we have two fixed points $Tx = x, Ty = y$. Therefore we have

$$d(x, y) = d(Tx, Ty) \leq \lambda d(x, y).$$

By the positive-definiteness in the definition of a metric, this implies $d(x, y) = 0$, hence $x = y$.

□

THE SECOND PROOF OF CAUCHY THEOREM. We let X be the space of continuous functions from $[t_0 - \delta, t_0 + \delta]$ to $[a, b]$ satisfying $x(t_0) = x_0$, endowed with the distance

$$d(x, y) = \sup_{t \in [t_0 - \delta, t_0 + \delta]} |x(t) - y(t)|.$$

To show (X, d) is complete, we need to show that each Cauchy sequence $\{x_n\}$ has a limit $x_*(t): [t_0 - \delta, t_0 + \delta] \to \mathbb{R}$ that is continuous. Note that the convergence

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under the distance d is the uniform convergence, the completeness follows from the Dini theorem.

We next introduce the map T

$$(Tx)(t) = x_0 + \int_{t_0}^{t} f(x(s), s)\, ds.$$

As in the first proof, we have that for δ sufficiently small,the image of Tx lies in $[a, b]$ if $x \in X$, so Tx is in X and T is a map from X to X.

To apply the contractign mapping theorem, we need to verify $d(Tx, Ty) \leq \lambda d(x, y)$ for some $\lambda < 1$ and for all $x, y \in X$. We have

$$\begin{aligned} |(Tx - Ty)(t)| &= \left| \int_{t_0}^{t} f(x(s), s) - f(y(s), s)\, ds \right| \\ &\leq \int_{t_0}^{t} |f(x(s), s) - f(y(s), s)|\, ds \\ &\leq L \int_{t_0}^{t} |x(s) - y(s)|\, ds \\ &\leq L\delta d(x, y), \\ d(Tx, Ty) &\leq L\delta d(x, y). \end{aligned} \tag{2.4}$$

Therefore, The map T is a contraction if $L\delta < 1$. □

In the uniqueness proof of the first proof of the Cauchy theorem, we encountered an differential inequality of the form $Z' \leq LZ$. It turns out that this is a special case of the following more general Gronwall inequality. This is essentially the only inequality in ODE. This inequality is extremely useful since most of the ODEs are not solvable, and if we can compare it with a simple linear equation, then we can estimate its solution by the solution to the linear equation.

THEOREM 2.4 (Gronwall inequality). *Let $P, Q : [a, b] \to \mathbb{R}$ be continuous. Let $x : [a, b] \to \mathbb{R}$ be the solution of the IVP* $\begin{cases} x' = P(t)x + Q(t) \\ x(a) = x_0, \end{cases}$ *Suppose the function $y : [a, b] \to \mathbb{R}$ satisfies the inequality* $\begin{cases} y' \leq P(t)y + Q(t) \\ y(a) = x_0. \end{cases}$ *Then we have $y(t) \leq x(t)$ for all $t \in [a, b]$.*

PROOF. We can apply the method of integrating factor to the inequality, to get

$$(\mu(t)y(t))' \leq \mu(t)Q(t).$$

where $\mu(t) = e^{-\int_a^t P(s)\, ds} > 0$, $\mu(a) = 1$. Integrating this inequality over the interval $[a, b]$, we obtain

$$\mu(t)y(t) - y(a) \leq \int_a^t \mu(s)Q(s)\, ds.$$

Therefore we get

$$y(t) \leq \mu(t)^{-1} \left(y(a) + \int_a^t \mu(s)Q(s)\, ds \right).$$

Replacing in the RHS $y(a) = x(a) = x_0$, we note that the RHS is exactly $x(t)$, the solution to the IVP in the statement. This completes the proof. □

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There is a more general comparison principle.

THEOREM 2.5. *Suppose* $x, y:\ [t_0-\delta, t_0+\delta] \to \mathbb{R}$ *satisfy the differential inequalities*

$$\begin{cases} x' & = F(x,t) \\ x(t_0) & = x_0 \end{cases}, \quad \begin{cases} y' & = G(y,t) \\ y(t_0) & = x_0 \end{cases}$$

where $F, G:\ [a,b]\times[t_0-\delta, t_0+\delta] \to \mathbb{R}$ *are continuous and satisfies* $F \geq G$ *pointwise. Then we have*

$$x(t) \geq y(t),\ \forall\ t \in [t_0, t_0+\delta], \quad x(t) \leq y(t),\ \forall\ t \in [t_0-\delta, t_0],$$

PROOF. We first prove that if $F > G$ pointwise, then $x(t) > y(t)$ for $t \in (t_0, t_0+\delta]$. Similarly, we can prove that $x(t) < y(t),\ \forall\ t \in [t_0-\delta, t_0)$. In the case of $F \geq G$, we modify the x-equation to $\begin{cases} x' & = F(x,t)+\varepsilon \\ x(t_0) & = x_0 \end{cases}$ for $\varepsilon > 0$, and obtain that $x_\varepsilon(t) > y(t)$ if $t \in (t_0, t_0+\delta]$. Then we let $\varepsilon \to 0$, to get that $x(t) = \lim_\varepsilon x_\varepsilon(t) \geq y(t)$ using the continuous dependence of solutions on the parameters in the next section.

Now let us assume $F > G$ pointwise. So we get $F(x_0, t_0) > G(x_0, t_0)$. By the continuity of F, G and x and y, we get there is ϵ sufficiently small such that

$$F(x(t), t) > G(y(t), t), \quad t \in [t_0, t_0+\epsilon].$$

From the integral equations $x(t) = x_0 + \int_{t_0}^t F(x(s), s)\, ds, \quad y(t) = x_0 + \int_{t_0}^t G(y(s), s)\, ds$, we get

$$x(t) > y(t), \quad t \in (t_0, t_0+\epsilon].$$

We next prove that the inequality $x(t) > y(t)$ holds on the interval $(t_0, t_0+\delta]$. Suppose $t_0+\varepsilon < \hat{t} < t_0+\delta$ is the first time when $x(t) = y(t)$. Then for $t = \hat{t}$, we again have $F(x(\hat{t}), \hat{t}) > G(x(\hat{t}), \hat{t})$. However, since $x(t) - y(t)$ decreases to zero as $t \to \hat{t}$ from the left, we see that $x' - y' = F(x,t) - G(y,t) \leq 0$ at $t = \hat{t}$. This is a contradiction. So such $\hat{t}$ does not exists. The proof is then complete. □

3. Continuous dependence on the initial conditions

The third aspect in the wellposedness is the continuous dependence of solutions on the initial conditions. In general, there are always errors in observations determining the initial conditions. We expect that the error in the initial conditions does not lead to big difference in the solutions. Moreover, the physical laws sometimes are only an approximation. For instance, Newton's mechanics is only an approximation of Einstein's general relativity in the low velocity regime. Therefore, we expect that when the differential equation is perturbed slightly, the solution with given initial condition is also perturbed slightly.

Suppose we are given a differential equation of the form $x' = f(x, t, \mu)$ where $\mu \in D$ is a parameter and D is the compact set $\{\mu \mid |\mu - \mu_0| \leq c\}$ in $\mathbb{R}^n$ for some constant c and μ_0, and given initial condition $x(t_0) = x_0$. We hope to show that the solution $x_{\mu,x_0}(t)$ with given initial condition $x(t_0) = x_0$ and parameter μ is a continuous function of x_0 and μ under certain conditions. Note that by making a change of variables $s = t - t_0$ and $y = x - x_0$, the initial condition becomes $y(0) = 0$ and the equation becomes $\frac{d}{ds}y = f(y + x_0, s + t_0, \mu)$. Now the initial conditions t_0, x_0 becomes parameters in the equation and can be absorbed into

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μ. Therefore, we consider only the continuous dependence on parameter μ in the equation $x' = f(x,t,\mu)$ with fixed initial condition $x(t_0) = x_0$.

THEOREM 3.1. *Let $f(x,t,\mu) : [a,b]\times[c,d]\times D \to \mathbb{R}$ be continuous in t and the parameter μ, and is Lipschitz in x (i.e. there exists L such that $|f(x,t,\mu) - f(y,t,\mu)| < L|x-y|$ for all $x, y \in [a,b]$, $t \in [c,d]$ and $\mu \in D$). Then for each given $x_0 \in (a,b)$ and $t_0 \in (c,d)$, there exists δ such that on the time interval the solution $x_\mu : [t_0-\delta, t_0+\delta] \to \mathbb{R}$ to the IVP $\begin{cases} x' &= f(x,t,\mu) \\ x(t_0) &= x_0 \end{cases}$ is continuous in μ.*

PROOF. We give the proof using the fixed point theorem. For each μ, we introduce the following operator

$$T_\mu : \ X \to X, \quad (T_\mu(x))(t) = x_0 + \int_{t_0}^{t} f(x(s), s, \mu)\, ds,$$

where X is the set of continuous functions from $I = [t_0-\delta, t_0+\delta]$ to $[a,b]$ with $x(t_0) = x_0$, endowed with the sup-distance. In the second proof of Cauchy theorem, we have shown that if $L\delta < 1$, then T_μ has a fixed point denoted by x_μ, i.e. $T_\mu x_\mu = x_\mu$. To show that x_μ is continuous in μ, we need to show that for any ε, there exists Δ, if $|\mu - \mu'| < \Delta$, then we have $d(x_\mu, x_{\mu'}) < \varepsilon$. We have

$$\begin{aligned} d(x_\mu, x_{\mu'}) &= d(T_\mu x_\mu, T_{\mu'} x_{\mu'}) \\ &\le d(T_\mu x_\mu, T_\mu x_{\mu'}) + d(T_\mu x_{\mu'}, T_{\mu'} x_{\mu'}) \\ &\le \lambda d(x_\mu, x_{\mu'}) + d(T_\mu x_{\mu'}, T_{\mu'} x_{\mu'}). \end{aligned} \tag{3.1}$$

So we get

$$d(x_\mu, x_{\mu'}) \le \frac{1}{1-\lambda} d(T_\mu x_{\mu'}, T_{\mu'} x_{\mu'}).$$

Now the problem is reduced to showing that T_μ is uniformly continuous in μ, i.e. for all ε, there exists Δ, such that for all $|\mu - \mu'| < \Delta$, we have $d(T_\mu x, T_{\mu'} x) < \varepsilon$ for all $x \in X$.

By assumption on f, we have that f is continuous in μ, i.e. for all ε, there exists Δ, such that for all $|\mu - \mu'| < \Delta$, we have $|f(x,t,\mu) - f(x,t,\mu')| < \varepsilon$ for all $(x,t) \in [a,b]\times[c,d]$ and $\mu, \mu' \in D$.

$$\begin{aligned} d(T_\mu x, T_{\mu'} x) &= \sup_{t\in I} \left| \int_{t_0}^{t} f(x(s), s, \mu) - f(x(s), s, \mu')\, ds \right| \\ &\le \sup_{t\in I} \int_{t_0}^{t} \varepsilon\, ds \le \delta\varepsilon. \end{aligned} \tag{3.2}$$

We thus prove that T_μ is continuous in μ, hence that x_μ is continuous in μ.

□

REMARK 3.2. *Here the continuous dependence on the initial condition is only proved in a short time interval. In general, there is no guarantee that if the initial conditions differ a little bit, the solutions also differ a little on the whole time interval where the solutions exist. In the long term, the solution may diverge significantly. This is the so-called chaos. It is one theme of research of the field of dynamical systems.*

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4. Differentiable dependence on the initial conditions

Continuing the theme from the previous section. We consider the IVP $\begin{cases} x' &= f(x,t,\mu) \\ x(t_0) &= x_0 \end{cases}$. We have shown that the solution $x_\mu(t)$ of the IVP depends on μ continuously. In this section, we will show that if f is C^1 in x and μ, i.e. the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial \mu}$ are continuous. Then the solution x_μ is actually differentiable with respect to μ, i.e. $\frac{dx_\mu}{d\mu}$ exists. More precisely, we will prove the following theorem

THEOREM 4.1. *Suppose $f(x,t,\mu):\ [a,b]\times[c,d]\times D\to\mathbb{R}$ is continuous and C^1 in x and μ. Then there exists $\delta>0$ such that the solution $x_\mu:\ [t_0-\delta,t_0+\delta]\to[a,b]$ of the IVP $\begin{cases} x' &= f(x,t,\mu) \\ x(t_0) &= x_0 \end{cases}$ is differentiable in μ.*

With this theorem, we can derive an equation governing the motion of $\frac{\partial x_\mu}{\partial \mu} := \xi_\mu$. Indeed, we simply differentiate the equation $x' = f(x,t,\mu)$ with respect to μ to obtain

$$\xi'_\mu = \frac{\partial f}{\partial x}(x_\mu(t),t,\mu)\xi_\mu + \frac{\partial f}{\partial \mu}(x_\mu(t),t,\mu).$$

The equation can be understood as follows. Let $x_\mu(t)$ be a solution to the IVP with parameter value μ. Then $\xi_\mu = \lim_\varepsilon \frac{1}{\varepsilon}(x_{\mu+\varepsilon} - x_\mu)$. In the above equation for ξ'_μ, we fix the value of μ and substitute the information of $x_\mu(t)$, so that $\frac{\partial f}{\partial x}(x_\mu(t),t,\mu)$ and $\frac{\partial f}{\partial \mu}(x_\mu(t),t,\mu)$ are only functions of t. Therefore, the equation is actually a linear equation for ξ_μ.

As a special case that we explained in the previous section (the initial condition can be viewed as a parameter), we obtain that in an IVP $\begin{cases} x' &= f(x,t) \\ x(t_0) &= x_0 \end{cases}$, if we treat x_0 and parameter to obtain solutions $x_{x_0}(t)$. Then the solution has differentiable dependence on x_0. Denoting $\xi = \frac{\partial x}{\partial x_0}$, we obtain the following equation

$$\xi' = \partial_x f(x(t),t)\xi,$$

governing the motion of ξ. This equation is called the variational equation associated to the ODE $x' = f(x,t)$.

The study of how nearby orbits move along a given orbit is an important source of ideas and concepts in mathematics and physics. For example, on the plane, given a geodesic, i.e. a line, if we perturb the initial condition slightly, a nearby geodesic will converge or diverge linearly. On a sphere, a geodesic is a big circle. If at the south pole we perturb the initial condition slightly, a nearby geodesic will meet the original big circle at the north pole. Therefore, the dynamics of nearby orbits reflects the curvature of the underlying space. In the sphere example, it looks as if there is a force to push the two geodesic to meet. In geometry, this is called curvature, while in physics, this is called the tidal force. This basic observation gives rise to the general theory of relativity.

In variational principle, the equations of motion appears as the first variation (first order derivative) of the action functional while the stability information of an orbit is contained in the second variation (second order derivative). The resulting equation is also called the variational equation.

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PROOF. We give a proof using Picard iteration. We first form the equivalent integral equation

$$x(t) = x_0 + \int_{t_0}^{t} f(x(s), s, \mu)\, ds$$

and the Picard sequence

$$x_\mu^{n+1}(t) = x_0 + \int_{t_0}^{t} f(x_\mu^n(s), s, \mu)\, ds$$

We start with the constant function $x_\mu^0(t) \equiv x_0$. As we have shown in the proof of Cauchy theorem, on the interval $I = [t_0 - \delta, t_0 + \delta]$ for some sufficiently small δ, we have that for each given μ, the sequence $\{x_\mu^n\}$ converges uniformly.

Next, we can show that inductively $\frac{dx_\mu^n}{d\mu}$ exists and we have the equation

$$\frac{dx_\mu^{n+1}}{d\mu} = \int_{t_0}^{t} \frac{\partial f}{\partial x}(x_\mu^n(s), s, \mu)\frac{dx_\mu^n}{d\mu} + \frac{\partial f}{\partial \mu}(x_\mu^n(s), s, \mu)\, ds.$$

To prove that x_μ is differentiable in μ, it is enough to show that the sequence $\frac{dx_\mu^n}{d\mu}$ converges uniformly.

By assumption on f, there exists $\alpha > 0$ such that on the domain we have $|\frac{\partial f}{\partial x}| \leq \alpha$ and $|\frac{\partial f}{\partial \mu}| \leq \alpha$. Since $\frac{dx_\mu^0}{d\mu} = 0$, we get $|\frac{dx_\mu^1}{d\mu}| \leq \int_{t_0}^{t} |\frac{\partial f}{\partial \mu}(x_\mu^n(s), s, \mu)|\, ds \leq \alpha|t - t_0|$. Inductively, we have

$$|\frac{dx_\mu^n}{d\mu}| \leq \sum_{k=1}^{n} \frac{(\alpha|t - t_0|^k)}{k!} \leq e^{\alpha\delta}. \tag{4.1}$$

We next introduce

$$v_{k,s} := \left| \frac{dx_\mu^{k+s}}{d\mu} - \frac{dx_\mu^k}{d\mu} \right|.$$

To show that the sequence $\{\frac{dx_\mu^n}{d\mu}\}$ converges uniformly, it is enough to show that for all ε, there exists K such that when $k > K$, we have $\sup_{t\in I, s\in\mathbb{N}} v_{k,s} < \varepsilon$ (This is an equivalent formulation of the Cauchy sequence).

We derive the estimate

$$v_{k+1,s} \leq \left| \int_{t_0}^{t} \left| \frac{\partial f}{\partial \mu}(x_\mu(s), s, \mu) \right| v_{k,s}\, ds \right| + d_{k,s}(t, \mu),$$

where

$$\begin{aligned} d_{k,s}(t, \lambda) = &\left| \int_{t_0}^{t} \left(\frac{\partial f}{\partial x}(x_\mu^{k+s}(s), s, \mu) - \frac{\partial f}{\partial x}(x_\mu(s), s, \mu) \right) \frac{dx_\mu^{k+s}}{d\mu}\, ds \right| \\ &+ \left| \int_{t_0}^{t} \left(\frac{\partial f}{\partial x}(x_\mu(s), s, \mu) - \frac{\partial f}{\partial x}(x_\mu^k(s), s, \mu) \right) \frac{dx_\mu^k}{d\mu}\, ds \right| \\ &+ \left| \int_{t_0}^{t} \left(\frac{\partial f}{\partial \mu}(x_\mu^{k+s}(s), s, \mu) - \frac{\partial f}{\partial \mu}(x_\mu^k(s), s, \mu) \right) ds \right|. \end{aligned} \tag{4.2}$$

Using the bound (4.1) and $|\partial f/\partial y| \leq \alpha$ as well as the uniform convergence of x_μ^k to x_μ, we get $d_{k,s}(t, \mu)$ converges to 0 uniformly in μ, s, t as $k \to \infty$. So there exists

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$\varepsilon_k \to 0$ such that $d_{k,s} < \varepsilon_k$ hence

$$v_{k+1,s} \le \alpha \left| \int_{t_0}^{t} v_{k,s}\, ds \right| + \varepsilon_k.$$

By (4.1), we get $v_{k,s} \le 2e^{\alpha\delta}$, so $v_{k+1,s} \le 2\alpha e^{\alpha\delta}|t-t_0| + \varepsilon_k$. Inductively, we get

$$v_{k+m,s} \le 2e^{\alpha\delta}\frac{(\alpha|t-t_0|)^m}{m!} + \sum_{j=0}^{m-1} \varepsilon_{k+m-1-j}\frac{(\alpha|t-t_0|)^j}{j!}.$$

Letting $\delta_k = \sup\{\varepsilon_k, \varepsilon_{k+1}, \ldots\} \to 0$, we get

$$v_{k+m,s} \le 2e^{\alpha\delta}\frac{(\alpha\delta)^m}{m!} + e^{\alpha\delta}\delta_k.$$

This shows that $v_{k,s}(t,\mu)$ converges to zero as $k \to \infty$ uniformly in s, t, μ as desired. The proof is then complete. □

5. The extension of solutions

Besides the wellposedness theory, the next basic question that one generally ask about a differential equation is the maximal existence time interval for a solution. We have seen that there are two drastically different behaviors for the linear equation $x' = ax$ and the Riccati equation $x' = x^2 + 1$, where in the former each solution is defined on the whole real line, while in the latter each solution is defined only on an open interval of length π. A solution defined on the half line $[0, \infty)$ is called a global solution.

Given a solution to an ODE $x' = f(x,t)$, to determine that it is a global solution or a finite time blow up solution, we in general use the comparison principle. For instance, if bound $|f(x,t)| \le a|x| + b$, then we know that each solution is global. On the other hand, we we bound $f(x,t) \ge 1 + x^2$, then each solution blows up within finite time.

THEOREM 5.1. *Let $f : G \to \mathbb{R}$ be continuous, where $G \subset \mathbb{R}^2$ is open, and is locally Lipschitz in x (for each $(x,t) \in G$, there is a square q containing (x,t) in its interior and $L > 0$, such that f restricted to q is L-Lipschitz. Note that L depends on q). Then passing through every point in G, there is a solution to the ODE $x' = f(x,t)$ and the solution extends to the boundary of G (i.e. for any compact domain $G' \subset G$, the solution extends outside G').*

PROOF. Given $(x_0, t_0) \in G$, by the local Lipschitz condition and Cauchy theorem, we see that there is a solution Γ passing through (x_0, t_0). We next show that the solution extends to the boundary of G. We consider only the case of $t > t_0$, and the case of $t < t_0$ is similar.

If the solution Γ exists on the interval $[t_0, \infty)$, the statement is obvious.

If the maximal interval where the solution Γ exists is $[t_0, t_1]$. Then the point $(x(t_1), t_1)$ lies in G. We apply Cauchy theorem to find a solution passing through $(x(t_1), t_1)$ defined on $[t_1 - \delta, t_1 + \delta]$ for some small δ. On the interval $[t_1 - \delta, t_1]$, by the uniqueness, the solution coincide with Γ. Therefore, we have extended Γ to $[t_0, t_1 + \delta]$ contradicting to the maximality of $[t_0, t_1]$.

The last possibility is that the maximal interval is $[t_0, t_1)$. Suppose the curve Γ is contained in a compact set $G' \subset G$. By the compactness, we have that

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$|f|$ is bounded by some constant M on G'. Therefore we get from $x(t) = x_0 + \int_{t_0}^{t} f(x(s), s)\, ds$ that

$$|x(t) - x(t')| \leq M|t - t'|.$$

As $t \to t_1$, the limit $\lim_{t\to t_1} x(t) = x_0 + \lim_{t\to t_1} \int_{t_0}^{t} f(x(s), s)\, ds$ exists. We thus have defined a solution to the integral equation on the time interval $[t_0, t_1]$, contradicting to the maximality of $[t_0, t_1)$. □

We next give two examples to show how to use the comparison principle to prove some results about the finite time blow up or global solution.

PROPOSITION 5.2. *Suppose the equation $x' = f(x, t)$ satisfies that $|f(x, t)| \leq a|x| + b$. Then for each initial condition (x_0, t_0), the solution is defined on $[t_0, \infty)$.*

PROOF. We write down the integral equation $x(t) = x_0 + \int_{t_0}^{t} f(x(s), s)\, ds$. Therefore, we get

$$|x(t)| \leq |x_0| + \int_{t_0}^{t} a|x(s)| + b\, ds.$$

Next we introduce $y(t) = \int_{t_0}^{t} a|x(s)| + b\, ds$, hence $y' = a|x(t)| + b$, so we get

$$y' \leq a|x_0| + b + ay.$$

Gronwall inequality implies that $y(t) \leq \frac{a|x_0|+b}{a}(e^{at} - 1)$. The RHS is a function defined on $[t_0, \infty)$, so y is defined on $[t_0, \infty)$, so is x. □

PROPOSITION 5.3. *Each solution of $x' = x^2 + t^2$ blows up within finite time.*

PROOF. We want to show that for each given initial condition (x_0, t_0), the solution is defined only on a finite time interval $[t_0, t_1)$. Suppose now $t_1 \leq 0$, then the solution is defined only on an interval $[t_0, t_1) \subset [t_0, 0)$ which is finite. By the previous theorem in this section, each solution extends to the boundary of the domain where f is defined, since it cannot be extended beyond $t = 0$, the only possibility is that the solution $x(t) \to \infty$ as $t \to t_1$. In this case, we are done.

Suppose now $t_1 > 0$. We pick $0 < \bar{t} < t_1$ and consider the solution on the interval $[\bar{t}, t_1)$, where we have $x_2 + t_2 \geq x^2 + t_1^2$. By the comparison principle, we get that the solution to the IVP $\begin{cases} x' & = x^2 + t^2 \\ x(\bar{t}) & = x(\bar{t}) \end{cases}$ is bounded from below by the solution to the IVP $\begin{cases} x' & = x^2 + \bar{t}^2 \\ x(\bar{t}) & = x(\bar{t}) \end{cases}$ The latter can be integrated explicitly similar to $x' = x^2 + 1$, so we know that its solution is defined on a finite interval and blows up to infinity close to the boundary of the interval. The comparison principle thus shows that the solution to the former IVP has the same behavior. □

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CHAPTER 3

Higher order equations

In this chapter, we study two types of equations: higher order linear equations and first order linear systems. The former has the form

$$a_n(t)x^{(n)} + a_{n-1}(t)x^{(n-1)} + \ldots + a_1(t)x' + a_0(t)x = b(t),$$

and the latter has the form

$$\begin{cases} x_1' &= a_{11}(t)x_1 + a_{12}(t)x_2 + \ldots + a_{1n}(t)x_n + b_1(t) \\ &\ldots \\ x_n' &= a_{n1}(t)x_1 + a_{n2}(t)x_2 + \ldots + a_{nn}(t)x_n + b_n(t) \end{cases}.$$

For the latter, we can introduce $X = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$, $A(t) = (a_{ij}(t)) \in \mathbb{R}^{n\times n}$ and $b(t) = (b_1(t), \ldots, b_n(t))$ to abbreviate the equation as $X' = A(t)X + b(t)$.

It is enough to illustrate the ideas by considering only the case of $n = 2$.

First, a higher order linear equation can be converted into a linear system. Consider $x'' + a_1(t)x' + a_0(t)x = b(t)$, we introduce $x' = y$, so

$$y' = x'' = -a_1(t)x' - a_0(t)x + b(t) = -a_1(t)y - a_0(t)x + b(t).$$

So we obtain a linear system in x and y In the matrix form, we can write

$$\begin{bmatrix} x \\ y \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ -a_0(t) & -a_1(t) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ b(t) \end{bmatrix}.$$

So the theory of linear systems is slightly more general than higher order linear equations.

1. Higher order equations and linear systems with constant coefficients

In this section, we first study higher order equations and linear systems with constant coefficients.

As we have shown before, such equations has a superposition principle, so we study the homogeneous equations first, i.e. we set $b = 0$.

Again, we study only the case of $n = 2$, which already captures all the main ideas.

1.1. Second order linear equation with constant coefficients. We study the following equation

$$ax'' + bx' + cx = 0.$$

The idea is to look for solutions of the form $x(t) = e^{\lambda t}$. Substituting it into the equation, we obtain the following algebraic equation

$$a\lambda^2 + b\lambda + c = 0.$$

There are three cases depending on the signs of the discriminant $\Delta = b^2 - 4ac$.

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If $\Delta > 0$, there are two distinct roots, $\lambda_{1,2} = \frac{-b \pm \sqrt{\Delta}}{2a}$. So both $e^{\lambda_1 t}$ and $e^{\lambda_2 t}$ are solutions, and all the linear combinations $x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t},\ c_1, c_2 \in \mathbb{R}$ is a solution.

If $\Delta = 0$, there is only one root $\lambda = -\frac{-b}{2a}$, so we get a solution to the ODE $x(t) = e^{\lambda t}$. However, there is a second solution $te^{\lambda t}$, and all the linear combinations $x(t) = c_1 e^{\lambda t} + c_2 te^{\lambda t},\ c_1, c_2 \in \mathbb{R}$ is a solution. The extra solution can be understood as follows. Suppose the algebraic equation has two roots λ_1 and λ_2 satisfying $\lambda_1 \to \lambda$ and $\lambda_2 \to \lambda$. Then

$$e^{\lambda_1 t} - e^{\lambda_2 t} = (e^{(\lambda_1 - \lambda_2)t} - 1)e^{\lambda_2 t} = (\lambda_1 - \lambda_2)te^{\lambda_2 t} + O((\lambda_1 - \lambda_2)^2)$$

is also a solution. Then we divide the expression by $\lambda_1 - \lambda_2$ and let it go to zero.

If $\Delta < 0$, then there are two complex roots, denoted by $\lambda_{1,2} = \alpha \pm i\beta$. So we get two solutions to the ODE $e^{(\alpha \pm \beta)t} = e^{\alpha t}(\cos \beta t \pm i \sin \beta t)$. We can substitute the solution to the equation and separate real and imaginary parts to get two solutions to the ODE $e^{\alpha t} \cos \beta t$ and $e^{\alpha t} \sin \beta t$, hence all the linear combinations $x(t) = c_1 e^{\alpha t} \cos \beta t + c_2 e^{\alpha t} \sin \beta t,\ c_1, c_2 \in \mathbb{R}$, is a solution.

1.2. Euler equation. Euler equation has the following form

$$at^2 x'' + btx' + cx = 0.$$

In this case, we look for solutions of the form $x(t) = t^\lambda$. Substituting this back to the equation, we obtain

$$a\lambda(\lambda - 1) + b\lambda + c = 0.$$

Again there are three cases, depending on the discriminant $\Delta = (b - a)^2 - 4ac$.

If $\Delta > 0$, then there are two roots $\lambda_{1,2} = \frac{-(b-a) \pm \sqrt{\Delta}}{2a}$. Then there are two solutions t^{λ_1} and t^{λ_2}, hence all the linear combinations $x(t) = c_1 t^{\lambda_1} + c_2 t^{\lambda_2},\ c_1, c_2 \in \mathbb{R}$ is a solution.

If $\Delta = 0$, then we get one root $\lambda = -\frac{(b-a)}{2a}$. Then $x = t^\lambda$ is a root and the second root is $t^\lambda \log t$, hence all the linear combinations $x(t) = c_1 t^\lambda + c_2 t^\lambda \log t,\ c_1, c_2 \in \mathbb{R}$ is a solution.

If $\Delta < 0$, then there are two complex roots $\lambda_{1,2} = \alpha \pm i\beta$. Then we get two solutions to the ODE

$$x = t^{\alpha \pm i\beta} = t^\alpha t^{\pm i\beta} = t^\alpha e^{\pm i\beta \log t} = t^\alpha(\cos(\beta \log t) \pm i \sin(\beta \log t)).$$

Separating the real and complex parts, we get two solutions $t^\alpha(\cos(\beta \log t)$ and $t^\alpha(\sin(\beta \log t)$, hence all the linear combinations $x(t) = c_1 t^\alpha(\cos(\beta \log t) + c_2 t^\alpha(\sin(\beta \log t),\ c_1, c_2 \in \mathbb{R}$ is a solution

1.3. Linear systems with constant coefficients. The simplest linear system is the following $\begin{cases} x_1' &= ax_1 \\ x_2' &= bx_2 \end{cases}$ The equations can be integrated explicitly as $\begin{cases} x_1(t) &= e^{at} x_1(0) \\ x_2(t) &= e^{bt} x_2(0) \end{cases}$. In general, to solve an equation of the form $X' = AX$ with $X = (x_1, x_2)$ and $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Formally, we obtain the solution of the initial

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value problem $X' = AX$ with initial data $X(t_0) = (c_1, c_2)^T$ as $e^{A(t-t_0)}X(t_0)$, where for a given matrix A, the exponential e^A is defined as

$$e^A = I + A + \frac{A^2}{2} + \ldots + \frac{A^n}{n!} + \ldots.$$

We remark that expoentiation can only be used to solve linear systems with constant coefficients. If the coefficient matrix A depends on t, then $e^{\int A(s)ds}$ does not give the solution to the ODE which is essentially due to the fact that $A(t)$ does not commute with $\int A(s)ds$.

However, the solution given by the exponentiation is not explicit. In the following we show how to obtain the solutions explicitly. We invoke the theory of Jordan normal form. There is a nondegenerate complex matrix C such that $C^{-1}AC = J$ where J is of the following three forms

$$\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}, \quad \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}, \quad \begin{bmatrix} \alpha + i\beta & 0 \\ 0 & \alpha - i\beta \end{bmatrix}$$

where $\lambda, \lambda_1, \lambda_2, \alpha, \beta$ are real, and λ_1 and λ_2 may or may not be equal. In the first two cases, the matrix C is real, while in the last case, C is complex.

We consider case 1. Let $Y = C^{-1}X$. Then $Y' = C^{-1}ACY = JY$. Since J is diagonal, we can obtain. $y_1 = y_1(0)e^{\lambda_1 t},\ y_2 = y_2(0)e^{\lambda_2 t}$. Therefore $X = CY = y_1(0)e^{\lambda_1 t}v_1 + y_2(0)e^{\lambda_2 t}v_2$ where v_i is the eigenvector corresponding to the eigenvalue $\lambda_i,\ i = 1, 2$ and $C = (v_1, v_2)$.

Next for case 2, writing $C = (v_1, v_2)$, we get

$$AC = A(v_1, v_2) = (v_1, v_2)J = (\lambda v_1, v_1 + \lambda v_2).$$

This gives $(A - \lambda I)v_1 = 0,\ (A - \lambda I)v_2 = v_1$. As before, $e^{\lambda t}v_1$ is a solution. Next, it can be verified that $X = te^{\lambda t}v_1 + e^{\lambda t}v_2$ is a solution. So all linear combinations of the form

$$c_1 e^{\lambda t}v_1 + c_2(te^{\lambda t}v_1 + e^{\lambda t}v_2)$$

is a solution. There is another way to see the same conclusion. Again, we study $Y' = JY$ with $Y = C^{-1}X$. Formally, we get $Y(t) = e^{Jt}Y(0)$. Here take the Taylor expansion, we get

$$e^{Jt} = I + Jt + \frac{(Jt)^2}{2} + \ldots + \frac{(Jt)^n}{n!} + \ldots.$$

Denoting $J = \lambda I + E$ where $E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. It can be verified that $IE = EI$, $E^2 = 0$. So we get

$$J^n = (\lambda I + E)^n = \sum_{k=0}^{n} \binom{n}{k} (\lambda I)^k E^{n-k} = (\lambda I)^n + n\lambda^{n-1}E.$$

This gives

(1.1)

$$\begin{aligned} e^{Jt} &= I + (\lambda I + E)t + \frac{1}{2}(\lambda^2 I + 2E)t^2 + \ldots + \frac{1}{n!}(\lambda^n I + nE)t^n + \ldots \\ &= (I + (\lambda It) + \frac{1}{2}(\lambda It)^2 + \ldots + \frac{(\lambda It)^n}{n!} + \ldots) + Et(1 + t + \frac{t^2}{2} + \ldots + \frac{t^n}{n!}) + \ldots \\ &= e^{I\lambda t} + Ete^{\lambda t} \end{aligned}$$

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So we get $Y(t) = (e^{I\lambda t} + Ete^{\lambda t})Y_0$. Therefore

$$X(t) = C(e^{I\lambda t} + Ete^{\lambda t})C^{-1}X_0 = (e^{\lambda t}v_1x_1(0) + (te^{\lambda t}v_1 + e^{\lambda t}v_2)x_2(0)).$$

In the third case, we denote by $u = v + iw$ the eigevector corresponding to $\alpha + i\beta$. So we have $Au = (\alpha + i\beta)u$ and $A\bar{u} = (\alpha - i\beta)\bar{u}$, so $\bar{u} = v - iw$ is the eigenvector corresponding to the eigenvalue $\alpha - i\beta$. As before, we get

$$Y = (y_1, y_2) = (y_0(0)e^{(\alpha+i\beta)t}, y_0(0)e^{(\alpha-i\beta)t}).$$

Hence

$$X = CY = (u, \bar{u})(y_1, y_2) = y_1u + y_2\bar{u} = y_1(0)e^{(\alpha+i\beta)t}u + y_2(0)e^{(\alpha-i\beta)t}\bar{u}.$$

We can rewrite

$$\begin{aligned} e^{(\alpha+i\beta)t}u &= e^{\alpha t}(\cos\beta t + i\sin\beta t)(v + iw) \\ &= e^{\alpha t}[(\cos\beta tv - \sin\beta tw) + i(\sin\beta tv + \cos\beta tw)]. \end{aligned}$$

So any linear combination of the form

$$c_1e^{\alpha t}(\cos\beta tv - \sin\beta tw) + c_2e^{\alpha t}(\sin\beta tv + \cos\beta tw),\ c_1, c_2 \in \mathbb{R},$$

is a solution.

2. The general theory

2.1. The wellposedness theory for systems. We consider the general nonlinear system of equations of the following form

$$\begin{cases} x_1' &= f_1(x_1, \ldots, x_n, t), \\ x_2' &= f_2(x_1, \ldots, x_n, t), \\ \ldots & \\ x_n' &= f_n(x_1, \ldots, x_n, t). \end{cases}$$

We introduce $X = (x_1, \ldots, x_n)$, $F = (f_1, \ldots, f_n)$ to abbreviate the system as $X' = F(X, t)$.

We assume that $F: D \times [c, d] \to \mathbb{R}$ where D is a domain in $\mathbb{R}^n$. For instance, we assume that $D = [a_1, b_1] \times \ldots \times [a_n, b_n]$.

We further assume that F satisfies the Lipschitz condition: there exists some constant $L > 0$ such that

$$\|F(X_1, t) - F(X_2, t)\| \le L\|X_1 - X_2\|, \quad \forall\ X_1, X_2 \in D.$$

Here $\|\cdot\|$ means the Euclidean norm.

Then we can carry out the Picard iteration completely verbally in the new setting as we did for the case of a single equation.

In this setting, we have the existence and uniqueness theorem, and the theorem on the continuous dependence on initial conditions. If F is known to be C^1 in all its variables, we can also establish the C^1 dependence on the initial conditions.

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2.2. The general theory for linear systems. In all the three cases that we have solved explicitly, each equation has two special solutions. There are some basic questions: can we find more? How different are the solutions.

We consider first the first order homogeneous linear system of the form

$$X' = A(t)X$$

where $A:\ [a,b] \to \mathbb{R}^{n\times n}$ is a time-dependent $n \times n$ matrix. We assume that A is continuous in t.

First, we know that for each initial condition $X(t_0) = X_0 \in \mathbb{R}^n$, the solution exists and is unique.

We next denote by $\mathcal{C}$ the set of all solutions to the system. By the superposition principle, we know that $\mathcal{C}$ is a linear space.

We next have the following result.

THEOREM 2.1. *$\mathcal{C}$ is a linear space of dimension n.*

PROOF. We already know that $\mathcal{C}$ is a linear space. To show that it has dimension n, it is enough to build a linear map that is both injective and surjective to $\mathbb{R}^n$.

We pick $t_0 \in (a,b)$. We define a map $H:\ \mathcal{C} \to \mathbb{R}^n$ as $H(X) = X(t_0) \in \mathbb{R}^n$. The map is injective by the uniqueness of solutions. The map is surjective, by the existence of solutions. It is also clear that the map is linear. Then the statement follows. □

Let $X_1, \ldots, X_n$ be n linearly independent solutions in $\mathcal{C}$. By linearly independent, we mean that for any numbers $c_1, \ldots, c_n$ with

$$\sum_{i=1}^{n} c_i X_i = \mathbf{0},$$

where $\mathbf{0}$ means zero solution in $\mathcal{C}$, we have $c_1 = c_2 = \ldots = c_n = 0$. Since $\mathcal{C}$ has dimension n, we know that $X_1, \ldots, X_n$ form a basis for $\mathcal{C}$. That means any $X \in \mathcal{C}$, there exists unique $(c_1, \ldots, c_n) \in \mathbb{R}^n$ such that $X = \sum_{i=1}^n c_i X_i$.

We form a matrix $(X_1, \ldots, X_n)$ by listing the solutions $X_1, \ldots, X_n$ as columns. We denote its determinant by W and call it the Wronskian, which is a function of t.

THEOREM 2.2 (Liouville formula). *We have the following equality*

$$W(t) = e^{\int_{t_0}^{t} TrA(s)ds} W(t_0).$$

PROOF. We give the proof in the case of $n = 2$. The general case is similar. Suppose $X_1(t) = (x_{11}(t), x_{12}(t))^T$ and $X_2(t) = (x_{21}(t), x_{22}(t))^T$ are solutions, so we get $X_1' = A(t)X_1$ and $X_2' = A(t)X_2$. Also denote $A(t) = \begin{bmatrix} a(t) & b(t) \\ c(t) & d(t) \end{bmatrix}$. We

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have $W = \det(X_1, X_2)$ so

$$\begin{aligned} W' &= \begin{vmatrix} x_{11}' & x_{21}' \\ x_{12} & x_{22} \end{vmatrix} + \begin{vmatrix} x_{11} & x_{21} \\ x_{12}' & x_{22}' \end{vmatrix} \\ &= \begin{vmatrix} ax_{11} + bx_{12} & ax_{21} + bx_{22} \\ x_{12} & x_{22} \end{vmatrix} + \begin{vmatrix} x_{11} & x_{21} \\ cx_{11} + dx_{12} & cx_{21} + dx_{22} \end{vmatrix} \\ &= a \begin{vmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \end{vmatrix} + d \begin{vmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \end{vmatrix} \\ &= (TrA(t))W \end{aligned} \tag{2.1}$$

□

Since the exponential function cannot be zero, we get that either $W(t)$ is constantly zero for all t or never zero for all t. This means that the solution set $X_1, \ldots, X_n$ are linearly independent as points in the linear space $\mathcal{C}$, if and only if $X_1(t), \ldots, X_n(t)$, for each fixed t, are linearly independent as vectors in $\mathbb{R}^n$.

REMARK 2.3. *Given a matrix A, we have $\det e^A = e^{TrA}$. The proof goes as follows. The exponential e^A can be equivalently defined as*

$$e^A = \lim_{m\to\infty} \left(I + \frac{A}{m}\right)^m.$$

The determinant $\det : \mathbb{R}^{n\times n} \to \mathbb{R}$ *is a continuous function, so we get*

$$\det e^A = \det \lim_{m\to\infty} \left(I + \frac{A}{m}\right)^m = \lim_{m\to\infty} \det \left(I + \frac{A}{m}\right)^m.$$

We next notice that

$$\det \left(I + \frac{A}{m}\right)^m = (\det(I + \frac{A}{m}))^m = (1 + \frac{TrA}{m} + O(m^{-2}))^m \to e^{TrA}.$$

Here the main contribution to $\det(I + \frac{A}{m})$ *is the product of the diagonal entries*

$$(1 + a_{11}/m) \ldots (1 + a_{nn}/m) = 1 + \frac{TrA}{m} + O(m^{-2}),$$

for all the other summands in $\det(I + \frac{A}{m})$*, since it misses at least one diagonal entries, it has to have at least two off-diagonal entries, so it can at most be* $O(m^{-2})$.

We introduce the notion of fundamental solutions.

DEFINITION 2.4. *Given the IVP* $\begin{cases} X' & = A(t)X \\ X(t_0) & = X_0 \end{cases}$*, we say that the matrix function* $\Phi : [a, b] \to \mathbb{R}^{n\times n}$ *is the fundamental solution if it satisfies*

$$\Phi' = A(t)\Phi,\ \Phi(t_0) = I.$$

Given the fundamental solution Φ, we can write the solution to the IVP as $X(t) = \Phi(t)X_0$.

In physics, the fundamental solution is also called the propagator, or Green's function, or heat kernal in various situations.

For linear system with constant coefficients $X' = AX$, the fundamental solution is e^{At}.

Given an equation $X' = A(t)X$ with initial time t_0. There are two ways to work out the fundamental solution. The first way is to work out the solutions with initial

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conditions $X(t_0) = X_0 = (1, 0, \dots, 0), (0, 1, 0, \dots, 0), \dots, (0, \dots, 0, 1)$. We put these solutions as columns together to form a matrix, which is exactly the fundamental matrix.

For the second method, we can pick any n linearly independent solutions to form a matrix $(X_1(t), \dots, X_n(t))$. It is known that this matrix is nondegenerate for all t. We then define $\Phi(t) = (X_1(t), \dots, X_n(t))(X_1(t_0), \dots, X_n(t_0))^{-1}$. We can verify that $\Phi(t_0) = I$ and $\Phi(t)' = A(t)\Phi(t)$ since we have $X_i' = A(t)X_i$ for all $i = 1, \dots, n$.

Suppose A is constant. Then the equation $X' = AX$ has global solution. The fundamental solution $\Phi(t) = e^{At}$ satisfies the properties

(1) $\Phi(0) = I$,
(2) $\Phi(t)\Phi(s) = \Phi(t + s), \quad \forall\, t, s \in \mathbb{R}$,
(3) $\Phi(-t) = \Phi(t)^{-1}$.

Therefore $\{\Phi(t) \mid t \in \mathbb{R}\}$ is an Abelian group.

When A depends on time, the fundamental solution is in general not a group but a semi-group.

2.3. Linear inhomogeneous equations. As we have leaned in the superposition principle, once the fundamental solution is known for a linear homogeneous system, to solve the inhomogeneous equation, it is enough to find one special solution.

There are in general two ways to solve inhomogeneous equations.

The first method works for all the inhomogeneous equations, that is the method of variations of parameters.

But for equations with special inhomogeneous terms of the forms, we can use the method of undetermined coefficients.

2.3.1. *The method of undertermined coefficients.* We take the equation $x'' + 3x' + 2x = g(t)$ for an example. The case of linear systems are similar.

The method of undetermined coefficients method are useful when g is of the following functions

(1) polynomials $\sum_{i=0}^{n} a_i t^i$;
(2) trigonometric functions $A\cos\omega t + B\sin\omega t$;
(3) exponential function $e^{\lambda t}$
(4) combinations of the above cases such as $t^2 e^t, t\cos t$, $e^t \cos t$, etc.

In the first case, we guess that a solution of the inhomogeneous equation has the form $\sum_{i=0}^{n} b_i t^i$. Then substitute it to the equation to obtain $n+1$ linear equations for the coefficients $b_0, \dots, b_n$ by comparing coefficients with the RHS. Then we solve for $b_0, \dots, b_n$.

In the second case, we guess that a solution is $\alpha\cos\omega t + \beta\sin\omega t$. Then we substitute it to the equation to obtain linear equations for α and β by comparing the coefficients. Then we solve for α and β.

In the third case, we guess a solution is $Ae^{\lambda t}$ and solve for A. We may be cautious that the first guess may fail, which in general occurs when λ is exactly a solution to the quadratic equation $a^2 + 3a + 2 = 0$. This is called resonance. When a resonance occurs, the guess would be $Ate^{\lambda t}$. The resonance may also occur in the second case. For example, the equation $x'' + x = \sin t$ has resonance. We have to guess the solution has the form $At\cos t + Bt\sin t$.

The fourth case is similar to the above cases.

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2.3.2. *The method of variation of parameters.* The method of variation of parameters works for all the linear equations provided we know the fundamental solution. Let us consider the IVP

$$\begin{cases} X' & = A(t)X + g(t) \\ X(t_0) & = X_0 \end{cases}.$$

Suppose Φ is the fundamental solution to the homogeneous equation $X' = A(t)X$. So we have $\Phi(t_0) = I$ and $\Phi' = A(t)\Phi$.

For the homogeneous equation, $\Phi(t)C$ is a solution for any $C \in \mathbb{R}^n$. The idea of variation of parameters is to suppose now C is a vector depending on t, instead of a constant vector. We then substitute $\Phi(t)C(t)$ to the inhomogeneous equation to get

$$(\Phi(t)C(t))' = \Phi(t)'C(t)+\Phi(t)C(t)' = A(t)\Phi(t)C(t)+\Phi(t)C(t)' = A(t)\Phi(t)C(t)+g(t).$$

This gives $g(t) = \Phi(t)C(t)'$, so

$$C(t) = \int_{t_0}^{t} \Phi^{-1}(s)g(s)\,ds.$$

Therefore all the solutions of the the inhomogeneous equation has the form

$$X(t) = \Phi(t)C + \int_{t_0}^{t} \Phi(t)\Phi^{-1}(s)g(s)\,ds$$

For our IVP, the solution is

$$X(t) = \Phi(t)X_0 + \int_{t_0}^{t} \Phi(t)\Phi^{-1}(s)g(s)\,ds.$$

This formula is called the DuHamel principle.

In case the equation has constant coefficients, i.e. $X' = AX$, the fundamental solution $\Phi(t) = e^{At}$, then we have $\Phi(t)\Phi(s)^{-1} = e^{A(t-s)} = \Phi(t-s)$.

2.4. The physical meaning of second order linear ODEs. Consider a massive particle with mass m, connected to the wall through a spring. By Hooke's law, the force that the spring gives the particle is proportional to the position denoted by $F = -kx$, where the minus sign means that the direction of the force is opposite to the position.

By Newton's second law, we get the equation

$$mx'' = -kx,$$

provided there is no other forces involved. This is called the harmonic oscillator. We expect that the motion of the oscillator is periodic since there is no friction so there is no loss of energy.

The general solution of this equation is $x(t) = A\cos\omega t+B\sin\omega t = \alpha\sin(\omega t+\beta)$ where $\omega = \sqrt{\frac{k}{m}}$ is called the frequency, $\alpha = \sqrt{A^2+B^2}$ is called the amplitude and $\beta = \arctan\frac{B}{A}$ is called the initial phase. Here the frequency is an intrinsic quantity depending on the mass and elastic constant k only. However, the quantities α and β are extrinsic, depending on the initial position and velocity of the oscillator.

We see that the mathematical solution reflects very well the physics.

We next introduce the friction, which is in general proportional to the negative velocity, denoted by $f = -\gamma x'$.

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Taking friction into account, we get the equation of motion by Newton's second law

$$mx'' = -\gamma x' - kx.$$

Here the coefficients m, k, γ are assumed to be positive constants.

We expect that physically the oscillator will damp and also oscillate when the friction is small, and it will damp without oscillating if the friction is too large. Let us now see how the mathematics reflects the physics.

Recall that the solution of the equation has the form $x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$ where $c_1, c_2 \in \mathbb{R}$ and $\lambda_{1,2} = \frac{-\gamma \pm \sqrt{\Delta}}{2m}$, $\Delta = \gamma^2 - 4mk$.

If γ is so small that $\Delta < 0$, then the roots $\lambda_{1,2}$ are complex, denoted by $\lambda_{1,2} = \alpha \pm i\beta$ with $\alpha = \frac{-\gamma}{2m} < 0$. Then the real solutions to the ODE is given by

$$x(t) = c_1 e^{\alpha t} \cos \beta t + c_2 e^{\alpha t} \sin \beta t.$$

We can see from the solution that no matter what constants c_1, c_2 are given, the solution damps exponentially due to $\alpha < 0$ and on the other hand it also oscillates because of the presence of sin and cos.

If γ is so large that $\Delta > 0$, then the two solutions $\lambda_{1,2}$ are real. We also notice that $\lambda_{1,2} < 0$ due to $\Delta < \gamma^2$. So for any c_1, c_2, the solution $x(t) = c_1 e^{-\lambda_1 t} + c_2 e^{-\lambda_2 t}$ damps to zero exponentially fast without having any oscillations.

There is also a critical case of $\Delta = 0$.

We may also consider an oscillator with an external force. Suppose for simplicity that there is no friction. We also assume that the external force $f = \sin(\Omega t)$ is periodic in t with a period $\Omega > 0$.

The equation of motion is

$$mx'' + kx = \sin(\Omega t).$$

Recall the way we solve the inhomogeneous equations. The general solution to the ODE has the form

$$x(t) = A \cos(\omega t) + B \sin(\omega t) + q(t), \quad \omega = \sqrt{\frac{m}{k}},$$

where $q(t)$ is a particular solution to the inhomogeneous equation. We use the method of undermined coefficient to find q. We make a guess $q(t) = C \cos \Omega t + D \sin \Omega t$ in case when $\omega \neq \Omega$. In case when $\omega = \Omega$, such a guess does not work, instead, we need to guess $q(t) = Ct \cos \Omega t + Dt \sin \Omega t$.

The behavior of the solutions are different. In the case of $\omega \neq \Omega$, the solution is still periodic. Though the constant $D = \frac{1}{-m\Omega^2 + k}$ ($C = 0$) depends strongly on the difference $\Omega - \omega$. In the case of $\omega = \Omega$, which we call resonance, the solution grows linearly while oscillating. Resonance is a phenomena that is due to the coincidence of the internal and external frequency, which can be very detrimental.

3. Linear system with periodic coefficients, Floquet theory

When a linear system $X' = A(t)X$ has nonconstant coefficient matrices $A(t)$, the solution is in general not solvable explicitly. There is a special case when A is periodic in t, for which there is a special theory called Floquet theory, on which we now focus.

The main theorem is as follows.

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THEOREM 3.1. *Suppose $A: \mathbb{R} \to \mathbb{R}^{n\times n}$ is T-periodic in t. The fundamental solution $\Phi(t)$ can be expressed into the form*

$$\Phi(t) = X(t)e^{Kt},$$

where X is T-periodic in t.

PROOF. First we know that $\Phi(t+kT) = \Phi(t)\Phi(T)^k$ by the group property of Φ. Indeed, for all $X_0 \in \mathbb{R}^n$, we can verify that both $\Phi(t+T)X_0$ and $\Phi(t)\Phi(T)X_0$ are solutions to the equation with the same initial condition $\Phi(T)X_0$ for $t = 0$. By the uniqueness theorem and the arbitrariness of X_0, we get $\Phi(t+T) = \Phi(t)\Phi(T)$.

We first find K by the equation $\Phi(T) = e^{KT}$. Note that this is equivalent to saying that we can take log of $\Phi(T)$. We will discuss this after the proof of the main theorem. We then introduce $X(t) = \Phi(t)e^{-Kt}$. We next prove that $X(t)$ is periodic. We have

$$X(t+T) = \Phi(t+T)e^{-K(t+T)} = \Phi(t)\Phi(T)e^{-KT}e^{-Kt} = \Phi(t)e^{-Kt} = X(t).$$

□

LEMMA 3.2. *Let $R \in \mathbb{R}^{n\times n}$ be a nondegenerate matrix. Then there exists Q such that $R = e^Q$. If R is real and has a square root, then Q can also be taken to be real.*

For example, the matrix $\begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$ has no real square root, then it has no real log.

PROOF. We give the proof in the case of $n = 2$. The general case is completely similar.

We consider first the case when $R \in \mathbb{R}^{2\times 2}$ is diagonalizable in $\mathbb{R}$. Then there is some $C \in \mathbb{R}^{2\times 2}$ nondegenerate such that

$$R = C^{-1}\Lambda C$$

where $\Lambda = \operatorname{diag}(\lambda_1, \lambda_2)$. When both λ_1 and λ_2 are nonnegative, we can take $Q = C^{-1}\operatorname{diag}(\log\lambda_1, \log\lambda_2)C$. When λ_1 or λ_2 is negative, there is also a way to define the log (simply define $\log(-1) = i\pi(2k+1),\ k \in \mathbb{Z}$), though the function is now multiple valued.

When $R = \begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix}$ with $\alpha^2 + \beta^2 = 1$, we have

$$\log R = \begin{bmatrix} 0 & -\theta \\ \theta & 0 \end{bmatrix}$$

with θ is such that $\alpha = \cos\theta$ and $\beta = \sin\theta$.

The remaining case is Jordan block. Note that this a a rare case. It occurs only when two eigenvalues are the same. The log of this case is more involved so we skip it. Please refer to [**C**] for more details. □

In general, the matrices X and K obtained in the theorem are complex. There is a way to compensate this.

Note that the matrix $\Phi(2T) = \Phi(T)^2$ has a real square root. Therefore, $\Phi(2T)$ has a real log. Define $K = \frac{1}{2T}\log\Phi(2T)$ and $X(t) = \Phi(t)e^{-Kt}$, then X is $2T$-periodic.

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Such a theory is important if we want to study the stability of the solution $\Phi(t)$. Since X is periodic, it is always bounded. Then as $t \to \pm\infty$, if $\Phi(t)$ is bounded or not, is completely determined by the matrix K. For instance, if $K = \text{diag}(\kappa_1, \kappa_2)$ with $\kappa_1 < 0$ and $\kappa_2 < 0$, then e^{Kt} converges to zero exponentially as $t \to \infty$ but grows to infinity exponentially as $t \to -\infty$. If $K = \begin{bmatrix} 0 & -\theta \\ \theta & 0 \end{bmatrix}$, then e^{Kt} is periodic.

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CHAPTER 4

The qualitative theory

The main problem in the study of differential equations beyond the wellposedness theory is the dynamics. By the dynamics, we mean that the asymptotic behavior of all the solutions as $t \to \infty$ in the case when the global solution exists. When there is a singularity, i.e. the solution blows up as $t \to t^* < \infty$, we may also study how the singularity is formed. In this course, let us focus on the former. As we know that most equations are not solvable. In general, it is not possible to understand the dynamics by finding the solution then study its asymptotic behavior.

The reason why we study the limiting behavior is that the limiting behavior is relatively easy to study than the behavior of all the solutions. In general, when all the limiting behaviors are understood, then we expect that the large time behavior of solutions can be classified.

We have been studying linear systems of the form $X' = AX$, where X is defined on $\mathbb{R}^n$. We say that $\mathbb{R}^n$ is the phase space. In the case of higher order equations, for instance $ax'' + bx' + cx = 0$, we introduce $y = x'$ and study the linear system for (x, y). In this case, we say that the phase space is $\mathbb{R}^2$. In general, we study equation of the form $X' = F(X)$, where F has no explicit dependence on t. We call such a system autonomous system. When there is t dependence on the RHS, we call it nonautonomous. The phase space is the space where X is defined. It can be $\mathbb{R}^n$ and can also be a manifold. When the phase space is a manifold M, then F would be a vector field defined on M. For instance, we consider $M = \mathbb{T}^n := (\mathbb{R}/\mathbb{Z})^n$ is the torus and F is a constant vector field on M.

Consider the equation $X' = F(X)$ with phase space $\mathbb{R}^n$. Suppose F is C^1 with uniformly bounded derivative. For each initial condition $X(t_0) = X_0 \in \mathbb{R}^n$, the corresponding solution of the IVP is denoted by $\varphi(t, t_0, X_0)$. The key property for an autonomous equation is that if $X(t)$ is a solution to the equation, then $X(t+s)$ is also a solution. By the uniqueness of solution we get that

$$\varphi(t, t_0, X_0) = \varphi(t - t_0, 0, X_0).$$

This means that a solution starting from time t_0 position X_0 traveling time t is the same as the solution starting from time 0 position X_0 traveling time $t - t_0$. We thus introduce the map $\phi^t : \mathbb{R}^n \to \mathbb{R}^n$ defined by $\phi^t(X_0) := \varphi(t, 0, X_0)$. Then we have the differentiable dependence on the initial condition. Moreover, by the uniqueness of solutions, we see that ϕ^t is a diffeomorphism (the map being one-to-one comes from the uniqueness theorem, being onto comes from the uniqueness theorem applied to $X' = -F(X)$ with solution ϕ^{-t}, $\phi^{\pm t}$ being continuous comes from the continuous dependence on the initial conditions of the equation $X' = \pm F(X)$, $\phi^{\pm t}$ being differentiable comes from the differentiable dependence on initial conditions on the initial conditions).

Moreover, ϕ^t generates an abelian group

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(1) ϕ^0 =identity;
(2) $(\phi^t)^{-1} = \phi^{-t}$;
(3) $\phi^t\phi^s = \phi^{t+s}$.

There are some features of Poincaré approach:

(1) It study the diffeomorphism ϕ^t rather than solve each individual IVP, so it is global;
(2) It in general does not care about the quantitative aspect of the solutions, but cares only the topological information.

The basic idea is to first find those invariant objects, including mainly fixed points and periodic orbits, next find orbits connecting these invariant objects. These are the skeleton of the dynamics. When these are done, then in general the behavior of all the other orbits are understood.

Poincaré invented topology (homology theory and homotopy theory) trying to organize the invariant objects to get global information. Both the first homology group and the first homotopy group are defined using loops. Morse theory is a good example in which the global topology is captured by the critical points.

1. Poincaré classification of planar phase portraits

The first step in the qualitative theory is to classify fixed points for linear systems. Let us revisit the linear system with constant coefficients from the viewpoint of Poincaré.

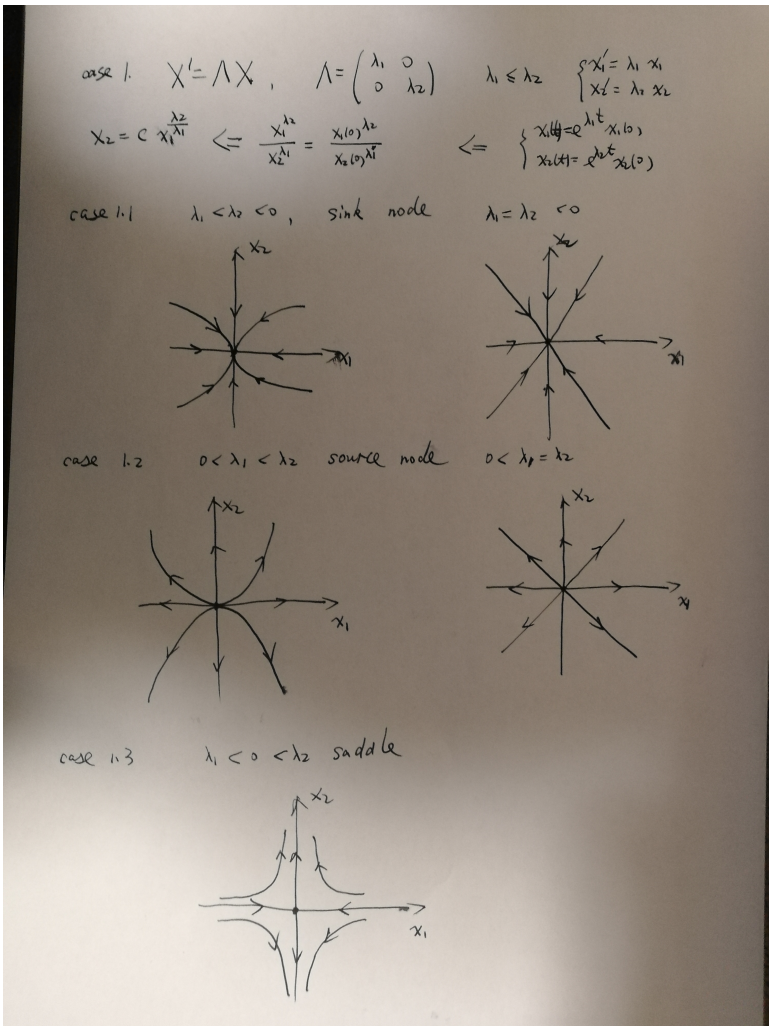
Case 1: diagonal case, assume $\lambda_1 \le \lambda_2$:
case 1.1: $\lambda_1 \le \lambda_2 < 0$, sink;
case 1.2: $0 < \lambda_1 \le \lambda_2$, source;
case 1.3: $\lambda_1 < 0 < \lambda_2$, saddle
Case 2: complex eigenvalues $\alpha \pm i\beta$:
case 2.1: $\alpha > 0$, unstable focus;
case 2.2: $\alpha = 0$, center;
case 2.3: $\alpha < 0$, stable focus.
Case 3: Jordan block $A = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$.
case 3.1: $\lambda < 0$;
case 3.2: $\lambda > 0$.

1.1. The diagonal case. For case 1, we first draw the orbits on the x- and y-axis. The arrows are determined by the signs of the corresponding eigenvalues. If the eigenvalue is negative, then the arrows point to the origin, otherwise, the arrows point opposite to the origin.

If the matrix A is not diagonal, but can be diagonalized in $\mathbb{R}^{2\times 2}$ by a real matrix C with column vectors v_1, v_2. In this case, we know that the vectors v_1 and v_2 are eigenvectors. The solution space is spanned by $e^{\lambda_1 t}v_1$ and $e^{\lambda_2 t}v_2$. So in this case, the x- and y-axis in the diagonal case should be replaced by the directions $v_1\mathbb{R}$ and $v_2\mathbb{R}$ respectively.

1.2. The complex case. For case 2, we study the equations of the form $X' = AX$ where $A = \begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix}$ We can introduce the polar coordinates $r = (x^2+y^2)^{1/2}$

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and $\theta = \arctan \frac{y}{x}$. Then we have

$$r' = \frac{1}{r}(xx' + yy') = \frac{1}{r}(x(\alpha x - \beta y) + y(\beta x + \alpha y)) = \alpha r$$

$$\theta' = \frac{y'x - x'y}{x^2 + y^2} = \frac{1}{r^2}((\beta x + \alpha y)x - (\alpha x - \beta y)y) = \beta.$$

The sign of β determines if the spiral is spiraling clockwise ($\beta < 0$) or counter-clockwise ($\beta > 0$). The sign of α determines if the spiral is shrinking ($\alpha > 0$) or expanding ($\alpha > 0$). When $\alpha = 0$, each orbit is a circle.

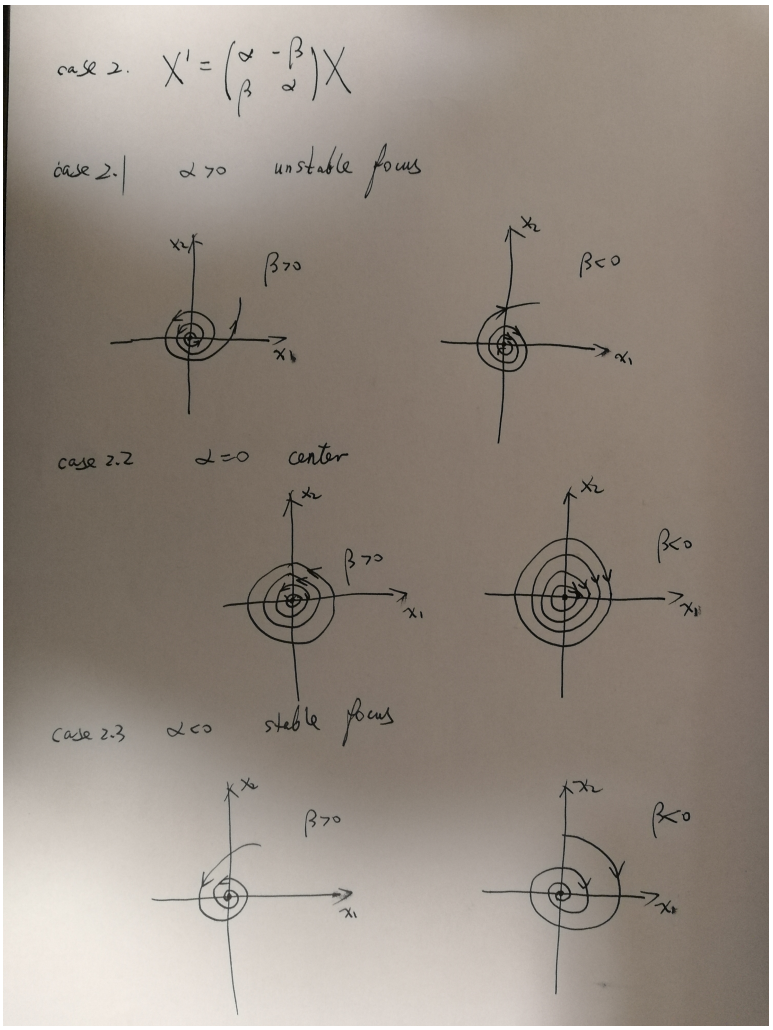
1.3. The Jordan block case. The general solutions to the ODE are

$$c_1 e^{\lambda t}(1, 0) + c_2(e^{\lambda t}(0, 1) + te^{\lambda t}(1, 0)).$$

So we have

$$x_1 = c_1 e^{\lambda t} + c_2 te^{\lambda t}, \quad x_2 = c_2 e^{\lambda t}.$$

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Hence

$$x_1 = \frac{c_1}{c_2} x_2 + x_2 \frac{\log \frac{x_2}{c_2}}{\lambda}.$$

The pictures are drawn as follows.

1.4. The nonlinear case. In general, for a nonlinear equation of the form $X' = F(X)$, we can first find all the fixed points, denoted by $X_1, \ldots, X_n$, by solving the equation $F(X) = 0$.

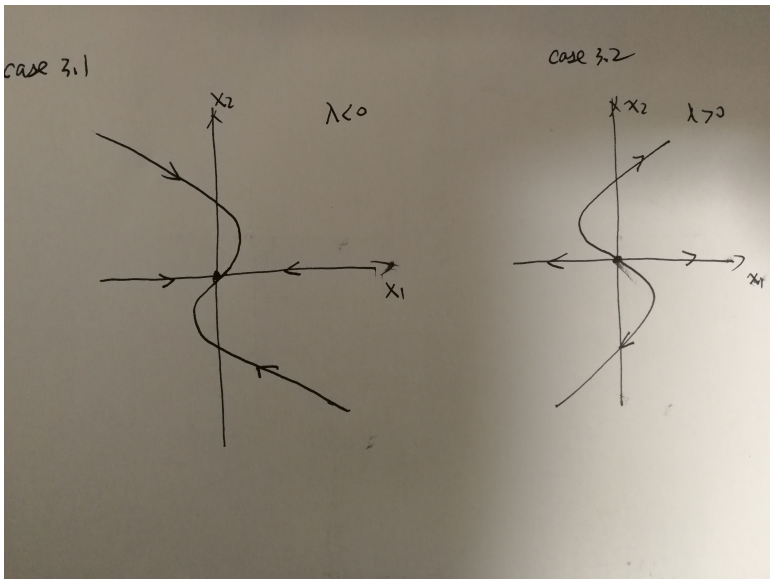
Then we take Taylor expansion of $F(X)$ near X_i to get $F(X) = DF(X_0)(X - X_i) + o(|X - X_i|)$. Discarding the nonlinear term $o(X - X_i)$, we get the linearized equation of $X' = F(X)$ at X_i as

$$X' = DF(X_i)(X - X_i).$$

If we further denote $Y = X - X_i$, since X_i is a constant, we get

$$Y' = DF(X_i)Y.$$

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We can apply the above techniques to analyze the local dynamics of the linearized equation near each fixed point.

We take the pendulum as an example. This is an important prototypical case of second order nonlinear ODE.

The equation is

$$\ddot{x} = \sin x.$$

Introducing $y = \dot{x}$, so we get the system $\begin{cases} \dot{x} &= y \\ \dot{y} &= \sin x \end{cases}$. Since $\sin x$ is 2π-periodic in x, the system is defined on a cylinder $(2\pi\mathbb{T}^1) \times \mathbb{R}^1$.

There are two fixed points given by $(0,0)$ and $(\pi,0)$. Since we have $\sin x = x+o(x)$ for x near 0 and $\sin x = \sin((x-\pi)+\pi) = -\sin(x-\pi) = -(x-\pi)+o(|x-\pi|)$ for x close to π. The linearized system is $\begin{cases} \dot{x} &= y \\ \dot{y} &= x \end{cases}$ near 0 and $\begin{cases} \dot{z} &= y \\ \dot{y} &= -z \end{cases}$ near $z = x - \pi = 0$. In the former case, the fixed point is a saddle and in the latter case the fixed point is a center.

We next multiply the equation by $\dot{x}$ and obtain $\ddot{x}\dot{x} = \sin x\dot{x}$, equivalently we have $(\frac{1}{2}y^2)' = (-\cos x)'$. Integrating this we get $\frac{1}{2}y^2 + \cos x$ is a constant denoted by E.

This is a case which we can get a complete understanding of the phase portrait. This is because our system has an additional structure: there exists a conserved quantity. This kind of systems are called Hamiltonian system, that we will study in details later.

2. The stability theory of Lyapunov

We introduce the notion of stability after Lyapunov. Let $X_*(t)$ be a solution to the equation $X' = F(X),\ X \in \mathbb{R}^n$. Suppose F is uniformly Lipschitz in X so that we always have global solutions.

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DEFINITION 2.1 (Lyapunov stability). (1) *The solution $X_*(t)$ is said to be stable, if for all ε, there exists δ such that for all solution $X(t)$ with $|X(0) - X_*(0)| < \delta$, we have*

$$|X(t) - X_*(t)| < \varepsilon, \ \forall \ t > 0.$$

Otherwise, we say the solution is unstable.

(2) *The solution $X_*(t)$ is said to be asymptotically stable, if it is stable, and there exists δ' such that for all solution $X(t)$ with $|X(0) - X_*(0)| < \delta'$, we have*

$$\lim_{t \to +\infty} |X(t) - X_*(t)| = 0.$$

For linear systems $X' = AX$ on $\mathbb{R}^2$, let us analyze the stability of 0. The following theorem can be proved by reading the phase portraits.

THEOREM 2.2. *Let λ_1 and λ_2 be the two eigenvalues of A.*

(1) *If $Re\lambda_1 < 0$ and $Re\lambda_2 < 0$, then 0 is asymptotically stable;*
(2) *If $Re\lambda_1 > 0$ or $Re\lambda_2 > 0$, then 0 is unstable;*
(3) *If $Re\lambda_1 = Re\lambda_2 = 0$ and there is no Jordan block, then 0 is stable but not asymptotically stable.*
(4) *If $Re\lambda_1 = Re\lambda_2 = 0$ and there is a Jordan block, then 0 is unstable.*

When an equation is nonlinear, we still hope to study the stability of the equilibrium. There is an important method called the method of Lyapunov functions.

Consider the system $X' = F(X)$, or written in components

$$\begin{cases} x' = f(x, y) \\ y' = g(x, y) \end{cases}.$$

Suppose $F(0) = 0$.

THEOREM 2.3. *Suppose there is a C^1 function $V : \{|X| \leq 1\} \to \mathbb{R}$ satisfying $V(0) = 0$ and $V(X) > 0$ for $X \neq 0$.*

(1) *If $\dfrac{d}{dt} V(X(t)) = \nabla V \cdot F \leq 0$, then 0 is stable;*
(2) *If $\dfrac{d}{dt} V(X(t)) = \nabla V \cdot F < 0$ outside 0, then 0 is asymptotically stable.*

PROOF. Let us first prove the item (1). By the continuity of V, the preimage $V^{-1}([0, 1/n]) \subset \mathbb{R}^2$ is a closed set. Moreover the diameter of the set $V^{-1}([0, 1/n])$ goes to zero as $n \to \infty$. Suppose not, then there is a sequence of points x_n with $|x_n| > c$ for some constant and $V(x_n) \to 0$. Pick any accumulation point x_* of $\{x_n\}$, we get $V(x_*) = 0$ by the continuity but $|x_*| \geq c$. This contradicts $V(X) > 0$ for $X \neq 0$.

For any ε, we pick n large such that $V^{-1}([0, 1/n]) \subset B_\varepsilon(0)$. We pick any initial point $x_0 \in V^{-1}([0, 1/n])$, then by the condition $\dfrac{d}{dt} V(\phi^t x_0) \leq 0$ along the flow, we get $V(\phi^t x_0) \leq V(x_0)$ so $\phi^t x_0 \in V^{-1}([0, 1/n]) \subset B_\varepsilon(0)$. This shows that 0 is Lyapunov stable.

We next prove item (2). As above, for all ε, we pick n large such that $V^{-1}([0, 1/n]) \subset B_\varepsilon(0)$, and we pick any initial point $x_0 \in V^{-1}([0, 1/n])$. We need to show that $\phi^t x_0 \to 0$ as $t \to \infty$. It is enough to show that $V(\phi^t x_0) \to 0$ as $t \to \infty$, since the diameter of the set $V^{-1}([0, 1/n])$ converges to zero. We note

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that $V(\phi^t x_0)$ is monotonically decreasing. So the limit $\lim V(\phi^t x_0)$ exists. Suppose $\lim_{t\to\infty} V(\phi^t x_0) = c > 0$. Suppose there is a subsequence $t_n \to \infty$ such that $x_n := \phi^{t_n} x_0 \to x_*$. Then we get $V(x_*) = c$. We can pick a compact annulus region $\mathbb{A}$ such that $\mathbb{A} := V^{-1}([c, 1])$ contains the whole orbit $\{\phi^t x_0\}$ and does not contain 0. Then by the continuity, we get that there is $d < 0$ such that $\frac{d}{dt}V(X) = \nabla V \cdot F < d < 0$ on $\mathbb{A}$. Moreover, we have

$$V(x_n) - V(x_0) = \int_0^{t_n} \frac{d}{dt} V(\phi^t x_0) dt.$$

This shows that $\frac{d}{dt}V(\phi^t x_0) < d < 0$. The RHS goes to $-\infty$. This gives a contradiction. □

We may think that V is a measurement of the distance to the origin. Therefore, the typical way to construct such a Lyapunov function is to consider a quadratic form $X^t BX$ where B is a positive definite matrix. Moreover, in many cases (when the eigenvalues of A are complex), the Lyapunov function can be chosen to be $r^2 = x^2 + y^2$.

The notion of Lyapunov function is widely used in mathematics. Whenever we have some function that is decreasing along the flow, it is called a Lyapunov function or entropy.

Let us consider an example.

$$\begin{cases} x' = -y + x(x^2 + y^2 - 1) \\ y' = x + y(x^2 + y^2 - 1) \end{cases}.$$

We can take then $V = x^2 + y^2$, so we get

$$\frac{d}{dt}V(X) = 2xx' + 2yy' = 2(x^2 + y^2)(x^2 + y^2 - 1).$$

This is less than zero if we restrict to the region where $x^2 + y^2 \leq 1/2$. This shows that 0 is asymptotically stable.

We have the following theorem showing the the linearized dynamics can capture the stability property of the fixed point of the nonlinear equation. The main tool is the Gronwall inequality and the formula for variation of constants.

THEOREM 2.4. *Suppose in the equation $X' = AX + f(X)$, we have*

(1) *all eigenvalues of A has negative real part;*
(2) *$f(t, X) = o(\|X\|)$ uniformly for all $t \in \mathbb{R}$ as $\|X\| \to 0$.*

Then the zero solution is asymptotically stable.

We will leave the proof as an exercise.

3. Structural stability

There are mainly two notions of stability. The first one is about perturbing the *initial data* then see if the perturbed solution remains close to the unperturbed one or not. The Lyapunov stability that we discussed in the previous section is of this kind. The second type is called structural stability, which means that if we perturb the *equation* slightly, then see if the phase portrait of the perturbed equation remains topologically equivalent to the unperturbed one or not. I will give more precise definitions slightly later.

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Recall that when we studied the continuous or the differentiable dependence on the initial conditions, we treat the change of initial conditions in the same way as the change of equations by considering that the ODE depends on a parameter and we study the dependence of the solutions with respect to the parameter. In other words, in the section of wellposedness, we did not distinguish perturbing the initial condition and perturbing the equation.

However, here we use two notions, stability and structural stability, to distinguish perturbing initial conditions and perturbing the equations.

Compared to the wellposedness part, the main difference is that in the wellposedness theory, we only care about a fixed time interval $[0, \delta]$. However, in this chapter, we care about the behaviors of solutions as $t \to \infty$. This causes an essential difference between the wellposedness theory for differential equations and the dynamics.

3.1. Hopf bifurcation and Poincaré bifurcation. In this section, we give two classical examples of structural instability.

3.1.1. *Hopf bifurcation.* We consider the system

$$\begin{cases} x_1' &= \varepsilon x_1 - x_2 - x_1(x_1^2 + x_2^2) \\ x_2' &= x_1 + \varepsilon x_2 - x_2(x_1^2 + x_2^2) \end{cases}.$$

We can convert to the polar coordinates to get

$$\begin{cases} r' &= \frac{x_1 x_1' + x_2 x_2'}{r} = r(\varepsilon - r^2) \\ \theta' &= \frac{x_2' x_1 - x_1' x_2}{r^2} = 1 \end{cases}.$$

When $\varepsilon \leq 0$, we have $r' < 0$ if $r \neq 0$, so the phase portrait looks like a stable focus. When $\varepsilon > 0$ and $0 < |r|^2 < \varepsilon$ we have $r' < 0$, when $r^2 = \varepsilon$, we have $r' = 0$ and when $r^2 > \varepsilon$, we have $r' > 0$. So we can draw the phase portrait as follows.

The point here is that when we perturb the $\varepsilon = 0$ case of the equation, depending on the signs of ε, the phase portrait looks drastically different: the $\varepsilon \leq 0$ case looks like a stable focus while the $\varepsilon > 0$ case contains a limiting cycle.

The birth of this limiting cycle is because 0 is a stable focus for $\varepsilon < 0$ and is an unstable focus for $\varepsilon > 0$. However, away from 0 in the case of $\varepsilon > 0$, the flow is still pushing inward. This forces a limiting cycle according to the famous Poincaré-Bendixon theorem related to this. Here I give only a statement without proof.

THEOREM 3.1 (Poincaré-Bendixon). *Let D be an open annulus domain enclosed by two simple closed curves C_1 and C_2. Suppose the ODE $X' = F(X)$ has no fixed point in $\bar{D} = C_1 \cup D \cup C_2$. Suppose along the boundaries C_1 and C_2, the vector field F points inside of D. Then there is a closed orbit of the equation in D that is not contractible to a point.*

As a homework problem, please try to give a statement without proof to generalize the phenomenon.

3.1.2. *Poincaré bifurcation.* There is another type of bifurcation called Poincaré bifurcation. Let us consider the following example.

$$\begin{cases} x_1' &= -x_2 - \varepsilon x_1(x_1^2 + x_2^2 - 1 - \varepsilon) \\ x_2' &= x_1 - \varepsilon x_2(x_1^2 + x_2^2 - 1 - \varepsilon) \end{cases}.$$

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We can convert to the polar coordinates to get

$$\begin{cases} r' &= \varepsilon r(r^2 - (1-\varepsilon)) \\ \theta' &= 1 \end{cases}.$$

If $|\varepsilon|$ is sufficiently small, regardless the sign of ε, there is always a limiting cycle $r = \sqrt{1-\varepsilon}$.

However, when $\varepsilon = 0$, the phase portrait is of center type. Every circle centered at zero is invariant under the flow.

So the phase portrait again has a drastic change when ε goes from negative to positive. However, the appearance of the limiting cycle is not born from the origin but from a circle in the unperturbed system.

3.2. Grobman-Hartman theorem and Anosov Theorem. The above examples of drastic topological change of the phase portrait is essentially due to the fact that in the $\varepsilon = 0$ case, the linearized system has eigenvalues with zero real part.

In this section, we discuss the structural stability theory. The materials of this section follows from the book [**A2**], Section 10 and Section 13. We first give a general definition for the structural stability following [**A2**].

DEFINITION 3.2. *A system $X' = F(X)$ is called structural stable, if any small perturbation in the vector field gives rise to a system that is equivalent to the original one.*

This definition is a bit vague. Here we formulate two notions of equivalence.

DEFINITION 3.3. *A system ϕ^t, $t \in \mathbb{R}$, is said to be topologically (respectively diffeomorphically) equivalent to ψ^t, $t \in \mathbb{R}$, on $\mathbb{R}^n$, if there is a homeomorphism (respectively diffeomorphism) $h : \mathbb{R}^n \to \mathbb{R}^n$ such that*

$$h \circ \phi^t = \psi^t \circ h.$$

The following theorem is a classical result about the structural stability.

THEOREM 3.4 (Grobman-Hartman). *Consider the equation $X' = F(X)$ where F is C^1 in a small neighborhood of 0. Suppose 0 is a hyperbolic critical point of F, i.e. the linearization $DF(0)$ has all the eigenvalues with nonzero real part, then there exist neighborhoods U and V of 0 and a homeomorphism $h : U \to V$ such that the phase portrait of the equation $X' = F(X)$ is transformed into that of $X' = DF(0)X$ preserving direction.*

We give a few remarks.

REMARK 3.5. *The statement of the GH theorem only asserts that h sends the phase portrait of one equation to the other, which is slightly weaker than topological equivalence of the two systems. Suppose $X(t)$, $t \in \mathbb{R}$, is an orbit of $X' = F(X)$ and $\bar{X}(t)$, $t \in \mathbb{R}$, is an orbit of $X' = DF(0)X$. The theorem only asserts that h sends the whole orbit $\{X(t) \mid t \in \mathbb{R}\}$ to the whole orbit $\{\bar{X}(t) \mid t \in \mathbb{R}\}$, but does not claim that $h(X(t)) = \bar{X}(t)$ for each individual $t \in \mathbb{R}$. That means the speed of the flow on the two orbits may be different. To get an authentic topological equivalence, we need also introduce a time reparametrization into h. This is quite a subtle problem. So in the following, when talking about the structural stability, we do not work on flows but work on maps.*

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This theorem says that the hyperbolicity is a mechanism of having structural stability. Note that hyperbolicity in general does not have Lyapunov stability. This is indeed a rather general phenomenon.

Hyperbolicity is a crucial concept in dynamical system. In geometry, it is related to the negative Ricci curvature. In probability it is related to the randomness and the independence.

The proof is a bit involved. Instead, we prove the following version of Grobman-Hartman for maps.

We first consider the map $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} : \mathbb{T}^2 \to \mathbb{T}^2$, which is called Arnold's cat map(see Section 13 of [**A2**]). This is a crucial prototypical hyperbolic dynamical system. Let us first analyze its preliminary properties.

The matrix A has two eigenvalues $\lambda_1 = \frac{3+\sqrt{5}}{2} > 1 > \lambda_2 = \frac{3-\sqrt{5}}{2}$. The corresponding eigenvectors v_1 and v_2 both have irrational slopes. So each line $v_1\mathbb{R}$ and $v_2\mathbb{R}$ is dense on the torus $\mathbb{T}^2$. Moreover, if two points x_1 and x_2 satisfies $(x_1 - x_2) \parallel v_1$, then we know that $A^n x_1 - A^n x_2 = A^n(x_1 - x_2) = \lambda_1^n(x_1 - x_2)$ when this operation is done on $\mathbb{R}^2$. We see that the distances of the images of x_1 and x_2 under A^n as $n \to \infty$ grows exponentially. When projected to the $\mathbb{T}^2$ by quotient $\mathbb{Z}^2$, the two points $A^n x_1$ and $A^n x_2$ may be very chaotic, not predictable. So this map is very unstable in the sense that if we perturb the initial condition slightly, then after sufficiently long time, the two orbits will diverge exponentially fast and result in big differences.

There are some other interesting behaviors of this map such as periodic orbits being dense, being ergodic and mixing, etc, that I do not discuss in this course. Please refer to [**A2**] for more details.

Similar behavior is seen from the geodesic flow on the surface of negative curvature. Nearby orbits may diverge exponentially.

With these preparations, now it is remarkable to have the following structural stability theorem. Instead of perturbing the initial condition, we perturb the map, then the perturbed map is topologically equivalent to the original one.

THEOREM 3.6 (Structural stability, Anosov). *Suppose $f : \mathbb{T}^2 \to \mathbb{T}^2$ is a small C^1 perturbation of the Arnold cat map $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$. Then there exists $H : \mathbb{T}^2 \to \mathbb{T}^2$ homeomorphism, such that $H \circ f = A \circ H$.*

REMARK 3.7. *The map $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ can be replaced by any matrix $A \in \mathrm{SL}(2, \mathbb{Z})$ with $|trA| > 2$.*

PROOF. Suppose $f = A + g$, $g : \mathbb{T}^2 \to \mathbb{R}^2$. By assumption, we have

$$\|g\|_{C^1} := \max_{x \in \mathbb{T}^2} |g(x)| + \max_{x \in \mathbb{T}^2} |Dg(x)| \ll 1.$$

We will give explicit upper bound for $\|g\|_{C_1}$ in the proof. We suppose $H^{-1}(x) = x + h(x)$, where $h : \mathbb{T}^2 \to \mathbb{R}^2$ is expected to be small in the C^0 norm. We get from the cohomological equation $fH^{-1} = H^{-1}A$, the following equation.

$$h(Ax) - Ah(x) = g(x + h(x)).$$

The equation is nonlinear so as the first step we consider the linearized equation

$$h(Ax) - Ah(x) = g(x).$$

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We can abbreviate this expression as $Lh = g$. We next show that we can invert L to find h for each given g.

Suppose $g(x) = g_1(x)v_1 + g_2(x)v_2$, $h(x) = h_1(x)v_1 + h_2(x)v_2$. Here h_i, g_i are functions from $\mathbb{T}^2$ to $\mathbb{R}$. We have $Av_i = \lambda_i v_i,\ i = 1, 2$. Projecting to the eigen directions we get two equations

$$h_i(Ax) - \lambda_i h_i(x) = g_i(x).$$

We have assumed $\lambda_1 > 1 > \lambda_2 := \lambda$. To solve the equation we introduce an operator $S:\ C^0(\mathbb{T}^2, \mathbb{R}) \to C^0(\mathbb{T}^2, \mathbb{R})$ via $(S\phi)(x) := \phi(Ax)\, x \in \mathbb{T}^2$. The norm of S is defined as

$$\|S\| := \sup_{\|\phi\|_{C^0}=1} \|S\phi\|_{C^0} = 1,\ \|\phi\|_{C^0} = \max_{x\in\mathbb{T}^2} |\phi(x)|.$$

We also have $S^{-1}\phi = \phi \circ A^{-1}$ and $\|S^{-1}\| = 1$. We get the expression

$$(S - \lambda_i)h_i(x) = g_i(x).$$

Using the Taylor expansion $(1 - x)^{-1} = 1 + x + x^2 + \dots$, we have the inverse

$$(S - \lambda_1 I)^{-1} = -\lambda(I + \lambda S + \lambda^2 S^2 + \cdots).$$

We get the estimate

$$\|(S - \lambda_1 I)^{-1}\| \le \frac{\lambda}{1 - \lambda}.$$

Similarly, we have

$$\|(S - \lambda_2 I)^{-1}\| \le \frac{1}{1 - \lambda}.$$

In this way, we are able to invert L to get $h = L^{-1}g$ and get an estimate $\|L^{-1}\| \le \frac{1}{1-\lambda}$.

We next consider the nonlinear equation $Lh(x) = g(x + h(x))$. Denote by

$$\Phi(h)(x) = g(x + h(x)) - g(x).$$

So the conjugation equation is

$$Lh = \Phi h + g,$$

The remaining work is to find a fixed point in $C^0(\mathbb{T}^2, \mathbb{R}^2)$ for

$$h = L^{-1}(\Phi h + g).$$

It is enough to check that

$$\begin{aligned}\|L^{-1}(\Phi h_1 + g) - L^{-1}(\Phi h_2 + g)\|_{C^0} &\le \|L^{-1}\|\|\Phi h_1 - \Phi h_2\|_{C^0} \\ &\le \|L^{-1}\|\|Dg\|_{C^0}\|h_1 - h_2\|_{C^0}\end{aligned}$$

Since g is C^1 small, we see that the map is $L^{-1}(\Phi h + g)$ contracting, so we get a fixed point.

Now we get that h is small in $C^0(\mathbb{T}^2, \mathbb{R}^2)$. We need to show that $H^{-1} := Id + h$ is a homeomorphism. We consider the equation $fH^{-1} = H^{-1}A$ on $\mathbb{R}^2$. Suppose H^{-1} is not injective, then $H^{-1}(x) = H^{-1}(y)$ for some $x \ne y$. Then we have $f^n H^{-1}x = H^{-1}A^n x$ and $f^n H^{-1}y = H^{-1}A^n y$. Taking difference we get $H^{-1}A^n x = H^{-1}A^n y$, so $A^n x + h(A^n x) = A^n y + h(A^n y)$. Since h is C^0 bounded, by taking $n \to \infty$ or $n \to -\infty$, we have $A^n(x - y) \to \infty$, we get a contradiction.

We next show H^{-1} is surjective. Since h is C^0 bounded, we can pick a large ball in $\mathbb{R}^2$ whose image under $H^{-1} = Id + h$ contains a unit square, so when projected to the $\mathbb{T}^2$, the image covers the whole $\mathbb{T}^2$. We fix the unit square $[0, 1]^2 \subset \mathbb{R}^2$.

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Suppose a point $(x, y) \in [0, 1]^2$ is not covered by the image of the ball B under $H^{-1} = \mathrm{Id} + h$. Then we consider the function $G : B \to \mathbb{R}$ by restricting the image of H^{-1} to the slice $\mathbb{R} \times \{y\}$. The function G is continuous since H^{-1} is, so it attains the max and min and all the values between the max and min. When $h = 0$, the function G is identity from $B \cap (R \times \{y\})$ to itself, so the max is the right most point (denoted by M) of $B \cap (R \times \{y\})$ and the min is the left most point (denoted by m). Suppose both points have large modulus by choosing B large. Now since h is small in the C^0 norm, the max and min are slightly perturbed, so G attains all the values between $M - \|h\|_{C^0}$ and $m + \|h\|_{C^0}$ by the continuity.

□

It is important to notice that the above conjugacy h is only continuous and cannot be made C^1. There is an obvious obstruction for improving the regularity to C^1. Note that 0 is a fixed point of A hence $h^{-1}(0)$ is a fixed point of f. If h were differentiable, then differentiating the equation $hfh^{-1} = A$ then evaluate it at the point 0, we get that $Df(h^{-1}(0))$ as a matrix is similar to A. However, it is easy to perturb $Df(h^{-1})$ so that it is not similar to A.

4. The globalization, the indices of singularities

We have explained before that the qualitative theory consists of two parts: the local theory and the global theory. For the local theory, we need to find the special orbits in the phase space such as fixed points, limit cycles. These special orbits are usually those in the ω-limit set. Once the ω-limit set is well-understood, then the behaviors of all other orbits can be understood. In the local theory, we also study the stability of the fixed points and the structural stability. To summarize, when there is hyperbolicity, then the dynamics in a neighborhood of a hyperbolic fixed points is essentially given by the linearized system

The global theory is to relate the special orbits to the global topology of the phase space. In this section, we give a theorem relating the local information of each fixed point to the global topological information of the manifold. We state the following Poincaré-Hopf theorem.

THEOREM 4.1 (Poincaré-Hopf). *Let V be a smooth vector field on a compact manifold M with finitely many zeros, then the sum of the indices of all the zeros is the Euler characteristic $\chi(M)$ of M.*

In history, Poincaré proved the theorem in the case of surfaces and Hopf proved it for general manifolds.

The proof of this theorem is a bit involved (see [M1, A4]) using knowledge from differential topology. It may also need some algebraic topology.

In this section, we will introduce the index theory and the ODE on manifolds so that we can appreciate the statement of Poincaré-Hopf.

For example, we can consider the 2-sphere and two vector fields on it. The first vector field is tangent to each latitude and the second is tangent to each longitude. Each of the vector field has zeros at the south pole and the north pole. We will see below that each zero has index 1 so for each vector field, the sum of the indices of the zeros is 2, which is exactly the Euler characteristic of the sphere.

One more example, on the $\mathbb{T}^2$, we can give a constant vector field for each given fixed slope. There is no zero for such a vector field. So the indices of zeros of this vector field is 0 which is again the Euler characteristic of $\mathbb{T}^2$. The main feature of

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this theorem is that the local information provided by the fixed points adds up to yield the global topology of the manifold.

Before talking about the index theory and the manifold, we explain some theorems related closely to Poincaré-Hopf. One theorem is the famous Gauss-Bonnet formula, which states that on a closed surface Σ, we have $\int_\Sigma K(x)dx = 2\pi\chi(\Sigma)$, where $K(x)$ is the Gauss curvature at $x \in \Sigma$. There is a higher dimensional generalization of Gauss- Bonnet, which also gave birth to the theory of Chern class, etc. The relation to the Poincaré-Hopf is as follows. One may think that the curvature is concentrated on zeros of a vector field and vanishes at those nonzero points of the vector field.

Poincaré-Hopf is also related to the Morse theory. Let us give a brief introduction.

Let M be a closed manifold and $f: M \to \mathbb{R}$ be a C^2 function. Suppose all the critical points of f are nondegenerate, i.e. when $\nabla f(x) = 0$, we have $\nabla^2 f(x)$ is nondegenerate. Since $\nabla^2 f(x)$ is a symmetric matrix, we can introduce the Morse index for x defined as the number of negative eigenvalues of $\nabla^2 f(x)$.

THEOREM 4.2. *Let m_i be the number of critical points of f with Morse index i. Then we have*

$$\sum_{i=0}^{n} (-1)^i m_i = \chi(M).$$

The relation to Poincaré-Hopf is as follows. The gradient ∇f is a vector field on M and the critical points are zeros of the vector field. It turns out that the index of each zero is $(-1)^i$ where i is the Morse index. Morse theory has a lot more contents than we state here. It has a modern interpretation by Witten and has some important offspring such as the Floer theory in symplectic geometry.

4.1. The index of a fixed point. We first introduce the notion of rotation degree of a simple closed curve.

DEFINITION 4.3. *Let $V: \mathbb{R}^2 \to \mathbb{R}^2$ be a continuous vector field and $V(0) = 0$. Let $\gamma : \mathbb{S}^1 \to \mathbb{R}^2$ be a simple closed curve which does not pass through zero. We next introduce a function $G: \mathbb{S}^1 \to \mathbb{S}^1$ as $G(t) = \frac{V(\gamma(t))}{\|V(\gamma(t))\|}$. The image $G(\mathbb{S}^1)$ covers $\mathbb{S}^1$ for j times for some $j \in \mathbb{Z}$. We call j the rotation degree of the curve γ relative to the vector field V.*

REMARK 4.4. *Here the map G is the Gauss map in differential geometry.*

The definition can be understood as follows. At each point $\gamma(t)$, $t \in \mathbb{S}^1$, there is vector $V(\gamma(t))$. We then normalize it to a unit vector $G(t)$ lying on the unit circle $\mathbb{S}^1$. As t winds around $\mathbb{S}^1$, the unit vector $G(t)$ winds around $\mathbb{S}^1$ for some $j \in \mathbb{Z}$ times.

Suppose we have the differential equation $\begin{cases} x_1' &= P(x_1, x_2) \\ x_2' &= Q(x_1, x_2) \end{cases}$ on the plane with $P(0,0) = Q(0,0) = 0$.

LEMMA 4.5. *The the rotation degree of γ can be computed using the integral*

$$j = \frac{1}{2\pi} \oint_\gamma d \arctan \frac{Q(x_1, x_2)}{P(x_1, x_2)} = \frac{1}{2\pi} \oint_\gamma \frac{PdQ - QdP}{P^2 + Q^2}.$$

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PROOF. Denote $\theta(x_1, x_2) = \arctan \frac{Q(x_1,x_2)}{P(x_1,x_2)}$, which is the angle formed by the vector $(P(x_1, x_2), Q(x_1, x_2))$ with the x_1-axis. When the point (x_1, x_2) moves along γ, the integral $\frac{1}{2\pi} \oint_\gamma d\theta$ computes how much the angle changes, which is exactly the same as rotation degree. □

LEMMA 4.6. *Suppose the loop γ does not enclose a zero of the vector field, then $j = 0$.*

PROOF. Suppose γ is the boundary of a disk D. The assumption implies that $P^2 + Q^2 \neq 0$ on D. So we can apply Green's formula to get

$$j = \frac{1}{2\pi} \oint_\gamma d\theta = \frac{1}{2\pi} \oint_{\partial D} d\theta = \frac{1}{2\pi} \oint_D d^2\theta = 0.$$

□

The following fact is intuitively clear. However, its proof is a bit involved. We state it but skip the proof.

COROLLARY 4.7. *If γ is a closed orbit of a vector field, then the index of γ is not zero hence γ encloses a fixed point.*

LEMMA 4.8. *Suppose γ_1 and γ_2 are two simple closed curves enclosing an annulus region whose closure does not contain a zero of the vector field, then the rotation degrees of γ_1 and γ_2 are equal.*

PROOF. We draw a line segment L connecting the two boundaries of the annulus. Then we form a simple closed curve C by concatenating γ_1, γ_2 and the extra line segments $\pm L$. Note that C does not enclose any zero of the vector field by assumption, we get that the rotation degree of C is zero.

$$0 = \oint_C d\theta = \oint_{\gamma_1 - \gamma_2 + L - L} d\theta = \oint_{\gamma_1} d\theta - \oint_{\gamma_2} d\theta.$$

□

DEFINITION 4.9. *Suppose $V(0) = 0$ and $V \neq= 0$ in a small neighborhood of 0 except 0. Let γ be a simple closed curve enclosing 0 but does not enclose any other zero of the vector field V. We define the index of 0 as the rotation degree of γ relative to the vector field V.*

Since the index takes only integer values, we get the following important property of index.

LEMMA 4.10. *The rotation degree of a simple closed curve γ is invariant under a continuous deformation of the vector field, provided during the deformation process, no zero appears on γ.*

The lemma is again intuitively clear but its proof needs differential topology and we skip it.

LEMMA 4.11. *The index of a saddle is -1, and that of a node or a center is 1.*

PROOF. We suppose the equation has the form

$$\begin{cases} x_1' &= ax_1 + bx_2 + P_2(x_1, x_2) \\ x_2' &= cx_1 + dx_2 + Q_2(x_1, x_2) \end{cases}.$$

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where P_2, Q_2 consist of quadratic and higher order terms and $ad - bc \neq 0$. We claim that the index of zero equals $\frac{ad-bc}{|ad-bc|}$.

We first pick a sufficiently small circle $\{x_1^2 + x_2^2 = \varepsilon^2$, for $\varepsilon \ll 1$. Then we continuously deform the vector field to kill P_2 and Q_2 . For the remaining linear equation, one can either choose $x_1 = \varepsilon \cos\theta$ and $x_2 = \varepsilon \sin\theta$ and compute the index explicitly using the formula above, or we can perform a second continuous deformation as follows.

The general linear group $\mathrm{GL}(2,\mathbb{R}) := \{M \in \mathbb{R}^{2\times 2} \mid \det M \neq 0\}$ has two connected components corresponding to $\det > 0$ and $\det < 0$ respectively. The fact can be proved as follows. Pick $M \in \mathrm{GL}(2,\mathbb{R})$. We assume M has distinct eigenvalues, since otherwise, we can pick another M' sufficiently close to M with distinct eigenvalues. Then there exists $C \in \mathrm{GL}(2,\mathbb{R})$ such that $C^{-1}MC = \Lambda$ is either $\Lambda = e^{\alpha} \begin{bmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{bmatrix}$ or $\Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$. We may even choose C to satisfy $\det C > 0$ by multiplying -1 to a column if otherwise. In the first case, we continuously deform Λ to I by sending α and β to 0. During the process, the matrix M is deformed continuously to I.

In the second case, when $\lambda_1 > 0$ and $\lambda_2 > 0$, we continuously deform Λ to I along the path Λ^t , $t \in [0,1]$. So M is deformed to I. When $\lambda_1 < 0$ and $\lambda_2 < 0$, we first deform $-\Lambda$ to I, then deform $-I$ to I similar to the above first case.

When $\lambda_1\lambda_2 < 0$, without loss of generality, we assume $\lambda_1 > 0, \lambda_2 < 0$, we continuously deform C to identity since $\det C > 0$, and λ_1 to 1 via λ_1^t, $t \in [0,1]$ and λ_2 to -1 via $(-\lambda_2)^t$, $t \in [0,1]$. So M is deformed to $\mathrm{diag}\{1,-1\}$.

With this fact, one can either deform the vector field to $V(x_1,x_2) = (x_1,x_2)$ for nodes or centers or to $V(x_1,x_2) = (-x_1,x_2)$ for saddles. Then the index can be computed using the formula above. □

THEOREM 4.12. *Let γ be a simple closed curve, then its rotation degree relative to a vector field equals the sum of the index of zeros of the vector field enclosed by γ.*

COROLLARY 4.13.
(1) *If γ is a closed orbit and encloses only one fixed point, then the fixed point cannot be a saddle.*
(2) *If γ is a closed orbit and encloses finitely many nondegenerate fixed points (DV at each fixed point is nondegenerate), then the number of fixed points is odd and the number of saddles is one less than zeros of other types.*

4.2. ODE on a manifold. We briefly introduce the notion of a manifold. I try to avoid the abstract definition. Here I use some ideas from [**M1**]. For rigorous and formal definitions, please refer to [**A4, M1**] and any book on differentiable manifolds.

A subset $M \subset \mathbb{R}^k$ is called a smooth manifold of dimension m if for each point $x \in M$, there is a neighborhood $x \in U \subset M$ such that U is diffeomorphic to an open ball in the the Euclidean space $\mathbb{R}^m$.

Since we consider M as being embedded in the Euclidean space, we also introduce the notion of a tangent space utilizing the embedding.

For each $x \in M$, the tangent space T_xM is a copy of R^m that is tangent to M at x, in other words, sufficiently close to x, the copy of R^m intersects M at only one point x.

We define the tangent bundle as $TM := \cup_{x\in M} T_xM$.

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For a smooth curve $\gamma : (-\varepsilon, \varepsilon) \to M$ with $\gamma(0) = x$, the tangent vector $\gamma'(0)$ is a point in T_xM.

A vector field $V : M \to TM$ on M is a function that assigns to each $x \in M$ a point in T_xM.

Given a vector field, we may consider ODE $x' = V(x)$ defined by the vector field. For each initial condition $x(0) = x_0 \in M$, the solution to the IVP gives a curve $x(t)$ on M passing through $x_0 \in M$ and the tangent vector $x_0(t)$ at each point $x(t)$ is exactly $V(x(t))$.

5. Invariant manifolds

The most important case in the above picture is the saddle case. There is a rich theory around hyperbolicity. We now only give the tip of an iceberg.

Let us recall the equation $\begin{cases} x_1' &= x_1 \\ x_2' &= -x_2 \end{cases}$.

The x_1-axis is identified with the set

$$W^u := \{X \in \mathbb{R}^2 \mid \phi^t(X) \to 0, \quad t \to -\infty\},$$

and the x_2-axis is identified with the set

$$W^s := \{X \in \mathbb{R}^2 \mid \phi^t(X) \to 0, \quad t \to +\infty\}.$$

We call W^u the *unstable manifold* of 0 and W^s the *stable manifold* of 0. All points that is not on W^u or W^s escape to infinity as $t \to \pm\infty$.

A stable or an unstable manifold is also called an *invariant manifold*.

We prove a theorem about invariant manifolds in the simplest setting. It basically says that the stable and unstable manifold persists small perturbations. They also exists for nonlinear systems whose linearization is saddle type. In general, the stable and unstable manifold is as smooth as the RHS of the differential equation. However, we will only prove that they are Lipschitz. Note that this result does not follow from the Grobman-Hartman theorem since the conjugacy is only a homeomorphism.

We denote $e_1 = (1, 0) \in \mathbb{R}^2$ and $e_2 = (0, 1) \in \mathbb{R}^2$ and by $\mathbb{R}e_1$ the x-axis and by $\mathbb{R}e_2$ the y-axis.

THEOREM 5.1. *Consider the ODE $X' = AX + f(X)$ where the Lipschitz function $f(X) = o(X)$ as $X \to 0$ and $A = \operatorname{diag}\{\lambda_1, \lambda_2\}$ with $\lambda_1 < 0, \lambda_2 > 0$. Then There exist $\delta > 0$ small enough and stable and unstable manifolds W^s_δ and W^u_δ in a small neighborhood B_δ of 0 such that*

1. *W^s_δ is the graph of a Lipschitz function $g^s : \mathbb{R}e_1 \cap B_\delta \to \mathbb{R}e_2$ satisfying $Dg^s(0) = 0$.*
2. *W^s_δ is invariant under the flow ϕ^t, i.e. $\phi^t W^s_\delta \cap B_\delta = W^s_\delta$ for $t < 0$.*

The statements for W^u are similar.

REMARK 5.2.

1. *In fact, it can be proved that W^u and W^s are as smooth as f. However, we skip the proof here.*
2. *global stable and unstable manifolds can be constructed as*
$$W^u = \cup_{t \geq 0} \phi^t W^u_\delta, \quad W^s = \cup_{t \leq 0} \phi^t W^s_\delta.$$
3. *In the ball B_δ for δ sufficiently small, the local invariant manifolds W^u_δ and W^s_δ are very close to the y and x-axes. However, in general we do not have any control outside a neighborhood of 0.*

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PROOF. We assume that f is Lipschitz on $\mathbb{R}^2$ with Lipschitz constant δ. Otherwise, we replace $f(X)$ by $f(X)\chi(|X|/\delta)$ where $\chi(x) : \mathbb{R} \to \mathbb{R}$ is a C^∞ function that is 1 for $|x| \le 1$ and is 0 for $|x| \ge 2$. We are only interested in the pieces of invariant manifolds inside B_δ.

From the DuHamel principle we get

$$\phi^t(X_0) = X(t) = e^{At}X_0 + \int_0^t e^{A(t-s)} f(X(s))\, ds.$$

This gives

$$|x(t)|e^{-\lambda_1 t} \le |x(0)| + \delta \int_0^t (|x(t)| + |y(t)|)e^{-\lambda_1 s}\, ds, \quad t \ge 0,$$

$$|y(t+\tau)|e^{-\lambda_2(t+\tau)} \le |y(t)|e^{-\lambda_2 t} + \delta \int_\tau^0 (|x(t+s)| + |y(t+s)|)e^{-\lambda_2(t+s)}\, ds, \quad 0 \le t+\tau \le t.$$

We next apply Gronwall's inequality to get

(1) if $|y(s)| \le k_1|x(s)|$ for all $s \in [0,t]$, then

$$|x(t)| \le |x(0)|e^{(\lambda_1 + \delta(1+k_1))t}. \tag{5.1}$$

(2) if $|x(s)| \le k_2|y(s)|$ for all $s \in [0,t]$, then

$$|y(t)| \ge |y(0)|e^{(\lambda_2 - \delta(1+k_2))t}. \tag{5.2}$$

We introduce the cone given $\lambda > 0$

$$\mathcal{K}_\lambda := \{(x,y) \in \mathbb{R}^2 \mid \lambda|x| \le |y|\}.$$

We start with the lemma.

LEMMA 5.3. (1) *Given $\lambda > 0$, there exists δ_0 sufficiently small such that for all $\delta < \delta_0$, the cone $\mathcal{K}_\lambda$ is positively invariant, i.e. $\phi^t(\mathcal{K}_\lambda) \subset \mathcal{K}_\lambda$, $t \ge 0$. Moreover, if $(x(0), y(0)) \in \mathcal{K}_\lambda$, then*

$$|y(t)| \ge |y(0)|e^{(\lambda_2 - \delta(1+\lambda^{-1}))t}.$$

(2) *More generally, if $(x_2(0), y_2(0)) \in (x_1(0), y_1(0)) + \mathcal{K}_\lambda$, then the solution $(x_2(t), y_2(t)) \in (x_1(t), y_1(t)) + \mathcal{K}_\lambda$ and we have*

$$|y_1(t) - y_2(t)| \ge |y_1(0) - y_2(0)|e^{(\lambda_2 - \delta(1+\lambda^{-1}))t}.$$

PROOF. We prove only (1). The other statement is similar.

We consider $(x_0, y_0) \in \partial K_\lambda \setminus \{0\}$, so we have $|y_0| = \lambda|x_0| \ne 0$. For small t, by the continuity of the flow, we get for $s \in [0,t]$, we have

$$|y(s)| \le (\lambda + \varepsilon)|x(s)|, \quad |x(s)| \le (\lambda - \varepsilon)^{-1}|y(s)|.$$

Then (5.1) and (5.2) imply

$$\frac{|x(t)|}{|y(t)|} \le \frac{|x_0|}{|y_0|} e^{((\lambda_1 - \lambda_2) + O(\varepsilon))t} < \lambda^{-1}.$$

We then repeat the argument from time t. □

Fix $x_0 \in \mathbb{R} \setminus \{0\}$, we introduce

$$B(x_0) = \{y_0 \in \mathbb{R} \mid |y_0| \le \lambda|x_0|\}$$

$$G_t(x_0) = \{y_0 \in B \mid |y(t)| \le \lambda|x(t)|\}.$$

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By Lemma, we get $G_t \subset G_s$ for $0 \le s \le t$. We claim that $G_t \neq \emptyset$ for all $t \ge 0$, which then implies $G_\infty(x_0) := \cap_{t\ge 0} G_t(x_0) \neq \emptyset$.

Assuming this for the moment, we get for all $x_0 \in \mathbb{R}$, we can associate some $y_0 \in G_\infty(x_0)$ to it.

First we show that $G_\infty(x_0)$ is a singleton so that the map $x_0 \mapsto G_\infty(x_0)$ is well-defined. Suppose there are two points y_1 and $y_2 \in G_\infty(x_0)$, then

$$(x_0, y_2) \in (x_0, y_1) + \mathcal{K}_\lambda,$$

so

$$(x(t), y_2(t)) \in (x(t), y_1(t)) + \mathcal{K}_\lambda$$
$$|y_2(t) - y_1(t)| \ge |y_2 - y_1| e^{(\lambda_2 - \delta(1+\lambda^{-1})t},\ t \ge 0.$$

By construction we have

$$\begin{aligned} \lambda|x(t)| &\ge |y_i(t)|,\ i = 1, 2,\ t \ge 0, \\ |x(t)| &\le |x_0| e^{(\lambda_1 + \delta(1+\lambda))t}. \end{aligned} \tag{5.3}$$

Therefore we get for $t \ge 0$,

$$2|x_0| e^{(\lambda_1+\delta(1+\lambda))t} \ge 2|x(t)| \ge \lambda^{-1}(|y_1(t)| + |y_2(t)|)$$
$$\ge \lambda^{-1}|y_2(t) - y_1(t)| \ge \lambda^{-1}|y_1 - y_2| e^{(\lambda_2 - \delta(1+\lambda^{-1})t}.$$

Since $\lambda_1 < 0$ and $\lambda_2 > 0$ we conclude $y_1 - y_2 = 0$.

We denote by g^s the function $x_0 \mapsto G_\infty(x_0)$. Equation (5.3) implies $(x(t), y(t)) = \phi^t(x_0, g^s(x_0)) \to 0$ as $t \to \infty$. Therefore by definition of W^s, we get that $(x_0, g^s(x_0)) \in W^s$, i.e. the graph of g^s is a subset of W^s.

Suppose now W^s_δ contains graphg^s as a proper subset, i.e. there is $y_1 \neq g^s(x_0)$ such that $(x_0, y_1) \in W^s_\delta$. Then we get

$$(x_0, y_1) \in (x_0, g^s(x_0)) + \mathcal{K}_\lambda$$

And

$$|y(t) - y_1(t)| \ge \lambda |y_1 - g^s(x_0)| e^{(\lambda_2 - \delta(1+\lambda^{-1}))t}.$$

It is clear that $y_1(t)$ does not converge to 0 so (x_0, y_1) does not lie in W^s_δ. This proves graph$g^s = W^s_\delta$.

We next show that g^s is a Lipschitz function. Suppose there are x_0 and x_1 such that

$$|g^s(x_0) - g^s(x_1)| \ge \lambda |x_0 - x_1|.$$

Then for all $t \ge 0$, we have

$$(x_1, g^s(x_1)) \in (x_0, g^s(x_0)) + \mathcal{K}_\lambda \Longrightarrow \phi^t(x_1, g^s(x_1)) \in \phi^t(x_0, g^s(x_0)) + \mathcal{K}_\lambda.$$

which implies

$$|y_1(t) - y_0(t)| \ge \lambda |g^s(x_1) - g^s(x_0)| e^{(\lambda_2 - \delta(1+\lambda^{-1}))t},$$

which contradicts the definition of W^s. The contradiction implies

$$|g^s(x_0) - g^s(x_1)| \le \lambda |x_0 - x_1|, \quad \forall x_0, x_1 \in B_\delta \cap \mathbb{R}e_1.$$

Note that for all $\lambda > 0$, by choosing δ small, we can make the Lipschitz norm of the nonlinearity f as small as we wish in the ball B_δ. This implies that the Lipschitz norm λ of W^s_δ in B_δ can be as small as we wish provided we choose δ small. In the limit we find that W^s_δ at the origin is tangent to $\mathbb{R}e_1$.

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We finally prove that $G_t(x_0)$ is not empty. Suppose $G_t(x_0)$ were empty. Then the definition of $G_t(x_0)$ implies that ϕ^t induces a continuous map on the ball $B(x_0)$ retracting to the boundary, which is impossible by Brouwer's fixed point theorem. The continuous map can be seen as follows. Whenever $(x(t), y(t))$ leaves the set $\{|y| \leq \lambda |x(t)|\}$, we think the corresponding point $(x(0), y(0))$ escapes to the boundary of $B(x_0)$ (and stays there) and no longer belongs to $G_t(x_0)$. □

The theory of invariant manifolds became mature after the work of [**HPS, F**]. In fact, invariant manifolds can appear in some more complicated forms. For example, for the Arnold's cat map acting on $\mathbb{T}^2$, either the stable or unstable manifolds of the fixed point 0 is dense on $\mathbb{T}^2$. Moreover, in this case, passing through any point of $\mathbb{T}^2$, there is a stable manifold and also an unstable manifold, so we say that there are two foliations. The same thing is true if we perturb the Arnold's cat map slightly in the C^1 norm.

For an application of the invariant manifold theory to PDEs, please refer to [**NS**], which also contains two other methods of proving the invariant manifold theory (Chapter 3.2 of [**NS**]). For an application to geometric flows, please refer to [**CM3**].

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CHAPTER 5

Sturm-Liouville boundary value problem

In this chapter we study the boundary value problem, which is different from the initial value problem. For a second order equation of the form $x''+bx'+cx=0$, we need to specify an initial position and an initial velocity to determine a unique solution. Instead, we may also specify an initial position and a terminal position to determine a solution. Such solutions may not be unique. The theory of this chapter is a special case of the boundary value problems in elliptic PDE. So in this chapter, we put the materials into a more general framework to reveal the key properties of both elliptic PDE as well as the index theory. The presentation of this chapter following mostly from [**W**].

We recall the following basic fact in linear algebra.

THEOREM 0.1. *Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix with simple spectrum (distinct eigenvalues with one dimensional eigen spaces). Then*

(1) *if $\lambda_i \neq \lambda_j$ are eigenvalues, then the corresponding eigenvectors are orthogonal;*
(2) *there exists $C \in O(n)$ such that $C^{-1}AC = \Lambda \in$ where Λ is diagonal and real;*

PROOF. Suppose we have $Av_i = \lambda_i v_i$ and $Av_j = \lambda_j v_j$. Then we have

$$\lambda_i\langle v_j, v_i\rangle = \langle v_j, Av_i\rangle = \langle Av_j, v_i\rangle = \lambda_j\langle v_j, v_i\rangle,$$

Where in the second $=$, we use $A^t = A$. Since we assume $\lambda_i \neq \lambda_j$, we have $\langle v_j, Av_i\rangle = 0$.

Each eigenvector has to be real. Suppose not, then we get a complex eigenvector v. We have $Av = \lambda v$, $A\bar{v} = \bar{\lambda}\bar{v}$.

$$\lambda\langle v, \bar{v}\rangle = \langle Av, \bar{v}\rangle = \langle v, A\bar{v}\rangle = \bar{\lambda}\langle v, \bar{v}\rangle.$$

Since $\langle v, \bar{v}\rangle > 0$, we get $\lambda = \bar{\lambda}$ so λ is real.

Now we get n mutually orthogonal eigenvectors, after normalized into unit vectors and put together to form a matrix C, then C is in $O(n)$. We can also make $C \in \mathrm{SO}(n)$.

□

1. Sturm-Liouville boundary value problem

1.1. Sturm-Liouville boundary value problem. The Sturm-Liouville boundary value problem (BVP) is of the form

$$Lu = \lambda u \text{ in } J = [a, b], \quad R_1u = R_2u = 0,$$

where L and R_1, R_2 are the operators defined as follows

$$Lu := -(p(t)u')' + q(t)u,$$

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$$R_1 u := \alpha_1 u(a) + \alpha_2 p(a) u'(a),$$
$$R_2 u := \beta_1 u(b) + \beta_2 p(b) u'(b).$$

The condition $R_1 u = R_2 u = 0$ is called boundary condition.

(1) If $\alpha_2 = \beta_2 = 0$ and $\alpha_1 \neq 0, \beta_1 \neq 0$, then we call the boundary condition the *Dirichlet boundary condition.*
(2) If $\alpha_1 = \beta_1 = 0$ and $\alpha_2 \neq 0, \beta_2 \neq 0$, then the boundary condition is called *Neumann boundary condition.*

The problem is to look for pairs (λ, u) solving the above BVP. If such a pair exists, we call λ an eigenvalue and u an eigenfunction.

1.2. An example. A special case to keep in mind is the BVP

$$\begin{cases} -u'' &= \lambda u \\ u(0) &= u(\pi) = 0 \end{cases}.$$

The general solution to the ODE $u'' + \lambda u = 0$ is $u(t) = A\cos\sqrt{\lambda}t + B\sin\sqrt{\lambda}t$. The boundary condition $u(0) = 0$ implies $A = 0$ and the condition $u(\pi) = 0$ implies $B\sin(\sqrt{\lambda}\pi) = 0$. If we do not want a trivial solution with $A = B = 0$, then we should have $\sqrt{\lambda} = n$. So the solution to the BVP is given by

$$\{\lambda_n = n^2, \quad u_n(t) = \sin nt, \quad n = 1, 2, \dots, \}$$

From the simple example, we see some main features of the BVP: There is a λ_n going to infinity such that the problem admits a solution u_n. Actually, there are more deep facts about the BVP. For instance, we have

(1) The n-th solution u_n corresponds to λ_n has $n-1$ zeros on $(0, \pi)$. In particular, the first function $u_1 = \sin t$ is positive on $(0, \pi)$;
(2) The zeros of u_n and u_{n+1} alternates;
(3) For any C^2 u satisfying $u(0) = u(\pi) = 0$, there is a Fourier expansion: there exists a sequence $\{c_n\}$ such that

$$u(t) = \sum_{n=1}^{\infty} c_n \sin nt.$$

It turns out that the general Sturm-Liouville BVP shares all these features.

1.3. Statements of the main theorems. Let us state the main theorem of the Sturm-Liouville BVP. We make the following Sturm-Liouville assumption:

$$(\mathbf{SL}) \ : \ p(t) > 0, \ t \in J, \quad p \in C^1(J), \ q \in C^0(J),$$
$$\alpha_1^2 + \alpha_2^2 > 0, \quad \beta_1^2 + \beta_2^2 > 0.$$

THEOREM 1.1 (Existence Theorem). *Under the assumption (SL), the BVP has infinitely many simple real eigenvalues*

$$\lambda_0 < \lambda_1 < \ldots < \lambda_n \to \infty$$

and no other eigenvalues. The eigen function u_n corresponding to λ_n has exactly n zeros in the open interval $J^o = (a, b)$. Between two successive zeros and also between a and the first zero and between the last zero and b there is exactly one zero of u_{n+1}.

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THEOREM 1.2 (Generalized Fourier expansion). *The eigen functions can be normalized such that* $\int_J u_n(t)^2 dt = 1$. *They then form an orthonormal set of functions, i.e., one has additionally that* $\int_J u_n(t)u_m(t)dt = 0$ *for* $m \neq n$. *Each function* $\varphi \in C^2(J)$ *that satisfies boundary conditions can be expanded in terms of the eigenfunctions in a series*

$$\varphi(t) = \sum_{n=0}^{\infty} c_n u_n(t)$$

that converges absolutely and uniformly in J. *This series is called the Fourier series of* φ. *The Fourier coefficients are given by* $c_n = \int_J \varphi(t)u_n(t)dt$.

We next work on the proofs of the theorems.

1.4. Prüfer transformation. The first step is to introduce a coordinate change called Prüfer transformation.

The second order equation $-(pu')' + qu = 0$ can be expressed as a first order system for $\xi = pu'$ and $\eta = u$. On the ξ-η plane, we introduce polar coordinates

$$\xi = p(t)u'(t) = \rho(t)\cos\varphi(t), \quad \eta = u(t) = \rho(t)\sin\varphi(t).$$

The function φ will be called the argument function.

LEMMA 1.3. *The variables* (ρ, φ) *satisfies the ODE system*

$$\begin{cases} \varphi' &= \frac{1}{p}\cos^2\varphi - q\sin^2\varphi = \frac{1}{p} - \left(q + \frac{1}{p}\right)\sin^2\varphi \\ \rho' &= \left(\frac{1}{p} - q\right)\cos\varphi\sin\varphi \end{cases}.$$

Note that the φ' equation has no dependence on ρ. Once the φ' equation is solved, it can be substituted into the ρ' equation to solve for ρ. In the following, we use only the φ' equation.

PROOF. We have

$$\varphi' = \frac{d}{dt}\arctan\frac{\eta}{\xi} = \frac{d}{dt}\arctan\frac{u}{pu'} = \frac{(u')^2 p - (pu')'u}{u^2 + (pu')^2} = \frac{\frac{1}{p}(\xi)^2 - q\eta^2}{\rho^2} = \frac{1}{p}\cos^2\varphi - q\sin^2\varphi.$$

Similarly, we get the ρ' equation. □

We have the following observations:

(1) $u(t) = 0, \Leftrightarrow \varphi(t) = k\pi$, and $u'(t) = 0 \Leftrightarrow \varphi(t) = k\pi + \frac{\pi}{2}, \ k \in \mathbb{Z}$;
(2) function $\varphi(t) + k\pi$ also solves the φ' equation.

1.5. The eigenvalue problem. We consider the solution $u = u(t, \lambda)$ of the initial value problem for equation

$$Lu = \lambda u, \quad u(a) = \sin\alpha, \ p(a)u'(a) = \cos\alpha, \ \alpha \in [0, \pi).$$

The argument function $\varphi(t, \lambda)$ satisfies the equation

$$\varphi' = \frac{1}{p} + \left(-q - \frac{1}{p} + \lambda\right)\sin^2\varphi, \tag{1.1}$$

and $\varphi(t, \lambda) = \alpha$.

LEMMA 1.4. *The argument function has the following properties:*

(1) $\varphi(t, \lambda)$ *is strictly increasing in* λ *for* $a < x < b$.
(2) $\varphi(t, \lambda) \to 0$ *as* $\lambda \to -\infty$.

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(3) *There exist positive constants* d, D, λ_0 *such that*

$$d\sqrt{\lambda} \le \varphi(b,\lambda) \le D\sqrt{\lambda}, \quad \forall \lambda \ge \lambda_0.$$

(4) *If* $\varphi(t_0,\lambda_0) = k\pi (k \in \mathbb{Z})$*, then* $\varphi'(t_0,\lambda_0) > 0$*. This means that in the* t-φ *plane (not the phase plane) the curve* $\varphi(t,\lambda_0)$ *crosses the line* $\varphi = k\pi$ *at most once, and in a strictly increasing manner.*

We postpone the proof of this Lemma and solve the eigenvalue problem assuming it.

Proof of Theorem 1.1. We first interpret the boundary conditions. The first boundary condition $R_1 u = \alpha_1 u(a) + \alpha_2 p(a) u'(a) = 0$ implies $(p(a)u'(a), u(a)) \perp (\alpha_2, \alpha_1)$. Therefore by choosing $\alpha = \frac{\pi}{2} + \arctan\frac{\alpha_1}{\alpha_2} \in [0,\pi)$, we get $\alpha_1 \sin\alpha + \alpha_2 \cos\alpha = 0$. With this α in the above IVP, we see that every solution to the IVP satisfies the first boundary condition $R_1 u = 0$ up to a scalar multiple.

Similarly, there is a unique number β satisfying $\beta_1 \sin\beta + \beta_2 \cos\beta = 0$, $\beta \in (0,\pi]$. The solution to the IVP satisfies the second boundary condition $R_2 u = 0$ if and only if $(p(b)u'(b), u(b)) = C(\cos\beta, \sin\beta)$, hence if and only if

$$\varphi(b,\lambda) = \beta + n\pi, \ n \in \mathbb{Z}.$$

By Lemma 1.4, we get that for each $n \ge 0$, there is exactly one $\lambda = \lambda_n$ such that

$$\varphi(b,\lambda_n) = \beta + n\pi, \quad n = 0, 1, 2, \dots,$$

while for $n < 0$, there is no such λ exists.

Then λ_n is the eigenvalue and $u_n = u_n(\cdot, \lambda_n)$ is the eigen function.

Lemma 1.4(3) shows

$$d^2\lambda_n \le (\beta + \pi n)^2 < D^2\lambda_n.$$

This gives the Weyl law on the asymptotic behavior of eigenvalues: there exist $c, C > 0$ such that for n large

$$cn^2 \le \lambda_n \le Cn^2.$$

Finally, we study the distribution of zeros. We know that u_n has a zero at t if and only if $\varphi(t,\lambda_n) = k\pi$. Now we have

$$0 \le \varphi(a,\lambda_n) = a < \pi, \quad n\pi < \varphi(b,\lambda_n) = n\pi + \beta \le (n+1)\pi.$$

This inequality combined with Lemma 1.4(d) shows that $\varphi(t,\lambda_n)$ assumes the values $\pi, 2\pi, ..., n\pi$ exactly once in (a,b) and no other values of the form $k\pi$. Hence u_n has exactly n zeros in (a,b).

□

We now prove Lemma 1.4.

Proof of Lemma 1.4. Part (d) follows immediately from (1.1) since $p > 0$.

For part (a), we need the following comparison lemma, which follows from the comparison principle (Theorem 2.5) applied to the φ' equation.

Consider two equations $Lu = -(pu')' + qu = 0$ and $L_0 v = -(p_0 v')' + q_0 v = 0$.

Lemma 1.5 (Comparison lemma). *Let* $p \le p_0$, $q \le q_0$ *in* $J = [a,b]$*. Then the corresponding argument functions satisfy if* $\varphi(a) \ge \varphi_0(a)$*, then* $\varphi \ge \varphi_0$ *in* J*. More precisely, we have*

(1) $\varphi_0(a) < \varphi(a)$ *implies* $\varphi_0 < \varphi$ *in* J*.*

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(2) $\varphi_0(a) = \varphi(a)$ *and* $q_0 \neq q$ *implies* $\varphi_0(b) < \varphi(b)$.

For part (b), we again use the comparison principle. Denote by $f(t, \varphi)$ the RHS of (1.1). Then we will construct a function w satisfying

(1) $w' > f(x, w)$ for $\lambda < \lambda_0 < 0$;
(2) $w(b) \leq \varepsilon$.

Then the comparison principle implies $\varphi(b, \lambda) \leq \varepsilon$ for $\lambda < \lambda_0$.

We pick w to be a linear function $w(a) = \pi - \varepsilon,\ w(b) = \varepsilon$ where ε is taken to be small such that $w(a) \geq \alpha$. So we have $\sin^2 w \geq \sin^2 \varepsilon$ and

$$f(x, w) \leq \frac{1}{p} + \big(-q - \frac{1}{p} + \lambda \big) \sin^2 \varepsilon \to -\infty, \quad \text{as } \lambda \to -\infty.$$

Part (c) is another application of the Comparison lemma 1.5. We compare with the following system with constant coefficients

$$p_0 u'' + (-q_0 + \lambda)u = 0, \quad u(a) = 0,\ u'(a) > 0,$$

where $p_0 = \max p, q_0 = \max q$. The solution is given by

$$u_0(t, \lambda) = \sin \omega_0 (t - a), \quad \omega_0 = \sqrt{(q_0 + \lambda)/p_0}.$$

The argument function is

$$\varphi_0(b, \lambda) = \omega_0 (b - a) + c, \quad |c| < \pi/2.$$

This implies $\varphi_0(b, \lambda) \geq d(b - a)\sqrt{\lambda}$ for large λ. Since we have $\varphi_0(a) = 0 \leq \varphi(a, \lambda)$, the Comparision Lemma 1.5 implies $\varphi_0(b, \lambda) \leq \varphi(b, \lambda)$. This gives the first inequality of part (c).

The second inequality is proved similarly by comparing with the problem

$$p_1 u'' + (-q_1 + \lambda)u = 0, \quad u(a) = 0,\ u'(a) > 0,$$

where $p_1 = \min p, q_1 = \min q$. □

1.6. Distribution of zeros.

LEMMA 1.6. *A nontrivial solution u of $Lu = 0$ on $J = [a, b]$, $-\infty < a < b < \infty$, has only a finite number of zeros, and they are all simple.*

PROOF. A simple zero means that $u(t) = 0$ but $u'(t) \neq 0$. When $u(t) = 0$ and $u'(t) = 0$, we get a solution $u \equiv 0$ by uniqueness. This shows that all zeros are simple. Suppose there are infinitely many zeros $t_1 < t_2 < \ldots < t_n < \ldots$, and $t_n \to \xi$. Then $u(\xi) = 0$. Moreover, on the interval (t_{n-1}, t_n), there is s_n such that $u'(s_n) = 0$ and $s_n \to \xi$. So we get $u'(\xi) = 0$. So $u \equiv 0$. □

We next state two central theorems on the distribution of zeros.

THEOREM 1.7. (1) *Sturm Separation Theorem: The zeros of two linearly independent solutions of $Lu = 0$ separate each other. In other words, between two consecutive zeros of u there is a zero of v and vice versa.*

(2) *Sturm-Picone. Let $p_0 \geq p$, $q_0 \leq q$ in J. Let v be a nontrivial solution of $L_0 v = 0$ with $v(\alpha) = v(\beta) = 0 (\alpha < \beta)$. If $q \neq q_0$ in (α, β), then every solution u of $Lu = 0$ has a zero in (α, β).*

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PROOF. We may assume that α and β are consecutive zeros of v and $v > 0$ in (α, β) (v can be replaced by $-v$). Then $v'(\alpha) > 0,\ v'(\beta) < 0$, and the corresponding argument function satisfies $\varphi_0(\alpha) = 0,\ \varphi_0(\beta) = \pi$.

For Sturm's theorem, we set $L = L_0$ and suppose $u(\alpha) > 0$. Then the argument function φ of u satisfies $0 < \varphi(\alpha) < \pi$, and Lemma 1.5(1) gives $\varphi(\beta) > \pi$. Hence there is $t_0 \in (\alpha, \beta)$ with $\varphi(t_0) = \pi$, i.e. $u(t_0) = 0$. This proves Sturm's theorem.

Similar argument with Lemma 1.5(2) gives Sturm-Picone theorem. □

We have known that u_n has exactly n zeros in (a, b), which we label $a < t_1 < t_2 < \ldots < t_n < b$. The Sturm-Picone theorem tells us that between t_k and t_{k+1}, there is a zero of u_{n+1}. Now, using the monotonicity of φ in λ, we get

$$\alpha = \varphi(a, \lambda_n) = \varphi(a, \lambda_{n+1}) < \pi = \varphi(t_1, \lambda_n) < \varphi(t_1, \lambda_n)$$

Therefore, there is a zero of u_{n+1} between a and t_1. Similarly, one shows that u_{n+1} has a zero in (t_n, b). Since there are $n + 1$ zeros, there must be exactly one zero of in each of the $n + 1$ intervals (a, t_1), (t_k, t_{k+1}),(t_n, b).

2. Compact self-adjoint operators in Hilbert spaces

The presentation in the previous section is relating the phase space dynamics with the eigenvalue problem. The equation in the 2D phase space is nonlinear. In this section, we view the problem as a linear problem in the infinite dimensional space. We put the above Sturm-Liouville boundary problem in a more general framework.

Let us recall the discussion at the beginning of this chapter. Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix with simple spectrum. Then we know

(1) All the eigenvalues are real.
(2) The eigenvectors are orthogonal. So we can form an orthonormal basis of $\mathbb{R}^n$ using the eigenvectors.

One can think that each positive definite symmetric matrix A gives rise to an ellipsoid $\{X \in \mathbb{R}^n \mid X^T AX \leq 1\}$. The symmetric axes are orthogonal. The directions of the symmetric axes are directions of eigenvectors. The length of each axis is twice of reciprocal of the eigenvalue. After rotations, the ellipsoid can be placed in such a way that each symmetric axis lies on a coordinate axis.

In this section, we will see that the Sturm-Liouville boundary value problem can be considered as an infinite dimensional generalization of this finite dimensional phenomenon.

(1) There are infinitely many eigenvalues, all of which are real.
(2) The eigenvectors are orthogonal. They can be normalized to form an orthonormal basis of an infinite dimensional space.

So we need

(1) An infinite dimensional analogue of the Euclidean space–Hilbert space.
(2) An infinite dimensional analogue of the symmetric matrices–self-adjoint operators.

The general theorem in this section is as follows.

THEOREM 2.1. *Let A be a compact self-adjoint operator on a Hilbert space H. Suppose 0 is not an eigenvalue of A. Then*

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(1) *There is a sequence of eigenvalues* $\lambda_0, \lambda_1, \ldots, \lambda_n, \ldots$, *satisfying*
$$|\lambda_0| \geq |\lambda_1| \geq |\lambda_2| \geq \ldots \geq |\lambda_n| \to 0,$$
each λ_n *has finite multiplicity, i.e.* $Ker(A - \lambda_n)$ *is finite dimensional.*
(2) *The eigenvectors are orthogonal.*
(3) *The eigenvectors* u_k, $k = 0, 1, 2, \ldots$, *can be normalized to form an orthonormal basis such that for each* $u \in H$, *we have*
$$u = \sum_{k=0}^{\infty} c_k u_k, \quad c_k = \langle u, u_k \rangle, \quad Au = \sum_{n=0}^{\infty} \mu_n c_n u_n.$$
We also have the Parceval equality
$$\|u\|^2 = \sum_{k=0}^{\infty} |c_k|^2.$$

We remark that the framework in this section is more general than the Sturm-Liouville setting.

In quantum mechanics, all the physical quantities are self-adjoint operators. The physical measurement can only detect the eigenvalues.

DEFINITION 2.2. (1) *A linear space* H *with an inner product, and is complete in the norm induced by the inner product, is called a Hilbert space.*
(2) *An inner product is a binary operation* $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{R}$ (*or* $\mathbb{C}$) *satisfying the following properties: for all* $u, v, w \in H$ *and* $\lambda, \mu \in \mathbb{R}$ (*or* $\mathbb{C}$),
   (a) *positive-definiteness,* $\langle u, u \rangle \geq 0$ *with equality iff* $u = 0$.
   (b) *linearity,* $\langle \lambda u + \mu v, w \rangle = \lambda \langle u, w \rangle + \mu \langle v, w \rangle$,
   (c) *symmetry,* $\langle u, v \rangle = \overline{\langle v, u \rangle}$.
(3) *The inner product induces a norm* $\| \cdot \| : H \to \mathbb{R}_{\geq 0}$ *via*
$$\|v\| := \langle v, v \rangle^{1/2}, \quad \forall\ v \in H.$$

Let us give some examples of Hilbert spaces:

Example: $\ell^2(\mathbb{Z}) := \{\mathbf{x} = (x_n)_{n \in \mathbb{Z}},\ x_n \in \mathbb{C} \mid \|\mathbf{x}\|^2 = \sum_{k \in \mathbb{Z}} |x_k|^2 < \infty\}$ with inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{n \in \mathbb{Z}} x_n y_n$.

Example: $L^2(\mathbb{T}) := \{x : \mathbb{T} \to \mathbb{C} \mid \|x\|^2 := \int_{\mathbb{T}} |x(t)|^2\, dt < \infty\}$, with inner product $\langle x, y \rangle = \int_{\mathbb{T}} x(t) \bar{y}(t)\, dt$.

REMARK 2.3. *It turns out that the* $L^2(\mathbb{T})$ *is isometric to* $\ell^2(\mathbb{Z})$ *and the isometry is given by the Fourier transformation. Let* $x(t) \in L^2(\mathbb{T})$ *and* $x(t) = \hat{x}_n e^{in2\pi t}$. *Then* $\|x\|_{L^2(\mathbb{T})} = \|\hat{x}\|_{\ell^2(\mathbb{Z})}$ *where* $\hat{x} = (x_n) \in \ell^2(\mathbb{Z})$.

It is important to notice that in a Hilbert space, bounded closed sets may not be compact.

For example we consider $e_n \in \ell^2(\mathbb{Z})$ as the point all whose entries are zero except the n-th entry that is 1. It is clear that $\langle e_n, e_m \rangle = 0$ for $m \neq n$. Moreover, we have
$$\|e_n - e_m\|^2 = \langle e_n - e_m, e_n - e_m \rangle = \|e_n\|^2 + \|e_m\|^2 = 2.$$
The unit ball $\{\mathbf{x} \in \ell^2(\mathbb{Z}) \mid \|\mathbf{x}\| \leq 1\}$ is a bounded closed set. However it is not compact! The reason is that the sequence $\{e_n\}$ does not have a convergent subsequence since it is not a Cauchy sequence.

To talk about convergence in a Hilbert space, we will need compactness. So the following concept is crucial.

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DEFINITION 2.4. (1) *A set $S(\subset H)$ is called* precompact*, if every sequence $\{u_n\} \subset S$ has a Cauchy subsequence. i.e. for all ε, there is N such that when $m, n > N$, we have $\|u_n - u_m\| < \varepsilon$.*

(2) *A set $S(\subset H)$ is called* compact*, if every sequence $\{u_n\} \subset S$ has a convergent subsequence. i.e. there is a point $u_* \in H$ such that for all ε, there is N such that when $n > N$, we have $\|u_n - u_*\| < \varepsilon$.*

We also have the notion of compactness for an operator.

DEFINITION 2.5. (1) *A linear operator $A: H \to H$ is bounded if its operator norm is bounded, i.e.*

$$\|A\| := \sup_{\|v\|=1} \|Av\| < \infty.$$

(2) *A linear operator on a Hilbert space is called a compact operator if it maps bounded sets into pre-compact sets.*

(3) *A operator $A: H \to H$ is called self-adjoint (or Hermitian) if the following holds*

$$\langle Au, v\rangle = \langle u, Av\rangle, \quad \forall\ u, v \in H.$$

For example, the operator $A: \ell^2(\mathbb{Z}) \to \ell^2(\mathbb{Z})$ via $(a_n) \mapsto (\frac{1}{|n|+1} a_n)$ is compact. A compact set in a Hilbert space should look thinner and thinner and shrink to zero as the dimension gets higher and higher.

2.1. Sturm-Liouville boundary value problem. Let us now consider the Sturm-Liouville boundary value problem as before

$$\begin{cases} -(p(t)x')' + q(t)x = \lambda x, \\ x(a) = x(b) = 0. \end{cases}$$

This kind of boundary value is called Dirichlet boundary value. It can also be replaced by the the general $R_1 = R_2 = 0$.

The analogue to the above linear algebra case is as follows.

We introduce the Hilbert space

$$L^2 := \left\{ x: \ [a,b] \to \mathbb{R} \ \middle|\ \int_a^b |x(t)|^2 dt < \infty, \quad x(a) = x(b) = 0 \right\},$$

with inner product $\langle x, y\rangle_2 := \int_a^b x(t)y(t)\,dt$. We also introduce the following subset of L^2

$$\bar{H} := \left\{ x: \ [a,b] \to \mathbb{R} \ \middle|\ \int_a^b |x'(t)|^2 + |x(t)|^2 dt < \infty, \quad x(a) = x(b) = 0 \right\}.$$

The L operator $Lx = -(p(t)x')' + q(t)x$ when restricted to $\bar{H}(\subset L)$ is formally self-adjoint, i.e. $\langle x, Ly\rangle = \langle Lx, y\rangle$ by the following calculation

$$\langle y, Lx\rangle = \int_a^b y(t)(-(p(t)x'(t))' + q(t)x(t))\,dt = \int_a^b p(t)x'(t)y'(t) + q(t)x(t)y(t)\,dt,$$

where the last step is the integration by parts.

Note that the integration by parts is a crucial method of obtaining duality. Here the meaning of the boundary condition is also clear.

So the boundary value problem is really to find the eigenvalue and eigenvector of the operator L, i.e. find λ and u such that $Lu = \lambda u$.

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There are some subtlety about the second order derivatives, which is handled by introducing the theory of generalized functions, weak derivative and Sobolev spaces which we skip here.

2.2. Eigenvalues for compact operators on Hilbert spaces. In this section, we prove Theorem 2.1.

LEMMA 2.6. *Let $A: H \to H$ be bounded and self-adjoint, then $\langle Au, u\rangle$ is real for all $u \in H$ and*

$$\|A\| = \sup_{\|u\|=1} \langle Au, u\rangle.$$

PROOF. We denote $\sup_{\|u\|=1}\langle Au, u\rangle = \beta$. Then we have $|\langle Au, u\rangle| \le \beta\|u\|^2$ for all $u \in H$.

By definition of operator norm, we have $\|Au\| \le \|A\| \cdot \|u\|$. For $\|u\| = 1$, by Cauchy-Schwarz we have

$$|\langle Au, u\rangle| \le \|Au\| \le \|A\|.$$

This gives $\beta \le \|A\|$.

(Proof of Cauchy-Schwarz:

$$0 \le \langle u + \lambda v, u + \lambda v\rangle = \|u\|^2 + \lambda\langle v, u\rangle + \bar{\lambda}\langle u, v\rangle + |\lambda|^2\|v\|^2.$$

Then substitute $\lambda = -\frac{\langle u,v\rangle}{\|v\|^2}$.)

On the other hand we have

$$\langle Au + Av, u + v\rangle - \langle Au - Av, u - v\rangle = 2\langle Au, v\rangle + 2\langle Av, u\rangle.$$

We have

$$LHS \le \beta\|u + v\|^2 + \beta\|u - v\|^2 = 2\beta(\|u\|^2 + \|v\|^2).$$

Now choose $u = \lambda w$ and $v = Aw$ with $\lambda = \|Aw\|$, $\|w\| = 1$, one then obtains

$$RHS = 2\langle Au, v\rangle + 2\langle Av, u\rangle = 2\lambda\langle Aw, Aw\rangle + 2\lambda\langle A^2w, w\rangle = 4\lambda^3,$$

hence

$$4\lambda^3 \le 2\beta(\lambda^2 + \lambda^2) \Longrightarrow \lambda = \|Aw\| \le \beta.$$

Since w with $\|w\| = 1$ is arbitrary, we have $\|A\| \le \beta$. This completes the proof. □

LEMMA 2.7. *Let $A: H \to H$ be a compact self-adjoint operator on a Hilbert space, then A has an eigenvalue $\mu_0 \in \mathbb{R}$ with $|\mu_0| = \|A\|$.*

PROOF. We pick a sequence u_n with $\|u_n\| = 1$ and

$$|\langle Au_n, u_n\rangle| \to \|A\|.$$

By the compactness of A, we get that $\langle Au_n, u_n\rangle$ converges to $\mu_0 \in \mathbb{R}$ with $|\mu_0| = \|A\|$ and Au_n converges to some point in H denoted by $\mu_0 u$ with $\|u\| = 1$.

We next have

$$0 \le \|Au_n - \mu_0 u_n\|^2 = \|Au_n\|^2 - 2\mu_0\langle Au_n, u_n\rangle + \mu_0^2 \le 2\mu_0^2 - 2\mu_0\langle Au_n, u_n\rangle \to 0.$$

This shows the error $Au_n - \mu_0 u_n \to 0$ in H. Since $Au_n \to \mu_0 u$, we have $\mu_0 u_n \to \mu_0 u$, so $u_n \to u$ and $Au_n \to Au$. This gives $Au = \mu_0 u$. □

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PROOF OF THEOREM 2.1. We first find the first eigenvalue μ_0 and its associated eigenvector u_0 by Lemma 2.7. Then we introduce the new Hilbert space

$$H_1 = \{u \in H \mid \langle u, u_0\rangle = 0\} = u_0^{\perp}.$$

The operator A is also self-adjoint and compact in H_1, so we apply Lemma 2.7 again. Inductively we define

$$H_n = \{u \mid \langle u, u_i\rangle = 0,\ i = 0, 1, \dots, n-1\}$$

to obtain $Au_n = \lambda_n u_n$, $\|u_n\| = 1$.

We perform this procedure for infinitely many steps.

Let $\mu = \mu_n$, we then claim that $\mathrm{Ker}(A - \mu)$ is finite dimensional. Otherwise, we get infinitely many eigenvectors $\{v_n\}$ of A with eigenvalue μ. The set $A\{v_n\} = \{\mu v_n\}$ is not precompact. Contradiction to the compactness of A.

The sequence μ_n converges to zero since otherwise defining $\phi_n = \frac{u_n}{\mu_n}$ which is a bounded sequence, then $A\phi_n = u_n$ with $\|u_n\| = 1$. This contradicts the assumption that A is a compact operator.

Finally, we prove the Fourier expansion $u = \sum_{n=0}^{\infty} c_n u_n$. Given $u \in H$, suppose $v_n = \sum_{k=0}^{n-1} c_k u_k$, $c_k = \langle u, u_k\rangle$. We get $\|v_n\|^2 = \sum_{k=0}^{n-1} |c_k|^2$ by orthogonality and $0 \le \|u - v_n\|^2 = \|u\|^2 - \sum_{k=0}^{n-1} |c_k|^2$, so $\|v_n\| \le \|u\|$. Then we get $u - v_n \in H_n$ and

$$\|A(u - v_n)\| \le |\mu_n| \|u - v_n\| \le \mu_n 2\|u\| \to 0.$$

This gives $Au = \sum_{n=0}^{\infty} \mu_n c_n u_n$.

The partial sum v_n is a Cauchy sequence in H, since we have $\sum_{k=0}^{\infty} |c_k|^2 \le \|u\|^2$ (which is called Bessel's inequality) and for $m < n$, we have

$$\|v_n - v_m\|^2 \le \sum_{k=m+1}^{n} |c_k|^2.$$

Suppose v_n converges to $v \in H$. We have $c_k = \langle v, u_k\rangle$, so Au and Av has the same Fourier coefficients $\mu_k c_k$. Then we get $A(u - v) = 0$. Since 0 is not an eigenvalue, this gives $u = v$. This completes the proof. □

2.3. Green's function and the compactness. The L-operator in Sturm-Liouville boundary condition has eigenvalues going to infinity, so L cannot be a compact operator. Instead of the eigenvalue problem $Lu = \lambda u$, we consider the eigenvalue problem $L^{-1}u = \frac{1}{\lambda}u$. When $\lambda_n \to \infty$, we have $\frac{1}{\lambda_n} \to 0$, so it is hopeful to show that $L^{-1} :\ L^2 \to L^2$ is a compact operator and apply Theorem 2.1 to the operator L^{-1}.

In elliptic PDE, there are systematic tools to find the inverse of L (Lax-Milgram theorem) and prove its compactness (Sobolev embedding). However, in our case, we can construct L^{-1} and establish its compactness explicitly using the notion of Green's function.

2.3.1. *Green's function.* Let us introduce Green's function as follows.

DEFINITION 2.8. *Green's function of the Sturm-Louville boundary value problem is a function* $G :\ [a, b] \times (a, b) \to \mathbb{R}$ *such that for each* $s \in (a, b)$*, the function* $G(t, s)$ *as a function of* t *satisfies the following*

(1) *for* $t = a$ *and* b*, we have the boundary condition* $R_1 G(a, s) = R_2 G(b, s) = 0$.
(2) $LG(t, s) = \delta_s(t)$*, more explicitly,*

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(a) *for $t \neq s$, we have $LG(\cdot, s) = 0$;*
(b) *when $t = s$, G is continuous but $\partial_t G$ has a jump $\frac{1}{p(s)}$ in derivative*

$$G(s-0,s) = G(s+0,s), \quad \partial_t G(s-0,s) = \partial_t G(s+0,s) + \frac{1}{p(s)}.$$

Here a Dirac-δ function is $\delta_s(t) = \begin{cases} 0 & t \neq s \\ \infty & t = s \end{cases}$ and $\int_a^b \delta_s(t)\, dt = 1$. From the definition we see that $\int_a^b \delta_s(t) f(t)\, dt = f(s)$ for a continuous function f. We can treat $\delta_0(x)$ as the second order derivative of the function $\frac{1}{2}|x|$.

Given the definition, we have the following method of finding the Green's function of the BVP.

Let $x_i(t)$ be a solution to the equation $Lx = 0$ satisfying the boundary condition $R_i x_i = 0,\ i = 1, 2$ and $x_1 \neq x_2$. Then the Green's function can be expressed as

$$G(t,s) = \begin{cases} \alpha(s) x_1(t) & (a \leq t \leq s), \\ \beta(s) x_2(t) & (s \leq t \leq b), \end{cases}$$

where α and β are solved from the conditions

$$\beta(s) x_2(s) = \alpha(s) x_1(s), \quad \alpha(s) x_1'(s) = \beta(s) x_2'(s) + \frac{1}{p(s)}.$$

We remark that it may happen that there is one solution x satisfying $Lx = 0$ as well as two boundary conditions $R_i x_i = 0,\ i = 1, 2$ simultaneously. In this case, we have that x is an eigenfunction with eigenvalue 0 for L, i.e. $\mathrm{Ker} L \neq 0$, so L is not invertible. In this case the Green's function does not exist.

The main utility of Green's function is to given an integral representation of the solution to an inhomogeneous BVP.

PROPOSITION 2.9. *Let G be the Green's function, then the solution to the BVP* $\begin{cases} Lx & = f(t), \\ R_1 x(a) & = R_2 x(b) = 0, \end{cases}$ *, where $f(t), t \in [a, b]$, is continuous, is expressed as*

$$x(t) = \int_a^b G(t,s) f(s) ds.$$

PROOF. We split the integral into

$$x(t) = \int_a^t G(t,s) f(s) ds + \int_t^b G(t,s) f(s) ds.$$

So

$$x'(t) = G(t,t) f(t) - G(t,t) f(t) + \int_a^t \partial_t G(t,s) f(s) ds + \int_t^b \partial_t G(t,s) f(s) ds = \int_a^b \partial_t G(t,s) f(s) ds.$$

Here note that when differentiating the integrand of $\int_a^t G(t,s) f(s) ds$, we need only the one-sided derivative G with respect to t along the diagonal. In the following, we apply the same procedure to differentiate x'. Note that it is enough to have one-sided second order derivative $\partial_t^2 G$ along the diagonal.

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The second order derivative yields

(2.1)
$$x''(t) = \partial_t G(t+,t)f(t) - \partial_t G(t-,t)f(t) + \int_a^t \partial_t^2 G(t,s)f(s)ds + \int_t^b \partial_t^2 G(t,s)f(s)ds$$
$$= \int_a^t \partial_t^2 G(t,s)f(s)ds + \int_t^b \partial_t^2 G(t,s)f(s)ds - \frac{f(t)}{p(t)}$$

Since we have $LG(t,s) = 0$ for $s \neq t$. This gives

$$Lx = -px'' - p'x' + qx = \int_a^b LG(t,s)f(s)\,ds = f(s).$$

□

2.3.2. *The compactness.*

THEOREM 2.10. *Let $\{f_n\}$ be a sequence in $C(J)$ with $\|f_n\|_{L^2} < C$. Then the sequence*

$$g_n(t) = \int_a^b G(t,s)f_n(s)\,ds$$

is precompact in $C(J)$. i.e. the sequence $\{g_n\}$ has a subsequence uniformly converging to $g \in C(J)$, so the convergence is also in $L^2(J)$.

PROOF. We apply Arzela-Ascoli Theorem which states that a sequence $\{g_n\}$ in $C(J)$ has a convergent subsequence if g_n is uniformly bounded and equicontinuous, i.e. $\|g_n\|_{C^0} \leq C$ and for all ε, there exist δ such that if $|t-t'| < \delta$ then we have

$$|g_n(t) - g_n(t')| \leq \varepsilon, \quad \forall n.$$

(This is an essential theorem on the compactness of a sequence of functions, which essentially says that the only mechanism of lossing compactness, besides becoming unbounded, is the oscillation towards small scales, for instance, the sequence $\{\sin nx\}$ is not precompact $C([0,1])$ due to the loss of equicontinuity).

We first check the uniform boundedness. The Green's function is continuous, so we have $|G| < C$. Then Cauchy-Schwarz gives

$$|g_n(t)| = |\langle G(t,\cdot), f_n(\cdot)\rangle| \leq \|G(t,\cdot)\|_{L^2}\|f_n\|_{L^2} \leq C.$$

We next check the equi-continuity. By the continuity of G (G can be extended to a continuous function on $[a,b]^2$, hence is uniformly continuous), for all ε, there exists δ such that when $|t-t'| < \delta$, we have $|G(t,s) - G(t',s)| < \varepsilon$ for all $s \in J$. So we have by Cauchy-Schwarz

$$\begin{aligned} |g_n(t) - g_n(t')| &\leq \int_a^b |G(t,s) - G(t',s)||f_n(s)|ds \\ &\leq \varepsilon \int_a^b |f_n(s)|ds \\ &\leq \varepsilon |b-a|^{1/2}\|f_n\|_{L^2} \\ &\leq C\varepsilon. \end{aligned} \tag{2.2}$$

This proves the equi-continuity. □

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2.4. Outreach. The Sturm-Liouville boundary value problem is the one dimensional special case of the general eigenvalue problem for elliptic operators in PDE.

For instance, let Ω be a closed manifold or a bounded simply connected domain in $\mathbb{R}^n$. Denoting the Laplacian operator $\Delta = \sum_{k=1}^n \frac{\partial^2}{\partial x_k^2}$. We study the boundary value problem

$$\begin{cases} -\Delta u &= \lambda u \\ u &= 0 \text{ on } \partial\Omega \end{cases}.$$

If Ω is a bounded domain in $\mathbb{R}^2$. It is a classical problem, called Kac problem, to determine the shape of Ω by the eigenvalues of $-\Delta$. This problem is also known as "can we hear the shape of a drum"?

Let Ω be a closed surface of genus greater than 1. There is a formula called trace formula, relating the eigenvalues to the length of closed orbits on Ω. This formula is significant in both physics and number theory.

Please refer to [**SY**] for more results on the estimate of the λ_0, $\lambda_1 - \lambda_0$ and $\lambda_{n+1} - \lambda_n$ and more results and the relation of these estimates to the global topology of the manifolds.

3. The Morse index

We list the eigenvalues of the BVP

$$\begin{cases} -x'' = \lambda x, \\ x(0) = x(\pi) = 0. \end{cases}$$

as $\lambda_n = n^2$. The corresponding eigenfunctions are $\sin nt$. We note that

(1) the first eigenvector $\sin t$ has no zero in the open interval (a, b);
(2) the n-th eigenfunction $\sin nt$ has $n - 1$ zeros in the open interval (a, b).

We reformulate these facts in a slightly different way (c.f. [**RS**] Section 6).

Consider a one parameter family of Sturm-Liouville operators

$$(L_s u)(t) := -\frac{d}{dt}\left(p(s,t)\frac{d}{dt}u(t)\right) + q(s,t)u(t).$$

Suppose $s \in [-1, 1]$, p is $C^1([-1, 1] \times [0, 1])$ and q is $C^0([-1, 1] \times [0, 1])$. For each s, let $\phi = \phi(s, t)$ be the solution to the IVP

$$L_s\phi(s, \cdot) = 0, \quad \phi(s, 0) = 0, \quad \phi'(s, 0) = 1.$$

Assume that $\phi(s = \pm 1, t = 1) \neq 0$.

THEOREM 3.1. *Denote by $\mu(L_s)$ the number of negative eigenvalues of L_s and by $\nu(\phi(s, 1))$ the number of zeros of ϕ as a function of t. Then we have*

$$\mu(L_{+1}) - \mu(L_{-1}) = \nu(\phi(+1, 1)) - \nu(\phi(-1, 1)).$$

PROOF. Repeat the arguments in Section 1. For a deeper proof, see Section 6 of [**RS**] using homotopy. □

This is a baby version of the Atiyah-Singer index theory.

For example, we consider $p(s, t) = p(t)$ and $q(s, t) = q(t) + \lambda(s)$ where $\lambda(-1) < \lambda_0$ and $\lambda(1) \in (\lambda_{n-1}, \lambda_n)$ where λ_0, λ_n are eigenvalues of the Sturm-Liouville operator $Lu = \frac{d}{dt}\left(p(t)\frac{d}{dt}u(t)\right) + q(t)u(t)$.

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Then the theorem says the following. When s goes from -1 to 1. Then we count the how many times when the operator L_s becomes degenerate (when L_s becomes degenerate, that means $\mathrm{Ker} L_s \neq 0$, that means λ_s is an eigenvalue). This number is the same as the number counting how many times $\phi(s, \cdot)$ intersects the x-axis. Note that there are two intersection numbers, one computed for the infinite dimensional operator L_s, the other counted for the function $\phi(s, \cdot)$ in $\mathbb{R}^2$. They coincide!

The Morse theory for geodesics (see [**M2**]) is a higher dimensional generalization of this fact. In general, this gives an effective way to compute the number of negative eigenvalues for an elliptic operator using only dynamics in finite dimensional space.

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CHAPTER 6

Gradient flow system

Starting from this chapter, we study equations with special structures. There are mainly two types of such equations. The first type is the gradient flow system and the second type is the Hamiltonian systems. In this chapter, we study the gradient flow systems.

1. General properties

Let $f:\ \mathbb{R}^n \to \mathbb{R}$ be C^2, then we form the differential equation

$$x' = -\nabla f(x).$$

This is called a gradient flow system. Here we need $f \in C^2$ since we need ∇f to be at least Lipschitz to apply existence and uniqueness theorem. We remark that we can also study the gradient flow systems on a manifold by considering $f:\ M \to \mathbb{R}$.

We have learned in the multivariable calculus that ∇f is a vector pointing to the direction along which f grows fastest. So the meaning of a negative gradient flow is a flow along which the value of f decreases fastest.

Each gradient flow system naturally admits a Lyapunov function that is f itself since we have the following

$$|x'(t)|^2 = -\nabla f(x)\cdot x'(t) = -\frac{d}{dt} f(x(t)).$$

Integrating over time $[a,b]$, we get

$$\int_a^b |x'(t)|^2\,dt = -\int_a^b \frac{d}{dt} f(x(t))\,dt = f(x(a)) - f(x(b)) \geq 0. \tag{1.1}$$

So along the orbit $x(t)$ from time a to time b the value of f is non increasing. In particular, if $x(t)$ is not a fixed point, then f is strictly decreasing.

This means that along each orbit, the value of f decreases strictly until the orbit approaches a fixed point. So the final motion of each orbit is clear if the fixed points are all located. So the study of a gradient flow system is completely reduced to understanding the set of critical points $\{\nabla f = 0\}$. Each critical point is a fixed point of the flow.

It is known that the gradient ∇f is always normal to the level set $\{x \mid f(x) = c\}$ of f. So the negative gradient flow $x(t)$ is always perpendicular to the level set $f^{-1}(f(x(t)))$ at each point $x(t)$.

LEMMA 1.1. *The negative gradient flow system has no nontrivial periodic orbit.*

PROOF. Suppose $x(t)$ is a periodic orbit of period T, i.e. $x(t) = x(t+T),\ \forall\, t \in \mathbb{R}$ and $x(t)$ is not a fixed point. Then we have

$$\int_0^T |\dot{x}(t)|^2\,dt = f(x(0)) - f(x(T)) = f(x(0)) - f(x(0)) = 0.$$

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This shows $\dot{x}(t) = 0$ for all $t \in [0, T]$ so $x(t) \equiv$const. □

By similar argument, we can prove the following nonrecurrent property.

LEMMA 1.2. *The negative gradient flow system has no nontrivial recurrent point. In other words, suppose there is a point x_0 such that there is a subsequence $t_n \to \infty$ such that the orbit $x(t)$ starting from x_0 satisfies $|x(t_n) - x_0| \to 0$. Then x_0 is a fixed point.*

So the dynamics of the gradient flow systems is very simple. Each orbit either hits a fixed point, or goes to infinity.

Equation (1.1) often gives the so-called energy estimate in PDE since it gives L^2-bounds on the derivative of x along an orbit. In many areas of mathematics such as geometry, analysis, PDE etc, it is favorable to have some monotone quantity for a differential equation. As we have seen previously in the section of Lyapunov function, the existence of such a monotone quantity along the flow gives important information on the stability. This idea is crucial. For instance, for Boltzmann equation on gas dynamics, there is a natural monotone quantity called entropy. For Navier-Stokes equation in Fluid mechanics, energy is monotone. Perelman introduced an entropy for the Ricci flow [**P**]. For mean curvature flow, there is also an entropy [**CM1**] (for those who are interested in mean curvature flow, they are recommended to start with this paper).

Suppose $n = 2$ and $x_* \in \mathbb{R}^2$ is a fixed point so we have $\nabla f(x_*) = 0$ and $f = ax_1^2 + 2bx_1x_2 + cx_2^2 + o(|x|^2)$. Assume further that the matrix $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ has distinct eigenvalues that are nonzero. After a linear transformation, we assume A is of the form $\mathrm{diag}\{\lambda_1, \lambda_2\}$ with $\lambda_1 > 0, \lambda_2 < 0$, so the linearization of the gradient flow around x_* has the form

$$\begin{cases} x_1' &= -\lambda_1 x_1 \\ x_2' &= -\lambda_2 x_2 \end{cases}.$$

In this case, there are only three types of phase portrait that can occur: sink node ($\lambda_1 > 0, \lambda_2 > 0$), sourse node ($\lambda_1 < 0, \lambda_2 < 0$), saddle ($\lambda_1\lambda_2 < 0$).

With the above preparations, we get in general a way to understand the phase space dynamics of a negative gradient flow system by inspecting the function f directly. The idea is to first find the critical points of f, then draw the level sets. All other orbits will be perpendicular to the level sets and terminate at either the critical points or infinity.

Let us consider an example: $f : \mathbb{R}^2 \to \mathbb{R},\ f(x_1, x_2) = x_1^2(x_1 - 1)^2 + x_2^2$. The global dynamics of the negative gradient flow system can be seen directly from the graph of the function without solving the equation.

The phase portrait is drawn following the rules:

1. First draw the level set $\{f(x_1, x_2) = C\}$ of the function f. Then the orbit are drawn as curves perpendicular to all the level sets. For example, $x_1^2(x_1-1)^2+x_2^2 = C$ gives

$$x_2^2 = \pm\sqrt{C - x_1^2(x_1 - 1)^2}.$$

The graph of x_2 as a function of x_1, when choosing $-$, has the same monotonicity as $x_1^2(x_1 - 1)^2$. The $+$ case is a reflection of the $-$ case along the x_1-axis.

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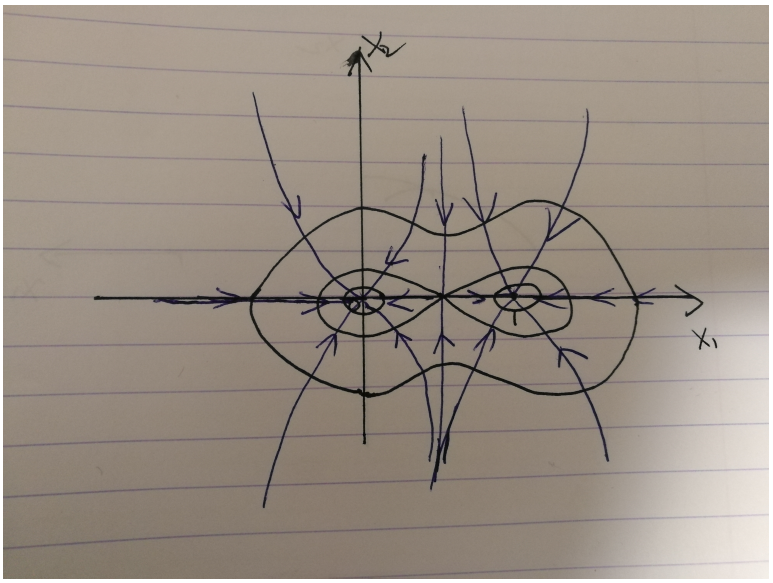


FIGURE 1. Phase protrait for $f = x_1^2(x_1 - 1)^2 + x_2^2$

2. The fixed points for the equation(or critical points for f) are important. If the critical points of f are nondegenerate. Then a local max corresponds to a source, a local min to a sink and a minimax to a saddle.

Sometimes it may happen that the linearization at the fixed point has a zero eigenvalue. We can easily add a perturbation to f to kill the zero eigenvalue. However, it may also happen that in some setting we are not allowed to add any perturbation. In this case, a strange phenomenon can happen. It may happen that for instance $f = 0$ along the unit circle $\mathbb{S}^1 = \{|x| = 1\}$ so each point on the unit circle is the fixed point of the gradient flow. Moreover, an orbit starting from a neighborhood of $\mathbb{S}^1$ may spiral around $\mathbb{S}^1$ for infinitely many times before dying on $\mathbb{S}^1$ at $t \to \infty$.

This is in practice a nondesirable picture since a small perturbation of the initial point may result in a different limit on $\mathbb{S}^1$, which is a sign of instability. In the next section we introduce a technique ruling out the phenomenon for analytic gradient flow systems.

2. The Lojasiewicz inequality

The Lojasiewicz inequalities are important in real algebraic geometry. It dictates that the gradient flow for an analytic function satisfies two inequalities which have strong implications on the dynamics. For instance, there is no orbit of infinite length for such gradient flow systems.

The two inequalities have deep applications to geometric flows, in particular, the mean curvature flow, in recent years.

We following the presentation from [**CM2**]. We refer interested readers to [**Lo1, Lo2, DK, JKS, BDL**] for more details.

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Let $U \subset \mathbb{R}^n$ be a bounded connected open set and $f : U(\subset \mathbb{R}^n) \to \mathbb{R}$ be a *real analytic function*, that is f is infinitely differentiable at each point $x \in U$, moreover, in a neighborhood of x, the function f admits a complete Taylor expansion (with no remainder)

$$f(x+h) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(x) h^n.$$

Note that the function $f(x) = e^{-\frac{1}{x^2}}$ for $x \neq 0$ extends continuously to a function on $\mathbb{R}$. However, this f is only C^∞ but not analytic at 0.

THEOREM 2.1 (Lojasiewicz first inequality). *Let $f : U(\subset \mathbb{R}^n) \to \mathbb{R}$ be real analytic where U is open and let $Z = \{x \in U \mid f(x) = 0\}$ be the zero locus of f. Then for any compact set K, there exist α and C such that for all $x \in K$, we have*

$$\inf_{z \in Z} |x - z|^\alpha \leq C|f(x)|.$$

THEOREM 2.2 (Lojasiewicz second inequality). *Under the same assumption on f, for each $p \in U$ with $\nabla f(p) = 0$, there is a neighborhood $W(\subset U)$ of p and constants $\beta \in (0,1)$ and $C > 0$ such that for all $x \in W$, we have*

$$|f(x) - f(p)|^\beta \leq C|\nabla f(x)|.$$

The proof of the two Lojasiewicz inequalities are not easy. Here we prove two results relevant to it to give you some feeling about them.

PROPOSITION 2.3. *The first Lojasiewicz with $\alpha = 2$ implies the second Lojasiewicz with $\beta = 2/3$.*

PROOF. Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is a smooth function with $f(0) = 0$ and $\nabla f(0) = 0$. Without loss of generality, we assume that $D^2 f(0)$ is diagonal. We use the coordinates $x = (y, z)$ where y is the coordinates where the Hessian is nondegenerate. So by Taylor's formula in a small neighborhood of 0, we have

$$\begin{aligned} f(x) &= \sum_i \frac{a_i}{2} y_i^2 + O(|x|^3), \\ f_{y_i}(x) &= a_i y_i + O(|x|^2), \\ f_{z_i}(x) &= O(|x|^2). \end{aligned} \tag{2.1}$$

We consider two cases.

Case 1, $|z|^2 \leq \varepsilon |y|$ for some small $\varepsilon > 0$.

Then we have for some C

$$C|y| \leq |\nabla f(x)|, \quad |f(x)| \leq C^{-1}|y|^{3/2}.$$

Hence we get

$$|f(x)|^{2/3} \leq C|\nabla f(x)|.$$

This is the second Lojasiewicz inequality. Note that this does not need the first inequality.

Case 2, $|z|^2 > \varepsilon |y|$.

By the first Lojasiewicz inequality applied to ∇f, we have

$$|\nabla f(x)| \geq C|x|^2.$$

Next, by the Taylor expansion, we have

$$|f(x)| \leq C|y|^2 + C|z|^3 \leq C|z|^3 \leq C|x|^3.$$

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This gives

$$|f(x)|^{2/3} \le C|x|^2 \le C|\nabla f(x)|.$$

This completes the proof. □

THEOREM 2.4 (Lojasiewicz Theorem). *Suppose f satisfies the second Lojasiewicz with $\beta \in (1/2, 1)$ and let $x(t)$ be an orbit of the negative gradient flow $\dot{x} = -\nabla f(x)$. Suppose $x(t)$ has an accumulation point x_∞. Then we have $\lim_{t\to\infty} x(t) = x_\infty$ and the length of the curve is finite*

$$\int_0^\infty |x'(t)|\, dt < \infty.$$

PROOF. We assume $f(x_\infty) = 0$. We have

$$\frac{d}{dt} f(x(t)) = -|\nabla f|^2 \le -|f(x(t))|^{2\beta}.$$

This gives $\frac{d}{dt}(f(x(t)))^{1-2\beta} \ge (2\beta - 1)$, which integrates to

$$f(x(t)) \le Ct^{-\frac{1}{2\beta-1}}.$$

Next by Cauchy-Schwarz we have

$$\begin{aligned} \int_1^\infty |\dot{x}(t)|\, dt &= \int_1^\infty |\nabla f(x(t))|\, dt \\ &= \int_1^\infty \sqrt{-\frac{d}{dt} f(x(t))}\, dt \\ &\le \left(-\int_1^\infty \frac{d}{dt} f(x(t)) t^{1+\varepsilon}\, dt \right)^{1/2} \left(\int_1^\infty t^{-1-\varepsilon}\, dt \right)^{1/2}. \end{aligned} \tag{2.2}$$

So it suffices to show that the integral $-\int_1^T \frac{d}{dt} f(x(t)) t^{1+\varepsilon}\, dt$ is uniformly bounded for all $T > 1$. Integration by parts gives

$$\int_1^T \frac{d}{dt} f(x(t)) t^{1+\varepsilon}\, dt = f(x(t)) t^{1+\varepsilon}|_1^T - (1+\varepsilon) \int_1^T f(x(t)) t^\varepsilon\, dt.$$

For given β, we pick ε sufficiently small to make the integral converge. This completes the proof. □

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CHAPTER 7

Hamiltonian systems

Hamiltonian systems form another class of ODEs with special structure. This gives a framework to write the Newtonian mechanics.

In Newtonian mechanics, we typically have the following situation, the force that exerts on a particle is given by a potential. Suppose the mass is m and the potential is U, then the Newtonian equation is written as

$$mx'' = -\nabla U.$$

Note that this is not a gradient flow system. Instead, there is a conserved quantity $E = \frac{1}{2}mx'^2 + U$. We have $E' = mx' \cdot x'' + \nabla U \cdot x' = 0$.

For instance, the Newtonian N-body problem has potential $U = -\sum_{k>j} \frac{Gm_j m_k}{|x_j - x_k|}$ and equations of motion of the form

$$m_k x_k'' = -\sum_{j \neq k} G m_k m_j \frac{x_k - x_j}{|x_k - x_j|^3}.$$

There are classically two formalisms to rewrite the Newtonian equations: Lagrangian and Hamiltonian. The two ways are related through the Legendre transformation. The Hamiltonian formalism has a symplectic structure (anti-symmetric). The Lagrangian formalism has a variational structure.

1. Lagrangian, Hamiltonian and Legendre transformation

In this section, we introduce Lagrangian and Hamiltonian mechanics. They are related by Legendre transformation two equivalent ways of writing classical mechanics. With these new language, the study of classical mechanics does not start from Newtonian second law, instead, it starts from either a Lagrangian or Hamiltonian function and the equations of motion are then written down according to the Euler-Lagrange equation or Hamiltonian equation.

1.1. Lagrangian mechanics. Let $L : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ be a C^2 function (In general, $L : TM \to \mathbb{R}$). Let $\gamma : [0,1] \to \mathbb{R}^n$ be a C^1 curve with endpoints $\gamma(0) = x,\ \gamma(1) = y$. We introduce the action functional:

$$S(\gamma) := \int_0^1 L(\gamma(t), \gamma'(t))\, dt.$$

We think S is a function defined on the space of paths connecting x and y

$$\mathcal{P}_{x,y} := \{\gamma :\ [0,1] \to \mathbb{R}^n,\ C^1 \mid \gamma(0) = x,\ \gamma(1) = y\}.$$

Recall that in calculus, to understand a function, we typically first set its first order derivative to be zero to locate the critical points, next we evaluate the second order derivative at the critical points to determine if each critical point is local max or

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min or saddle. The situation is similar here in the infinite dimensional setting. We want to find the critical point of the action functional first.

What we are doing next is similar to taking the directional derivatives in multivariable calculus. We pick any C^1 vector field $v : [0,1] \to \mathbb{R}^n$ such that $v(0) = v(1) = 0$. Then we see that $\gamma + \varepsilon v$ is again a C^1 curve in $\mathcal{P}_{x,y}$. This is called a variation of γ.

Let us fix γ and v. Now the action

$$\tilde{S}(\varepsilon) := S(\gamma + \varepsilon v) = \int_0^1 L(\gamma(t) + \varepsilon v(t), \gamma'(t) + \varepsilon v'(t))\, dt$$

is a function of ε.

Now we want to show that $\varepsilon = 0$ is a critical point of the function $\tilde{S}(\varepsilon)$, so we take derivative with respect to ε, i.e., we compute $\lim_{\varepsilon\to 0} \frac{\tilde{S}(\varepsilon)-\tilde{S}(0)}{\varepsilon}$.

We have

$$\begin{aligned} \tilde{S}(\varepsilon) :&= S(\gamma + \varepsilon v) = \int_0^1 L(\gamma(t) + \varepsilon v(t), \gamma'(t) + \varepsilon v'(t))\, dt \\ &= \int_0^1 L(\gamma(t), \gamma'(t)) + \varepsilon \left(\frac{\partial L}{\partial x} v + \frac{\partial L}{\partial x'} v' \right) + O(\varepsilon^2)\, dt \\ \lim_{\varepsilon\to 0} \frac{\tilde{S}(\varepsilon) - \tilde{S}(0)}{\varepsilon} &= \int_0^1 \left(\frac{\partial L}{\partial x} v + \frac{\partial L}{\partial x'} v' \right) dt \\ &= \int_0^1 \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial x'} \right) v\, dt \end{aligned} \tag{1.1}$$

where in the last $=$, we apply integration by parts and use the fact that $v = 0$ at the endpoints.

We say γ is an extremal of the action S, or a critical point of S, if for all the variations, the derivative $\tilde{S}'(0) = 0$. Since v can be any C^1 function with zero boundary conditions, we conclude

$$\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial x'} = 0.$$

This equation is the so-called Euler-Lagrange equation.

Let us consider the special Lagrangian $L(x, x') = \frac{m}{2} x'^2 - U(x)$, which is the kinetic energy minus the potential energy. Then the Euler-Lagrange equation gives us

$$mx'' = -\nabla U(x),$$

which is exactly the Newtonian equation.

1.2. Legendre transformation. Legendre transformation is

DEFINITION 1.1. *Let $f : \mathbb{R} \to \mathbb{R}$ be a C^2 strictly convex function, we define the Legendre transformation to be*

$$f^*(y) := \sup_x (xy - f(x)).$$

Geometrically, this means that for each given slope y, the value $f^*(y)$ measures the largest distance between the line xy and the function $f(x)$. The convexity of f guarantees that the sup is always attained. So we get

$$y = f'(x).$$

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The variable x can be solved from this equation as a function of y, i.e. $x = (f')^{-1}(y)$. So the function f^* is solved explicitly as

$$f^*(y) = y \cdot (f')^{-1}(y) - f((f')^{-1}(y)).$$

PROPOSITION 1.2. *We have*

$$y = f'(x),\ x = (f^*)'(y), \quad f''(x) \cdot (f^*)''(y) = 1.$$

PROOF. We treat x as a function of y via $x = (f')^{-1}(y)$ and write $f^*(y) = xy - f(x)$. We then differentiate f^* to get

$$(f^*)'(y) = \frac{\partial f^*}{\partial x}\frac{dx}{dy} + \frac{\partial f^*}{\partial y} = x,$$

since we have $\frac{\partial f^*}{\partial x} = 0$.

For the second order derivative we have

$$(f^*)''(y) = \frac{d}{dy}(f^*)'(y) = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{\frac{d}{dx}f'(x)} = \frac{1}{f''(x)}.$$

□

Note that by definition, we naturally have

$$f^*(y) + f(x) \geq xy.$$

This inequality will give a number of concrete inequalities.

Let us consider an example. Let $f(x) = \frac{1}{p}x^p$, $p > 1$, then we can compute

$$f^*(y) = \frac{1}{q}y^q, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

We naturally obtain the Young's inequality

$$\frac{1}{p}x^p + \frac{1}{q}y^q \geq xy.$$

From the above proposition, we see that the Legendre transformation of a strictly convex function is still strictly convex, so we may think to apply the Legendre transformation for a second time.

PROPOSITION 1.3. *The Legendre transformation is involutive, i.e.* $(f^*)^* = f$.

PROOF. By definition, we have

$$(f^*)^*(x) = \sup_y(xy - f^*(y)),$$

where the sup is attained at the point $x = (f^*)'(y)$, and

$$f^*(y) = \sup_x(xy - f(x)),$$

where the sup is attained at the point $y = f'(x)$. Proposition 1.2 implies that the maps f' and $(f^*)'$ are mutual inverse.

For each x_0, we determine y_0 from the equation $x_0 = (f^*)'(y_0)$, so we have

$$(f^*)^*(x_0) = x_0y_0 - f^*(y_0).$$

For this y_0, we see that $y_0 = f'(x_0)$ since $(f^*)'$ is the inverse of f'. This means

$$f^*(y_0) = x_0y_0 - f(x_0).$$

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The two equations yield

$$(f^*)^*(x_0) = f(x_0).$$

□

We may allow f to be convex but not strictly convex, i.e. there may be flat part on the graph of f. We can see that the Legendre transformation of a flat piece becomes a corner and vice versa. In this case, we have the following result.

PROPOSITION 1.4. *f is strictly convex if and only if f^* is C^1.*

We can naturally generalize the definition of Legendre transformation to multivariable functions $f : \mathbb{R}^n \to \mathbb{R}$ strictly convex and C^2, i.e. $D^2 f$ is positive definite.

Suppose $f(x) = \frac{1}{2}\sum_{i,j} a_{i,j} x_i x_j$ is a positive definite quadratic form, then $f^*(y) = \frac{1}{2}\sum_{i,j} b_{i,j} x_i x_j$ where the matrices $(a_{i,j})$ and $(b_{i,j})$ are inverse.

1.3. Hamiltonian mechanics.

1.3.1. *The Hamiltonian equations.* Assuming the Lagrangian L is convex, then Hamiltonian is a function $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ (in general, $H : T^*M \to \mathbb{R}$) such that H is the Legendre transformation of L, i.e.

$$H(q,p) = \sup_{\dot{q}}(\langle p, \dot{q}\rangle - L(q,\dot{q})).$$

The Legendre transform does not involve the q variable.

From the Legendre transformation, we have

$$p = \frac{\partial L}{\partial \dot{q}}, \quad \dot{q} = \frac{\partial H}{\partial p}.$$

From the Euler-Lagrange equation we get

$$\dot{p} = \frac{d}{dt}\frac{\partial L}{\partial \dot{q}} = \frac{\partial L}{\partial q} = -\frac{\partial H}{\partial q}.$$

So we form the Hamiltonian equation

$$\begin{cases} \dot{q} &= \frac{\partial H}{\partial p} \\ \dot{p} &= -\frac{\partial H}{\partial q}. \end{cases}$$

We call the vector field $(\frac{\partial H}{\partial p}, -\frac{\partial H}{\partial q})$ the Hamiltonian vector field.

Example: Let us consider $H : \mathbb{R}^2 \to \mathbb{R}$, $H(p,q) = \frac{1}{2}(p^2+q^2)$. We get the equations of motion

$$\begin{cases} \dot{q} &= \frac{\partial H}{\partial p} = p \\ \dot{p} &= -\frac{\partial H}{\partial q} = -q. \end{cases}$$

So the phase portrait is a center. Note that a Hamiltonian system differs drastically from a negative gradient flow system.

Let us consider again the example $L(q,\dot{q}) = \frac{m}{2}\dot{q}^2 - U(q)$. We get

$$p = \frac{\partial L}{\partial \dot{q}} = m\dot{q}.$$

So p has the physical meaning of momentum.

Substituting this equation to the Legendre transformation to replace everywhere $\dot{q}$ by p/m, we get

$$H(q,p) = \langle p, \dot{q}\rangle - L(q,\dot{q}) = \frac{p^2}{m} - \frac{p^2}{2m} + U(q) = \frac{p^2}{2m} + U(q).$$

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So the physical meaning of the Hamiltonian H is the total energy. However, the letter H is introduced in memorial of Huygens who introduced the concept of energy.

We solve the Hamiltonian equation to obtain a flow which we call Hamiltonian flow. In physics we have the energy conservation. It turns out that H is a conserved quantity along the Hamiltonian flow. i.e. $H(q(t), p(t))$ =constant. We have

$$\frac{d}{dt}H(q(t),p(t)) = \frac{\partial H}{\partial q}\dot{q} + \frac{\partial H}{\partial p}\dot{p} = \frac{\partial H}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial H}{\partial q}\frac{\partial H}{\partial p} = 0.$$

1.4. Hamiltonian system with one degree of freedom. We call the dimension n the number of degrees of freedom of the system. Due to the presence of the special structure, the case of $n = 1$ is well-understood.

Let us consider $H : \mathbb{R}^2 \to \mathbb{R}$ via $H(q,p) = \frac{p^2}{2} + U(q)$ where U is C^2 and satisfies $U(q) \to \infty$ if $|q| \to \infty$. For example, we consider $U(q) = q^2(q-1)^2$.

Note that along an orbit H is constant. We consider the energy level $\{H(q,p) = E\}$. We get

$$\frac{p^2}{2} + U(q) = E.$$

So the solution is a one-dimensional curve in general, i.e. $p(q) = \pm\sqrt{2(E - U(q))}$.

So the phase portrait can be drawn as follows. The phase portrait is in general

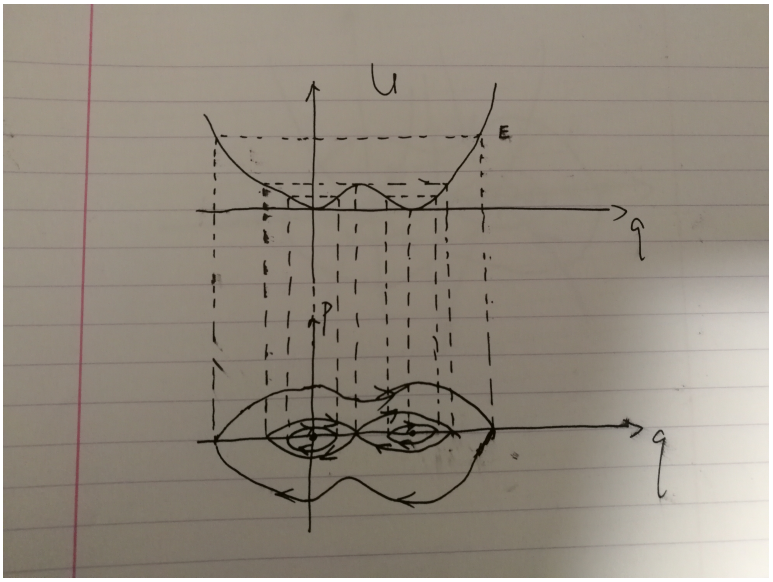


FIGURE 1. Phase portrait for $H = \frac{p^2}{2} + q^2(q-1)^2$

drawn following the rules:

1. The function $p(q) = -\sqrt{2(E - U(q))}$ has the same monotonicity as $U(q)$. The function $p(q) = \sqrt{2(E - U(q))}$ is the reflection of $p(q) = -\sqrt{2(E - U(q))}$ along the q-axis.

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2. In the upper half plane, by the Hamiltonian equation $\dot{q} = p > 0$, so q is increasing along the orbit and the arrows point to the right. In the lower half plane, the arrows point to the left.

3. If the E-energy level does not contain a critical point of U, then the level set is an entire periodic orbit.

4. If the E-energy level does contain a critical point of U and if the critical point is nondegenerate, then there are two cases depending on the sign of U'' at the critical point. If the point is a local max, then it gives rise to a saddle and if it is a local min, then it gives rise to a center. The statement can be seen as follows. Suppose the critical point is at $q = 0$. We linearize the Hamiltonian equation at the critical point to obtain

$$\begin{cases} q' &= p \\ p' &= -U''(0)q \end{cases}.$$

We see that when $U'' > 0$ (local minimum), then we get a center and when $U'' < 0$ (local maximum), we get a saddle.

When drawing the phase portrait, it is always important to draw all the fixed points and make clear their stability types.

Similarly, we also understand the Hamiltonian system $H : \mathbb{T}^1 \to \mathbb{R}^1 \to \mathbb{R}$. For example, the pendulum $H(q,p) = \frac{p^2}{2} + \cos q - 1$.

The case of one degree of freedom can in general be treated in this way. Please compare with the gradient flow systems that we studied in the previous chapter. However, the cases of $n \geq 2$ are much more complicated.

1.5. Poisson bracket and the symplectic form. We first introduce the commutator for a vector field.

Let A be a vector field on $\mathbb{R}^n$, then we form the ODE $x' = A(x)$. Denote by α^t the flow generated by this ODE.

DEFINITION 1.5. *For any $f : \mathbb{R}^n \to \mathbb{R}$, the Lie derivative of f along the vector field A is defined as*

$$(L_A f)(x) := \frac{d}{dt}|_{t=0} f(\alpha^t x) = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} \dot{x}_i = \sum_{i=1}^{n} A_i \frac{\partial f}{\partial x_i}.$$

So we have

$$L_A := A_1 \frac{\partial}{\partial x_1} + \cdots + A_n \frac{\partial}{\partial x_n}.$$

LEMMA 1.6. *The operator $L_B L_A - L_A L_B$ is a first order linear operator.*

PROOF.

$$L_B L_A \varphi = \sum_I B_i \frac{\partial}{\partial x_i} \sum A_j \frac{\partial}{\partial x_j} \varphi = \sum_{i,j} B_i \frac{\partial A_j}{\partial x_i} \frac{\partial}{\partial x_j} \varphi + \sum_{i,j} B_i A_j \frac{\partial^2}{\partial x_i \partial x_j} \varphi.$$

We compute $L_A L_B$ similarly and take the difference $L_B L_A - L_A L_B$. We find that the second order derivatives are canceled and we arrive at

$$(L_B L_A - L_A L_B)\varphi = \sum_{i,j} \left(B_i \frac{\partial A_j}{\partial x_i} - A_i \frac{\partial B_j}{\partial x_i} \right) \frac{\partial}{\partial x_j} \varphi$$

□

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So we denote by $[A, B]$ the resulting vector field from $L_B L_A - L_A L_B$. i.e. in coordinates we have

$$[A, B]_j = \sum_I \left(B_i \frac{\partial A_j}{\partial x_i} - A_i \frac{\partial B_j}{\partial x_i} \right).$$

We call $[A, B]$ the commutator of the vector fields A and B.

PROPOSITION 1.7. *The following properties are satisfied: for all vector fields A, B, C*

(1) *Linearity:* $[aA + bB, C] = a[A, C] + b[B, C], \quad \forall a, b \in \mathbb{R}$
(2) *anti-symmetry:* $[A, B] = -[B, A]$
(3) *Jacobi identity:* $[[A, B], C] + [[B, C], A] + [[C, A], B] = 0.$

This means that the set of C^∞ vector fields on $\mathbb{R}^n$ is a Lie algebra.

The main point of introducing the commutator is the following theorem.

THEOREM 1.8. *Denote by a^t and b^t the flows generated by the vector fields A and B. Then the flows commute if and only the commutator of the vector fields vanish. In other words, $a^t b^s = b^s a^t$ for all $s, t \in \mathbb{R}$, if and only if $[A, B] = 0$.*

PROOF. $\Longrightarrow$: this follows directly from the following lemma 1.9.
$\Longleftarrow$: If we have $[A, B] = 0$. The lemma 1.9 gives

$$\varphi(a^t b^s x) - \varphi(b^s a^t x) = o(s^2 + t^2).$$

Note that here the second order derivatives $\frac{\partial^2}{\partial s^2}$ and $\frac{\partial^2}{\partial t^2}$ also vanishes.

Let us next draw a rectangle with vertices $(0, 0), (0, s), (t, 0), (t, s)$. We want to show that going from zero to (t, s) along two different paths $(0, 0)$-$(0, s)$-(t, s) and $(0, 0)$-$(t, 0)$-(t, s) gives the same value for φ.

We partition the intervals $[0, s]$ and $[0, t]$ into N equal parts each, so the rectangle consists of N^2 small rectangles.

The key observation is that going from one path to the other can be accomplished in N^2 steps. Each step creates an error $o(\frac{1}{N^2})$ by the $o(t^2 + s^2)$ estimate. So as $N \to \infty$, the error converges to zero.

□

The proof of the following lemma is straightforward.

LEMMA 1.9. *We have*

$$\frac{\partial^2}{\partial s \partial t}|_{t=s=0} \left(\varphi(a^t b^s x) - \varphi(b^s a^t x) \right) = (L_B L_A - L_A L_B)\varphi.$$

We next introduce the Poisson bracket.

DEFINITION 1.10. *Given functions $F, H : \mathbb{R}^{2n} \to \mathbb{R}$, the Poisson bracket $\{F, H\}$ is defined as the derivative of the function F in direction of the flow of the Hamiltonian H:*

$$\{F, H\}(x) = \frac{d}{dt}|_{t=0} F(\phi_H^t(x)) = \frac{\partial F}{\partial q}\dot{q} + \frac{\partial F}{\partial p}\dot{p} = \frac{\partial F}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial F}{\partial p}\frac{\partial H}{\partial q}$$

for all $x = (q, p) \in \mathbb{R}^{2n}$.

PROPOSITION 1.11. *Poisson bracket has the following properties: for all $H_1, H_2, H_3 \in \mathcal{C}^\infty(\mathbb{R}^{2n})$*

(1) *Linearity:* $\{aH_1 + bH_2, H_3\} = a\{H_1, H_3\} + b\{H_2, H_3\}, \quad \forall a, b \in \mathbb{R}$

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(2) *anti-symmetry:* $\{H_1, H_2\} = -\{H_2, H_1\}$
(3) *Jacobi identity:* $\{\{H_1, H_2\}, H_3\} + \{\{H_2, H_3\}, H_1\} + \{\{H_3, H_1\}, H_2\} = 0$.

so the space of C^∞ functions on $\mathbb{R}^{2n}$ with the Poisson bracket form a Lie algebra.

THEOREM 1.12. *Let B and C be two Hamiltonian vector fields with Hamiltonian functions β and γ. Consider the commutator $[B, C]$ of the vector fields. Then this vector field is Hamiltonian and its Hamiltonian function equals to $\{\beta, \gamma\}$.*

PROOF. Set $\{\beta, \gamma\} = \delta$. Then the Jacobi identity gives for all α

$$\{\alpha, \delta\} = \{\{\alpha, \beta\}, \gamma\} - \{\{\alpha, \gamma\}, \beta\}.$$

By definition, we have $\{\alpha, \delta\} = L_D\alpha$. This means

$$L_D = L_C L_B - L_B L_C = L_{[B,C]}.$$

This proves the statement. □

This shows that the map that sends a Hamiltonian to its Hamiltonian vector field is a Lie algebra homomorphism if we endow the former with Poisson bracket and the latter with commutator.

2. Canonical transformation and Hamilton-Jacobi equation

2.1. Canonical transforms. We wish to introduce coordinate change in hopes of simplifying the Hamiltonian equation.

Let us consider an example: $H : \mathbb{R}^2 \to \mathbb{R}$, $H(q, p) = \frac{1}{2}(p^2 + q^2)$. The equations of motion are

$$\begin{cases} \dot{q} &= p \\ \dot{p} &= -q \end{cases}.$$

As before, for center type equation, we can switch to polar coordinates. Let us introduce $q = r\cos\theta$, $p = r\sin\theta$. Then as before we get $\frac{1}{2}(p^2 + q^2) = \frac{r^2}{2}$ and the equations of motion becomes

$$\begin{cases} \dot{r} &= 0 \\ \dot{\theta} &= 1 \end{cases}.$$

However, we notice that the polar coordinate change does not preserve the Hamiltonian equation. Otherwise, the new Hamiltonian in the (r, θ) coordinates is $H = \frac{r^2}{2}$ and the Hamiltonian equation is

$$\begin{cases} \dot{r} &= 0 \\ \dot{\theta} &= r \end{cases}$$

which is not correct.

However, there is a way to compensate this. Instead of the (θ, r) coordinates, we can use the (θ, I) coordinates where $I = \frac{r^2}{2}$. So $H(\theta, I) = I$ and

$$\begin{cases} \dot{I} &= 0 \\ \dot{\theta} &= 1 \end{cases},$$

which gives the correct equation.

Moreover, the Hamiltonian system in the coordinates (I, θ) is much simplified. The Hamiltonian H does not depend on θ, so $\dot{I} = \frac{\partial H}{\partial \theta} = 0$ so I is a constant along each orbit.

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In general, we hope to find a coordinate change that preserves the Hamiltonian structure, and moreover, after the transformation, the new Hamiltonian is eliminated as many variables as possible.

Suppose we have a coordinate change $(q,p) \mapsto (Q,P) = (Q(q,p), P(q,p))$. The Hamiltonian $H(Q,P) = H(Q(q,p),P(q,p)) = K(q,p)$. We hope that the Hamiltonian structure is preserved by the transform, in other words, we have

$$\begin{cases} \dot{Q} &= \frac{\partial H}{\partial P} \\ \dot{P} &= -\frac{\partial H}{\partial Q} \end{cases}, \quad \begin{cases} \dot{q} &= \frac{\partial K}{\partial p} \\ \dot{p} &= -\frac{\partial K}{\partial q} \end{cases}.$$

On the LHS of the first equation we have by chain rule

$$\begin{bmatrix} \dot{Q} \\ \dot{P} \end{bmatrix} = \frac{D(Q,P)}{D(q,p)} \begin{bmatrix} \dot{q} \\ \dot{p} \end{bmatrix}.$$

On the RHS, introducing $J = \begin{bmatrix} 0 & \mathrm{Id} \\ -\mathrm{Id} & 0 \end{bmatrix}$, we have

$$\begin{bmatrix} \frac{\partial H}{\partial P} \\ -\frac{\partial H}{\partial Q} \end{bmatrix} = J \begin{bmatrix} \frac{\partial H}{\partial Q} & \frac{\partial H}{\partial P} \end{bmatrix}^T = J \left[\begin{bmatrix} \frac{\partial K}{\partial q} & \frac{\partial K}{\partial p} \end{bmatrix} \frac{D(q,p)}{D(Q,P)} \right]^T = J \frac{D(q,p)}{D(Q,P)}^T \begin{bmatrix} \frac{\partial K}{\partial q} & \frac{\partial K}{\partial p} \end{bmatrix}^T.$$

Note that $\frac{D(q,p)}{D(Q,P)} = \frac{D(Q,P)}{D(q,p)}^{-1}$. In order to obtain the Hamiltonian equation in the (q,p) coordinates, we need to require

$$\frac{D(q,p)}{D(Q,P)} J \frac{D(q,p)}{D(Q,P)}^T = J,$$

or equivalently

$$\frac{D(Q,P)}{D(q,p)} J \frac{D(Q,P)}{D(q,p)}^T = J.$$

This is the kind of transformations that we are allowed to do. So we introduce the following definition.

DEFINITION 2.1. *A C^1 map $\phi : U(\subset \mathbb{R}^n \times \mathbb{R}^n) \to \mathbb{R}^n \times \mathbb{R}^n$ is called symplectic or canonical if*

$$D\phi(x) J D\phi(x)^T = J, \quad \forall x \in U.$$

Felix Klein once defined geometry as a mathematics that studying the invariance under some symmetry group, which is called the Erlangen program.

For example, $\mathbb{R}^n$ endowed with the Euclidean metric (or Euclidean inner product $\langle \cdot, \cdot \rangle$), the group that leaves invariant the Euclidean metric, or the inner product, is the group of orthogonal matrices

$$O(n) = \{M \in \mathbb{R}^{n \times n} \mid M^T M = \mathrm{Id}\}.$$

Any $M \in O(n)$ preserves the inner product $\langle u, v \rangle = \langle Mu, Mv \rangle$.

Analogously, we can consider the $\mathbb{R}^{2n}$ endowed with the symplectic form $\omega = \sum_{i=1}^n dx_i \wedge dy_i$. Here $\omega : \mathbb{R}^{2n} \times \mathbb{R}^{2n} \to \mathbb{R}$ is given by $\omega(u,v) = \langle u, Jv \rangle$.

So we can introduce the symplectic group

$$Sp(n) = \{M \in \mathbb{R}^{2n \times 2n} \mid M^T J M = J\}.$$

So we have

$$\omega(Mu, Mv) = \langle Mu, JMv \rangle = \langle u, M^T J M v \rangle = \langle u, Jv \rangle = \omega(u,v).$$

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So we can also define symplectic map as diffeomorphisms preserving the symplectic structure

$$\omega(D\phi\cdot, D\phi\cdot) = \omega(\cdot,\cdot),$$

or simply $\phi^*\omega = \omega$.

There is a remarkable result by Gromov, called nonsqueezing theorem.

THEOREM 2.2. *Let $\phi:\ B_R(\subset \mathbb{R}^{2n}) \to Z_r$ be a symplectic map where $B_R := \{p^2+q^2 \le R^2\}$ is the ball of radius R and $Z_r := \{(q,p)\in\mathbb{R}^n\times\mathbb{R}^n \mid p_1^2+q_1^2 \le r^2\}$. Then we have*

$$R \le r.$$

This result reflects that symplectic maps has certain rigidity property that is different from volume preserving maps.

The formula $\omega(D\phi\cdot, D\phi\cdot) = \omega(\cdot,\cdot)$, can be written in the following form. Suppose the map $\Phi:\ (q,p)\mapsto (Q(q,p),P(q,p))$ is symplectic, it is equivalent to saying that

$$dP\wedge dQ = dp\wedge dq.$$

This means that the symplectic form ω in the new coordinates (Q,P) has the same expression as that in the old coordinates. This expression should be understood as follows. Both the LHS and the RHS mean to apply two vectors to the expression $\langle\cdot, J\cdot\rangle$. However, the LHS should be applied to vectors chosen in a (tangent) space associated to the (Q,P) coordinate system while the RHS should be applied to (tangent) space associated to the (Q,P) coordinate system and the two spaces are related by the map $D\Phi = \frac{D(Q,P)}{D(q,p)}$.

The reason is as follows. Taking any $u,v\in\mathbb{R}^{2n}$, we get $dp\wedge dq = \langle u, Jv\rangle$. On the other hand, the map $D\Phi$ sends the vectors u and v to $U = D\Phi u$ and $V = D\Phi v$. So since the map Φ is symplectic, we get

$$dp\wedge dq(u,v) = \langle u, Jv\rangle = \langle D\Phi u, JD\Phi v\rangle = \langle U, JV\rangle = dP\wedge dQ(U,V).$$

2.2. Action and Hamilton-Jacobi equation. Besides the Lagrangian and Hamiltonian formalisms, there is a third equivalent way of writing the classical mechanics through a PDE called Hamilton-Jacobi equation.

Fix a point $q_0\in\mathbb{R}^n$ and a time t_0, we introduce the action function

$$S(q,t) = \int_{t_0}^{t} L(q(s), q'(s))\, ds.$$

Here the integration is along an orbit $(q(s))$ connecting q_0 and q from time t_0 to t, i.e. $(q(s), q'(s))$ solves the Euler-Lagrange equation. When q_0 and q are sufficiently close, the orbit $(q(s))$ can be found be minimizing the action $S(\gamma) = \int_\gamma L(\gamma(s),\gamma'(s))\, ds$ where γ is taken over all C^1 curves with $\gamma(t_0) = q_0$ and $\gamma(t) = q$. The existence and regularity of such a minimizing curve is the content of the Tonelli theorem, which is beyond the scope of this course. However, one could still get an idea of such an existence result by looking at shortest curves connecting two nearby points on a sphere, which is a segment on a big circle. When the two points q and q_0 are not close, there might be more than one curve solving the Euler-Lagrange equation that can be used to define the function S. In general, the nonuniqueness issue is reflected in the nonsmoothness of S.

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THEOREM 2.3. *We have*

$$\frac{\partial S}{\partial q} = p, \quad \frac{\partial S}{\partial t} = -H.$$

Therefore, the function S solves the following Hamilton-Jacobi PDE

$$\partial_t S + H(q, \partial_q S) = 0.$$

PROOF. Repeating the calculation of deriving the Euler-Lagrange equation (c.f. Section 1.1), we get

$$\delta S(q,t) = \frac{\partial L}{\partial q'}\delta q|_{t_0}^t - \int_{t_0}^t (\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial q'})\delta q\, dt.$$

Here we have boundary terms $\frac{\partial L}{\partial q'}\delta q|_{t_0}^t$ from integration by parts, compared to the previous calculation since we did not impose the condition of fixing the endpoints.

The integral term vanishes by the E-L equation. The t_0-boundary term also vanishes since the initial point is fixed ($\delta q = 0$). So we apply $p = \frac{\partial L}{\partial q'}$ to obtain $\frac{\partial S}{\partial q} = p$.

Next, from the definition of S, we get the full time derivative is

$$\frac{d}{dt}S = L = \langle p, q' \rangle - H.$$

Here the sup in the Legendre transform is attained since the differential is along an orbit. On the other hand, we have

$$\frac{d}{dt}S = \frac{\partial S}{\partial q}\dot{q} + \frac{\partial S}{\partial t} = \langle p, \dot{q} \rangle + \frac{\partial S}{\partial t}.$$

Comparing the two equations, we get

$$\frac{\partial S}{\partial t} = -H.$$

□

REMARK 2.4. (1) *If $H = 0$ and $\dot{q} = \frac{1}{m}p$, then restricted to the zero energy level, we have*

$$\dot{q} = \frac{1}{m}\frac{\partial S}{\partial q}.$$

This gives a gradient flow system.

(2) *The solution of the H-J PDE is a function S of the variables q and t, which is different from the solution of the ODEs that is a function of t only. In fact such a solution S can be considered a union of a family of curves each of which solves the Hamiltonian equation and is called a characteristic. In general, the H-J PDE only provides a new perspective of looking at the mechanics. It does not simplify the problem of solving the Hamiltonian equations.*

2.3. Liouville Theorem and Poincaré recurrence.

THEOREM 2.5. *The Hamiltonian flow preserves the volume of the phase space.*

PROOF. Proof 1. We need to prove that for any set $\Omega \subset \mathbb{R}^{2n}$, we have

$$\int_\Omega dpdq = \int_{\phi_H^t \Omega} dpdq = \int_\Omega \det(D\phi_H^t) dpdq.$$

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To prove the statement, it is enough to show that the Jacobian $\det(D\phi_H^t) = 1$. Then the statement follows from the following two lemmas.

LEMMA 2.6. *For each t, the Hamiltonian diffeomorphism ϕ_H^t is a symplectic map.*

LEMMA 2.7. *A symplectic map preserves volume.*

Proof 2. For linear equations $X' = A(t)X$, we have learned a Liouville theorem stating that $\det \Phi(t) = e^{-\int_0^t TrA(s)\,ds}$ where Φ is the fundamental solution of the equation. In this case, $\det \Phi(t) = 1$, i.e. the flow preserves volume if and only if $\mathrm{Tr}A = 0$. For a nonlinear equation $X' = V(X)$, we can consider very short Δt time interval and a small neighborhood on which the equation looks linear, i.e. $X' = V(X_0) + DV(X - X_0) + O((X - X_0)^2)$. The situation is similar. Being volume preserving is then reduced to the divergence free condition i.e. $Tr(DV) = \nabla \cdot V = 0$. Now that the Hamiltonian vector field is divergence free ($\nabla \cdot JDH = 0$). Please refer to the book [**A3**] for more details. □

PROOF OF LEMMA 2.6. We give two proofs.

Proof 1. We consider an orbit (q_t, p_t) and apply the transform ϕ_H^τ to get $\phi^\tau(q_t, p_t) = (q_{t+\tau}, p_{t+\tau})$.

By Theorem 2.3, we have

$$dS_1 = p_t dq_t - Hdt, \quad dS_2 = p_{t+\tau} dq_{t+\tau} - Hdt,$$

where S_1 is the action functional with initial point $(q_0, 0)$ and S_2 with initial point (q_τ, τ). Taking difference we get

$$d(S_2 - S_1) = p_{t+\tau} dq_{t+\tau} - p_t dq_t.$$

Taking the exterior differential we get

$$dp_{t+\tau} \wedge dq_{t+\tau} = dp_t \wedge dq_t.$$

This means that ϕ_H^τ is a symplectic map.

Proof 2. We have by definition $\omega(X_H, \cdot) = -dH$ where $X_H = JdH$ is the Hamiltonian vector field. Then by Cartan magic formula

$$L_{X_H}\omega = i_{X_H} d\omega + d i_{X_H}\omega = d(-dH) = 0,$$

where $i_X\omega = \omega(X, \cdot)$. Here L_{X_H} means to differentiate ω along the flow generated by the Hamiltonian vector field X_H. More explicitly, $L_{X_H}\omega := \frac{d}{dt}((\phi_H^t)^*\omega - \omega)$.

This shows that ω when differentiated along the Hamiltonian flow vanishes. So we get by Newton-Leibniz

$$(\phi_H^\tau)^*\omega - \omega = \int_0^\tau L_{X_H}\omega = 0.$$

□

PROOF OF LEMMA 2.7. Proof 1, this follows from the definition of a symplectic map

$$D\phi_H^t J (D\phi_H^t)^T = J.$$

Taking determinant, we get

$$\det D\phi_H^t \det J \det(D\phi_H^t)^T = \det J, \quad (\det D\phi_H^t)^2 = 1.$$

(Here we only have $\det D\phi_H^t = \pm 1$. It is not quite easy to exclude the case of -1. However, for the purpose of proving Liouville Theorem, it is easy to prove

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$\det D\phi_H^t = 1$, since $\det D\phi_H^t$ is a continuous function of t and when $t = 0$, we have $D\phi_H^t = \mathrm{id}$.)

Proof 2. by the previous lemma, we have $(\phi_H^t)^*\omega = \omega$. Taking n times wedge product we get

$$(\phi_H^t)^*(\omega^n) = (\phi_H^t)^*\omega \wedge \ldots \wedge (\phi_H^t)^*\omega = \omega^n.$$

Note that $\omega^n = \prod_i (dp_i dq_i)$ is the volume form.

□

The volume preserving property has the following important Poincaré recurrence. Note that it does not depend on any concrete information of the differential equation.

THEOREM 2.8 (Poincaré recurrence). *Let $f : D \to D$ be a volume preserving homeomorphism on a bounded region $D \subset \mathbb{R}^n$. Then for any point $x \in D$ and neighborhood $U(\subset D)$ of x, there is a point $y \in U$ and a number n, such that $f^n(y) \in U$.*

PROOF. Consider $f^k U,\ k = 0, 1, 2, \ldots$. If they never intersect, then we have that the union $\cup_k f^k U$ has infinite volume, which contradicts the fact that D has finite volume. So we get some k and j such that $f^k U \cap f^j U \neq \emptyset$. This gives $f^{k-j} U \cap U \neq \emptyset$. It is enough to pick a y in $f^{k-j} U \cap U$. □

For example, this applies for the mechanical system $H(q,p) = \frac{p^2}{2m} + U(q)$ where $U \to \infty$ as $|q| \to \infty$. The sublevel set $\{H \leq E\}$ for some E, is invariant under the Hamiltonian flow and is bounded.

The statement can be stronger. Actually, one can prove that for the U and y in the statement, there is a sequence $n_k \to \infty$ such that $f^{n_k}(y) \in U$.

3. Liouville-Arnold theorem

In this section we prove the Liouville-Arnold theorem, which is a fundamental theorem about integrable systems.

THEOREM 3.1. *Let $H_1 = H : \mathbb{R}^{2n} \to \mathbb{R}$ be a Hamiltonian and suppose there are $H_2, \ldots, H_n : \mathbb{R}^{2n} \to \mathbb{R}$ satisfying*

1. *$\{H_i, H_j\} \equiv 0$, for all $i, j = 1, \ldots, n$;*
2. *the level set $M_{\mathbf{a}} := \{(q,p) \in \mathbb{R}^{2n} \mid H_i(q,p) = a_i,\ i = 1, \ldots, n\}$ is compact;*
3. *At each point of $M_{\mathbf{a}}$, the n vectors $DH_i,\ i = 1, \ldots, n$ are linearly independent.*

Then

1. *$M_{\mathbf{a}}$ is diffeomorphic to $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ and is invariant under the Hamiltonian flow of each H_i.*
2. *$M_{\mathbf{a}}$ is a Lagrangian submanifold, i.e. for any $u, v \in T_x M_{\mathbf{a}},\ \forall\ x \in M_{\mathbf{a}}$, we have $\omega(u, v) = 0$.*
3. *in a neighborhood U of $M_{\mathbf{a}}$, there is a symplectic transform $\Phi(q,p) = (\theta, I)$ such that $\Phi(U) = \mathbb{T}^n \times (-\delta, \delta)^n$ for some $\delta > 0$.*
4. *In the new coordinates, each $K_i := H_i \circ \Phi^{-1}$ is a function of I only so the Hamiltonian equation is*

$$\dot{\theta} = \omega_i(I) := \frac{\partial K_i}{\partial I}, \quad \dot{I} = 0.$$

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REMARK 3.2. *This theorem gives a complete characterization of both the topology and the dynamics of the level set $M_{\mathbf{a}}$. The essential ingredient of the proof is the existence of an abelian group action.*

DEFINITION 3.3. *We call a Hamiltonian system satisfying the assumptions of the Liouville-Arnold theorem to be an* integrable system *and the mutually functions $H_i,\ i = 1, \dots, n$* first integrals.

We next devote the remaining part of this section to the proof of Liouville-Arnold theorem.

LEMMA 3.4. *On $M_{\mathbf{a}}$, there are n tangent vector fields which commute with one another and linearly independent at each point. Moreover, $M_{\mathbf{a}}$ is a Lagrangian submanifold that is invariant under the flow generated by each Hamiltonian $H_i,\ i = 1, \dots, n$.*

PROOF. We will show that the n Hamiltonian vector fields $JDH_i,\ i = 1, \dots, n$ are commutative, tangent to $M_{\mathbf{a}}$ and linearly independent at each point.

By assumption, the vector fields $DH_i,\ i = 1, \dots, n$ are linearly independent at each point, so do $JDH_i,\ i = 1, \dots, n$ since J is an isomorphism on $\mathbb{R}^{2n}$.

Next, we have learned that $[JDH_i, JDH_j]$ is a Hamiltonian vector field generated by the Hamiltonian $\{H_i, H_j\} \equiv 0$, so $[JDH_i, JDH_j] \equiv 0$.

By Definition of the Poisson bracket, we have the derivative of F_i along the flow of F_j is $\{F_i, F_j\} \equiv 0$. This means that the flow of F_j leaves F_i invariant so it lies entirely on $M_{\mathbf{a}}$. This means that the Hamiltonian vector field JDH_j is tangent to $M_{\mathbf{a}}$.

Finally we show that $M_{\mathbf{a}}$ is a Lagrangian manifold, we have

$$\omega(JDH_i, JDH_j) = \langle JDH_i, J^2 DH_j \rangle = \langle DH_i, JDH_j \rangle = \{H_i, H_j\} = 0.$$

Next, since $JDH_i,\ i = 1, \dots, n$ are linearly independent, the span the tangent space $T_x M_{\mathbf{a}}$ at each point $x \in M_{\mathbf{a}}$. This means that each $u, v \in M_{\mathbf{a}}$ can be written as a linear combination of $JDH_i,\ i = 1, \dots, n$, so we get $\omega(u, v) = 0$. □

REMARK 3.5. *Recall that the gradient of a function H_i is perpendicular to its level set $\{H_i = a_i\}$. However, in our case, by applying J to the gradient ∇H_i, the resulting Hamiltonian vector field becomes tangent to the level set.*

PROPOSITION 3.6. *Let M^n be a compact connected n dimensional manifold on which there are n mutually commutative vector fields linearly independent at each point. Then M^n is diffeomorphic to $\mathbb{T}^n$.*

PROOF. Denote by $\phi_i^{t_i},\ t_i \in \mathbb{R}$, the flow generated by i-th vector field. We have $\phi_i^{t_i} \phi_j^{t_j} = \phi_j^{t_j} \phi_i^{t_i}$ for all $t_i, t_j \in \mathbb{R}$ by commutativity assumption. So we introduce $\Phi^t : M \to M$ where $t = (t_1, \dots, t_n) \in \mathbb{R}^n$ by $\Phi^t = \prod_{i=1}^n \phi_i^{t_i}$. Let us now fix a point $x_0 \in M$ and introduce the map $\Psi : \mathbb{R}^n \to M$ via $t \mapsto \Phi^t(x_0)$.

We first show that there is a small neighborhood U of $0 \in \mathbb{R}^n$ and a small neighborhood V of x_0 such that Ψ is a diffeomorphism from U to V. By the implicit function theorem, it is enough to check that $D\Psi$ is nondegenerate. In fact we have the matrix $D\Psi$ is formed by putting the vector fields at x_0 as columns.

We next show that Ψ is onto. We let x_0 run over the whole M. For each x_0, we find a neighborhood U on which Ψ^{-1} is diffeomorphic U to an open set $(0 \in) V \in \mathbb{R}^n$. Then we get a covering of M, and by the compactness, we can select

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a finite cover. Now any two points x_0 and x_1 can be connected by a path γ. We break the path into segments $\{\gamma_i\}$ each of which lies entirely in an open set of the finite cover and record by d_i the difference of the endpoints of $\Psi^{-1}(\gamma_i)$. Then we have $\Psi(t) = \Phi^t(x_0) = x_1$ where $t = \sum d_i$.

REMARK 3.7. *The surjectivity can be seen more directly as follows. Starting from each point $x_0 \in M_{\mathbf{a}}$, we apply the flow Φ^t. Since the n vector fields $JDH_i,\ i = 1, \ldots, n$ span the tangent space $T_{x_0}M_{\mathbf{a}}$, we get that by choosing t in a small neighborhood of 0, the image $\Phi^t(x_0)$ covers a neighborhood of x_0 on $M_{\mathbf{a}}$. This is true for any point on $M_{\mathbf{a}}$ replacing x_0. We may think that the flow overwhelms a neighborhood of each point so one can go from one point to another.*

However, the map Ψ cannot be injective since $\mathbb{R}^n$ is noncompact but M is compact. We next introduce the stabilizer subgroup

$$\mathrm{Stab}(x_0) := \{t \in \mathbb{R}^n \mid \Phi^t(x_0) = x_0\}.$$

It is clear that $\mathrm{Stab}(x_0)$ is a group. Moreover, by the local diffeomorphism property of Ψ, we get that $\mathrm{Stab}(x_0)$ is a discrete subgroup of $\mathbb{R}^n$, i.e. there cannot be a sequence in $\mathrm{Stab}(x_0)$ accumulating to a point in $\mathbb{R}^n$.

We need the following algebraic result.

LEMMA 3.8. *Let Γ be a discrete subgroup of $\mathbb{R}^n$, then there exists k, $(0 \le k \le n)$ linearly independent vectors $e_1, \ldots, e_k \in \Gamma$ such that $\Gamma = \mathrm{span}_{\mathbb{Z}}\{e_1, \ldots, e_k\}$.*

We postpone the proof of this lemma and complete the proof of the Proposition first.

We have that $\mathbb{R}^n/\Gamma$ is isomorphic to $\mathbb{R}^k/\mathbb{Z}^k \times \mathbb{R}^{n-k}$. Indeed, we pick standard basis vectors $\hat{e}_i$ of $\mathbb{R}^n$ and introduce the linear map $A : \mathbb{R}^n \to \mathbb{R}^n$ which sends e_i to $\hat{e}_i$ and identity from $\mathbb{R}^n/(\mathbb{R}\Gamma) \simeq \mathbb{R}^{n-k}$ to $\mathbb{R}^{n-k}$. Note that we have $M = \mathbb{R}^n/\mathrm{Stab}(x_0)$. We already know that Ψ is surjective but not injective. By changing the domain from $\mathbb{R}^n$ to $\mathbb{R}^k/\mathbb{Z}^k \times \mathbb{R}^{n-k}$, we recover the injectivity. Therefore, the map A induces a diffeomorphism between M and $\mathbb{T}^k \times \mathbb{R}^{n-k}$. Finally, the compactness of M implies $k = n$. This completes the proof.

□

PROOF OF LEMMA 3.8. If $\Gamma = \{0\}$, then there is nothing to prove. Suppose $\Gamma \neq \{0\}$. Then we pick any $e(\neq 0) \in \Gamma$ and consider the line $\mathbb{R}e$. On the line, there is a e_1 that is closest to 0 by the discreteness (there cannot be a sequence in Γ accumulating on 0). Then we get $\mathbb{Z}e_1 = \Gamma \cap \mathbb{R}e_1$. Indeed, first it is clear that $\mathbb{Z}e_1 \subset \Gamma \cap \mathbb{R}e_1$. Suppose $e \in \Gamma \cap \mathbb{R}e_1$ but is not in $\mathbb{Z}e_1$, then there exists $m \in \mathbb{Z}$ such that e lies on the segment $(m, m+1)e_1$. So the point $e - me_1 \in \Gamma$ is closer to 0 than e_1, which contradicts to the choice of e_1.

If $\Gamma = \mathbb{Z}e_1$, then the lemma is proved with $k = 1$. Otherwise, there exists $e \in \Gamma \setminus \mathbb{Z}e_1$. We next project orthogonally e to $\mathbb{R}e_1$ to get $e' \in \mathbb{R}e_1$ with $e_1 \in [m, m+1)e,\ m \in \mathbb{Z}$, and draw a ball centered at e_1 containing e. We next pick in the ball the point e_2 that is closest to the line $\mathbb{R}e_1$ but not on the line. Then $\mathrm{span}_{\mathbb{Z}}(e_1, e_2)$ is a lattice of the plane $\mathrm{span}_{\mathbb{R}}(e_1, e_2)$. Similarly as before, we prove that $\mathrm{span}_{\mathbb{Z}}(e_1, e_2) = \Gamma \cap \mathrm{span}_{\mathbb{R}}(e_1, e_2)$.

We proceed by induction to complete the proof. □

We have completed the proof that the topology of $M_{\mathbf{a}}$ is a torus. The proof of the existence of the symplectic coordinates (I, θ) is slightly more involved. We

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refer readers to [**A3**] for more details. The procedure is in fact similar to going from (q,p) to $(I=\frac{r^2}{2},\theta)$ for the system $H(q,p)=\frac{1}{2}(p^2+q^2)$.

In reality, integrable systems are very rare. There are few instances that are integrable. Here are some of them,

1. Newtonian two-body problem.

2. Euler two-center problem, that is one particle moving in the gravitational field of two fixed massive particles.

3. Geodesics on the sphere of an ellipsoid.

4. Calogero-Moser systems.

$$H(p,q)=\frac{1}{2}\sum_{n=1}N(p_n^2+\omega^2q_n^2)+g^2\sum_{m,n=1;m\neq n}^{N}(q_n-q_m)^{-2}.$$

5. Schwartzchild solution and Kerr solution of the Einstein equation.

6. There are some remarkable infinite dimensional integrable systems that are both significant for PDE and representation theory including the KdV equation $u_t+u_{xxx}+6uu_x=0$, Nonlinear Schrodinger equation $iu_t+u_{xx}+|u|^2u=0$,etc.

4. The averaging method and Kolmogorov-Arnold-Moser theory

Readers are referred to [**A2**] (Section 12) and [**Po**] for more details. The proof in Section 12 of [**A2**] is an implementation of the KAM method in the simplest setting, with which we recommend to start. The proof of [**Po**] is for Hamiltonian systems.

With Liouville-Arnold theorem, we know that an integrable system can be written in the form

$$H(\theta,I)=h(I),\quad (\theta,I)\in\mathbb{T}^n\times D,$$

where $D\subset\mathbb{R}^n$.

In practice, most of the systems are not integrable. The easiest non-integrable systems are called nearly integrable systems of the form

$$H(\theta,I)=h(I)+\varepsilon f(\theta,I).$$

Nearly integrable systems already describe many important natural phenomena. For instance, in the Newtonian N-body problem governing the motion of our solar system, since the planets have small masses, we may treat each planet and the sun form a two-body problem and treat the mutual interactions of the planets as ε-perturbations. We have seen from an earlier homework that the two-body problem can be integrated (In fact, a two-body problem has two degrees of freedom, and we have two integrals, the Hamiltonian itself (or total energy) and the angular momentum), so we can write the motion of the planets in our solar system in terms of a nearly integrable Hamiltonian system.

We also see that each I-torus is invariant under the Hamiltonian flow and the motion on each I-torus is simply a translation flow by the vector $\omega(I):=\nabla h(I)$.

The KAM theorem named after Kolmogorov-Arnold-Moser is an important result on the dynamics of nearly integrable systems.

The KAM theorem basically says that if $\omega(I)$ is sufficiently irrational, then the corresponding invariant torus is robust under the small εf perturbation.

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DEFINITION 4.1. *A vector $\alpha \in \mathbb{R}^n$ is called Diophantine if there exist some $C > 0, \tau > n-1$ such that for all $k \in \mathbb{Z}^n \setminus \{0\}$, we have*

$$|\langle \alpha, k \rangle| \geq \frac{C}{|k|^\tau}.$$

We denote $\alpha \in \mathrm{DC}(C, \tau)$.

The next lemma shows that most points in $\mathbb{R}^n$ are Diophantine.

LEMMA 4.2. *For each $\tau > n-1$, the set $\cup_{C>0}\mathrm{DC}(C,\tau)$ has full Lebesgue measure in $\mathbb{R}^n$.*

We here prove a easy version of this lemma and the general one can be proved similarly. We say a number $\alpha \in \mathbb{R}$ is Diophantine, if there exist some $C > 0$ and $\tau > 1$ such that for all $\frac{p}{q} \in \mathbb{Q}$, we have

$$|\alpha - \frac{p}{q}| \geq \frac{C}{q^{\tau+1}}.$$

It is a well-known result in number theory that for each irrational α, there is a sequence p_n/q_n with $q_n \to \infty$ such that

$$|\alpha - \frac{p_n}{q_n}| \leq \frac{1}{q_n^2}.$$

Here we prove the following

LEMMA 4.3. *The set of Diophantine numbers on $[0,1]$ has full measure.*

PROOF. For each $q \in \mathbb{N}$, there are at most q rational numbers with denominator q, i.e. $1/q, 2/q, \ldots, q/q$. To satisfy the inequality $|\alpha - \frac{p}{q}| \geq \frac{C}{q^{\tau+1}}$, we should remove an open interval of length $2\frac{C}{q^{\tau+1}}$ around each point p/q and find α in the resulting set. So the total length of the removed intervals is

$$q \times 2\frac{C}{q^{\tau+1}} = 2\frac{C}{q^\tau}.$$

Next, we compute the total length of the removed intervals by letting q range over $\mathbb{N}$ to get an upper bound

$$\sum_{q=1}^{\infty} 2\frac{C}{q^\tau} < \infty.$$

In particular, the upper bound goes to zero as $C \to 0$. □

REMARK 4.4. *The procedure of removing countably many open intervals results in a closed set which is in general called a Cantor set. The most common example of a Cantor set is the middle-third Cantor set. That is divide the interval $[0,1]$ into three equal pieces and remove the middle open interval. In the second step, we divide each of the remaining two intervals $[0,1/3]$ and $[2/3,1]$ into three equal pieces, and remove the middle open intervals. Then keep repeating the procedure. The resulting set has total length 0. However, it has the same cardinality as $[0,1]$. The set of Diophantine numbers shares similar properties.*

DEFINITION 4.5. *We define the C^ℓ norm as*

$$\|f\|_{C^\ell} := \sup_{\theta \in \mathbb{T}^n, I \in D} \sum_{0 \leq \sum |\ell_k| \leq \ell} \|\partial_{\theta_1}^{\ell_1} \ldots \partial_{\theta_n}^{\ell_n} \partial_{I_1}^{\ell_{n+1}} \ldots \partial_{I_n}^{\ell_{2n}} f(\theta, I)\|$$

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This is to collect all the partial derivatives up to order ℓ with respect to both I and θ. By taking Fourier expansion

$$f(\theta, I) = \sum_{k\in\mathbb{Z}^n} \hat{f}_k(I) e^{i\langle k,\theta\rangle}.$$

The C^ℓ norm of f can be equivalently defined as follows.

$$\|f\|_{C^\ell} := \sup_{I\in D} \sum_{k\in\mathbb{Z}^n} \sum_{\sum \ell_i \le \ell} |\partial_{I_1}^{\ell_1} \dots \partial_{I_n}^{\ell_n} \hat{f}_k(I)| \cdot |k_1|^{\ell_{n+1}} \dots |k_n|^{\ell_{2n}}.$$

We next state the KAM theorem.

THEOREM 4.6. *Suppose $\det D^2 h(I) \neq 0$ for $I \in D$. Then for all $C > 0, \tau > n-1$ and all $I \in D$ with $\omega(I) \in \mathrm{DC}(C,\tau)$, there exists $\varepsilon_0 > 0$ and ℓ sufficiently large such that when $\varepsilon\|f\|_{C^\ell} < \varepsilon_0$, we have that there is a torus $\mathcal{T}_I$ that is invariant under the Hamiltonian flow generated by the Hamiltonian $H(\theta, I) = h(I) + \varepsilon f(\theta, I)$. Moreover, there is a symplectic transform Φ such that $\Phi(\mathcal{T}_I) = \mathbb{T}^n$ and the conjugated flow $\Phi \circ \phi_H^t \circ \Phi^{-1}$ is the translation $\theta \mapsto \theta + \omega(I)t$ on $\mathbb{T}^n$.*

REMARK 4.7.
(1) *The KAM theorem holds also for analytic functions. However, we prove now only a version of C^ℓ for ℓ sufficiently large to avoid the use of complex analysis, though we still need to assume something such as the interpolation estimates.*
(2) *The method of the proof is widely known as the KAM method. It is a method consisting of infinitely many iterations. The idea essentially is the Newton iteration method of solving a nonlinear equation.*
(3) *The same method appeared in various circumstance, for instance, Nash's embedding theorem, de Geogi-Nash-Moser theory for regularity of elliptic equations, Mouhot-Villani's work on Landau damping, etc.*

PROOF. The proof involves a procedure of constructing the symplectic transformation Φ by infinitely many steps of iterations such that the transformed Hamiltonian $H \circ \Phi$ has no dependence on θ.

The strategy of iteration.

Suppose we have a small Hamiltonian $\varepsilon F : \mathbb{T}^n \times D \to \mathbb{R}$, such that its Hamiltonian flow ϕ_F^t with $t = 1$ provides the symplectic map. We apply the Taylor expansion with an integral remainder to get $g(1) = g(0) + g'(0) + \frac{1}{2}\int_0^1 (1-t) g''(t)\, dt$. Replacing the derivative by Poisson bracket we get

(4.1)

$$\begin{aligned} H \circ \phi_F^1 &= H + \{H, \varepsilon F\} + \int_0^1 (1-t)\{\{H, \varepsilon F\}, \varepsilon F\} \circ \phi_{\varepsilon F}^t \, dt \\ &= h + \varepsilon f + \varepsilon\{h, F\} + \varepsilon^2\{f, F\} + \varepsilon^2 \frac{1}{2}\int_0^1 (1-t)\{\{H, F\}, F\} \circ \phi_{\varepsilon F}^t \, dt. \end{aligned}$$

We first try to find a proper F to kill the $O(\varepsilon)$ term, i.e. try to solve $f + \{h, F\} = 0$. We take the Fourier expansion

$$f(\theta, I) = \sum_{k\in\mathbb{Z}^n} \hat{f}_k(I) e^{i\langle k,\theta\rangle}, \quad F(\theta, I) = \sum_{k\in\mathbb{Z}^n} \hat{F}_k(I) e^{i\langle k,\theta\rangle}.$$

By comparing the Fourier coefficients we get

$$\hat{f}_k(I) = -i\omega(I) \cdot k \hat{F}_k(I), \quad \hat{F}_k(I) = i\frac{\hat{f}_k(I)}{\langle \omega(I), k\rangle}, \quad k \neq 0.$$

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When $k = 0$, we find $\hat{f}_0(I) = 0$. So we only solve the equation

$$f - \hat{f}_0 = \{h, F\}.$$

Suppose we are able to solve for F, then we set

$$H_1 = H \circ \phi_F^1, \quad h_1(I) = h(I) + \varepsilon \hat{f}_0(I)$$

and

$$f_1 = \{f, F\} + \frac{1}{2} \int_0^1 (1-t)\{\{H, F\}, F\} \circ \phi_F^t \, dt$$

to get the resulting Hamiltonian

$$H_1(\theta, I) = h_1(I) + \varepsilon^2 f_1(\theta, I).$$

In this way, we find that in the new Hamiltonian H_1, the dependence on θ is only in $\varepsilon^2 f_1$, which is supposed to be smaller than εf. We expect to repeat the above procedure infinitely many steps until in the limit the θ dependence is completely killed. When there is no θ dependence in the Hamiltonian, we know that the system is integrable and the motion is translation on a torus. The symplectic transformation Φ in the statement is thus the composition $\lim_n \phi^1_{F_n} \circ \cdots \circ \phi^1_{F_1}$.

Some difficulties.

1. Consider $f : \mathbb{T}^1 \to \mathbb{R}$ a C^r function, then the k-th Fourier coefficient $\hat{f}_k := \int_{\mathbb{T}^1} f(x) e^{2\pi i k x} \, dx$ decays like $\hat{f}_k \sim \frac{1}{|k|^r}$. This can be seen by performing r-steps of integration by parts. The multi-dimensional case is similar.

Suppose the vector $\omega(I)$ satisfies the Diophantine condition, then we have $|\hat{F}_k(I)| \leq \frac{|k|^\tau}{C} |\hat{f}_k(I)|$. So we get $\|F\|_{C^{\ell-\tau}} \leq c\|f\|_{C^\ell}$. This is the phenomenon of losing derivative. If f is only known to be finitely differentiable, then we can at most iterate the above procedure for at most ℓ/τ steps.

2. Moreover, since we add $\varepsilon \hat{f}_0(I)$ to $h(I)$ to form $h_1(I)$, the new frequency $\omega_1(I) = \nabla h_1(I)$ is different from the old frequency $\omega(I)$, so the Diophantine condition cannot in general hold for both.

The first problem is essential. The way to solve it is to use the quadratic convergence of the iteration scheme. Note that the error after one step becomes ε^2. Then after the second step, it is formally ε^4 and inductively ε^{2^n} after the n-th step.

Sketch of the solution.

To address these problems, we introduce the Fourier cut-off. In the above calculation, we consider only F with finitely many Fourier modes,

$$F(\theta, I) = \sum_{|k| \leq J} \hat{F}_k(I) e^{i\langle k, \theta \rangle}.$$

We also decompose f into $f = f_{\leq J} + f_{>J}$ where $f_{\leq J}$ consists of Fourier modes with $|k| \leq J$ and $f_{>J}$ consists of the rest. So we solve the equation $f_{\leq J} - \hat{f}_0 = \{h, F\}$ instead of $f - \hat{f}_0 = \{h, F\}$ as before. Note that the higher Fourier modes $f_{>J}$ is always small, for instance $\|f_{>J}\|_{C^r} \leq \frac{C}{|J|^{\ell-r}}$.

So after the first step of iteration we obtain instead of $H_1(\theta, I) = h_1(I) + \varepsilon^2 f_1(\theta, I)$ that

$$H_1(\theta, I) = h_1(I) + f_{>J}(\theta, I) + \varepsilon^2 f_1(\theta, I).$$

To proceed, we need to address the following questions.

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1. Control the frequency drift: we will allow I to vary in an open set Ω_1 such that

$$|\langle \omega(I), k\rangle| > \frac{C}{2|k|^\tau}, \quad |k| \le J.$$

Since there are only finitely many integer vectors k, we get that Ω_1 is an open set containing an I such that $\omega(I)$ is an Diophantine vector. This will allow the perturbation of $\omega(I)$ caused by $\varepsilon\nabla\hat{f}_0(I)$.

2. Control the C^r norms: to estimate the remainder f_1, we need to estimate the C^r norm of $\{\{H, F\}, F\}$, for some large r. Since F has finitely many Fourier modes, we can always estimate $\|F\|_{C^m} \le J^{m-r}\|F\|_{C^r},\ m > r$. We also need the interpolation inequality to estimate the intermediate norms $\|f\|_{C^r} \le C\|f\|_{C_0}^{1-\frac{r}{m}}\|f\|_{C^m}^{r/m}$ for $0 < r < m$.

When we perform the above procedure for infinitely many steps, the quadratic converges will garantee that $J \to \infty$ quickly, Ω_n converges quickly to a point I corresponding to a Diophantine vector and the remainders depending on θ converges quickly to zero.

□

5. Nonintegrability, Homoclinic intersection and Melnikov function

The materials of this section can be found in [**AKN**], Chapter 7.2.

In this section, we talk about the mechanism that is responsible for the nonintegrability. It is universal in the complement of the KAM set.

5.1. autonomous and non autonomous systems. Sometimes it is more convenient to work with time dependent Hamiltonian system $H(q, p, t), (q, p, t) \in \mathbb{T}^n \times \mathbb{R}^n \times \mathbb{T}$ where the t dependence is periodic. We still have the Hamiltonian equation $\dot{q} = \frac{\partial H}{\partial p}(q, p, t)$ and $\dot{p} = -\frac{\partial H}{\partial q}(q, p, t)$. This kind of nonautonomous equation was studied before as $x' = f(x, t)$.

We call such a system has n+1/2 degrees of freedom. It is easy to rewrite such a system as an autonomous system of $n + 1$ degrees of freedom. The method is to consider a new Hamiltonian $\tilde{H}(q, p, t, E) = H(q, p, t) + E$ by introducing a new variable E dual to t. We see that the equations for $\dot{q}$ and $\dot{p}$ are the same as before and we get $\dot{t} = 1$. Note that in a nonautonomous system, the Hamiltonian H is not a constant under the flow, but $\tilde{H}$ is.

Conversely, we can also pass from an autonomous system of n degrees of freedom to a nonautonomous system of $n + 1/2$ degrees of freedom. Suppose $H : \mathbb{T}^n \to \mathbb{R}^n \to \mathbb{R}$ is autonomous and $\frac{\partial H}{\partial p_1} \neq 0$. Then we can fix an energy level E and solve the equation $H(q_1, p_1, \hat{q}, \hat{p}) = E$ for p_1, where $q = (q_1, \hat{q})$ and $p = (p_1, \hat{p})$. Now we get $p_1 = p_1(q_1, \hat{q}, \hat{p})$. We may now think $-p_1$ as the new Hamiltonian and q_1 as the new time to obtain a system of $n + 1/2$ degrees of freedom. Implicit function theorem implies that

$$\frac{d\hat{q}}{dq_1} = \frac{\partial(-p_1)}{\partial\hat{p}} = \frac{\partial H}{\partial\hat{p}}\Big/\frac{\partial H}{\partial p_1},\ \frac{d\hat{p}}{dq_1} = -\frac{\partial(-p_1)}{\partial\hat{q}} = -\frac{\partial H}{\partial\hat{q}}\Big/\frac{\partial H}{\partial p_1}.$$

5.2. Invariant manifolds for autonomous and nonautonomous systems. The prototypical example that we consider in this section is a pendulum Hamiltonian

$$H_0(q, p) = \frac{1}{2}p^2 + (\cos 2\pi q - 1),$$

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and its time-dependent perturbation

$$H_\varepsilon(q,p,t) = \frac{1}{2}p^2 + (\cos 2\pi q - 1) + \varepsilon H_1(q,p,t).$$

We may choose simply $H_1 = \varepsilon(\cos 2\pi q - 1)\sin 2\pi t$.

There is a saddle fixed point on the zero energy level of H_0 so that the Liouville-Arnold theorem is not applicable there. Nonintegrability is created by adding a small perturbation.

We denote by $\phi^t_{H_0}$ and $\phi^t_{H_\varepsilon}$ the flow generated by the Hamiltonians H_0 and H_ε respectively.

The Hamiltonian H_0 is autonomous, so $\phi^t_{H_0},\ t\in\mathbb{R}$, is a one-parameter diffeomorphism group, i.e. $\phi^{t+s}_{H_0} = \phi^t_{H_0}\phi^s_{H_0},\ s,t\in\mathbb{R}$.

The Hamiltonian H_ε is nonautonomous and the time dependence is 1-periodic. We have $\{\phi^n_{H_\varepsilon},\ n\in\mathbb{Z}\}$ is a group, i.e. $\phi^{m+n}_{H_\varepsilon} = \phi^m_{H_\varepsilon}\phi^n_{H_\varepsilon},\ m,n\in\mathbb{Z}$. In fact given initial condition (q_0,p_0), we find that $\phi^{t+n}_{H_\varepsilon}(q_0,p_0) = \phi^t_{H_\varepsilon}\phi^n_{H_\varepsilon}(q_0,p_0),\ t\in\mathbb{R},\ n\in\mathbb{Z}$, by the uniqueness of solutions. We have seen this argument in Floquet theory.

The next remark is that the invariant manifold theorem 5.1 actually holds for maps. Namely, let 0 be a hyperbolic fixed point for the map $\phi_0:\ \mathbb{R}^2\to\mathbb{R}^2$, i.e. $D\phi(0)$ has two eigenvalues satisfying $\lambda_1 < 1 < \lambda_2$ (or equivalently, the linearized flow $X' = AX$ has A with two eigenvalues with opposite signs). Then it has stable manifold and unstable manifold $W^s = \{X\in\mathbb{R}^2 \mid \phi^n(X)\to 0,\ n\to\infty\}$ and unstable manifold $W^u = \{X\in\mathbb{R}^2 \mid \phi^n(X)\to 0,\ n\to-\infty\}$ satisfying the conclusion of Theorem 5.1. One can check that the proof of Theorem 5.1 goes through if we use discrete time $n\in\mathbb{Z}$.

Therefore, we will consider the phase space $((q,p)\in)\mathbb{R}^2$ and the maps $\phi^1_{H_0}$ and $\phi^1_{H_\varepsilon}$, which are called time-1 maps. We can talk about invariant manifolds for both maps.

The phase portrait of $\phi^n_{H_0},\ n\in\mathbb{Z}$, is the same as that of $\phi^t_{H_0},\ t\in\mathbb{R}$. In particular, since the hyperbolic fixed point $(0,0)$ is identified with $(2\pi,0)$, the stable manifold of $(0,0)$ actually is simultaneously also its unstable manifold. However, for $\phi^n_{H_\varepsilon}$, the situation is different, as we will see below.

Finally, we remark that the stable and unstable manifold of a hyperbolic fixed point for the time-1 map of a Hamiltonian (autonomous or nonautonomous) are Lagrangian submanifolds.

5.3. Poincaré theorem. We consider a Hamiltonian $H_0:\ \mathbb{R}^2\to\mathbb{R}$ satisfying the following assumptions (the H_0 in the previous subsection is a typical example):

(1) there are two critical points (q_-,p_-) and (q_+,p_+) of H_0 at which the eigenvalues of the linearized Hamiltonian equations are real and nonzero. This means that the eigenvalues have opposite sign and same modulus since the flow preserves volume (Liouville theorem).
(2) $W^s((q_-,p_-)) = W^u((q_+,p_+))$, so we get $H_0(q_-,p_-) = H_0(q_+,p_+)$. We assume $H_0(q_-,p_-) = H_0(q_+,p_+) = 0$ by adding a constant to H_0 if necessary.
(3) There is a solution $(q_0(t),p_0(t))$ such that $(q_0(t),p_0(t))\to(q_\pm,p_\pm),\ t\to\pm\infty$.

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In the case of $H_0 : \mathbb{R}^2 \to \mathbb{R}$, either the stable or the unstable manifold consists of a solution to the Hamiltonian equation. So the Hamilton-Jacobi equation is satisfied

$$H_0(q, \partial_q S_0) = 0,$$

where the graph $\{(q, \partial_q S_0(q)),\ q \in \mathbb{T}^1\}$ of $\partial_q S$ is the stable and unstable manifold. In this case, we call the merging stable and unstable manifold as the separatrix.

We remark that the same conclusion holds for a Hamiltonian $H_0 : \mathbb{R}^{2n} \to \mathbb{R}$, which we do not discuss here.

We next add a small time dependent perturbation to H_0 to get the Hamiltonian $H_\varepsilon : \mathbb{R}^2 \times \mathbb{T}^1 \to \mathbb{R}$, $H_\varepsilon(q,p) = H_0(q,p) + \varepsilon H_1(q,p,t)$. In general, the stable and unstable manifolds of the hyperbolic fixed point do not coincide and complicated dynamics is created. The next theorem discovered by Poincaré gives a characterization of such phenomenon. This is the main mechanism responsible for the nonintegrability.

THEOREM 5.1 (Poincaré). *If $H_1(q_-, p_-, t) = H_1(q_+, p_+, t) = 0$ for all $t \in \mathbb{T}^1$ and the Melnikov function does not vanish:*

$$\mathcal{M}(\alpha) := \int_{\mathbb{R}} \{H_0, H_1\}(q(t+\alpha), p(t+\alpha), t)\, dt \neq 0,$$

Then for $\varepsilon > 0$ sufficiently small, the perturbed stable and unstable manifolds $W^s_\varepsilon(q_-, p_-)$ and $W^u_\varepsilon(q_+, p_+)$ do not coincide. Actually they intersect transversally.

PROOF. We have Hamilton-Jacobi equation for the unperturbed system

$$H_0(q, \partial_q S_0) = 0.$$

Similarly, we also have the Hamilton-Jacobi equation for the perturbed system

$$\partial_t S_{\varepsilon,\pm} + H_\varepsilon(q, \partial_q S_{\varepsilon,\pm}) = 0,$$

where the graph $(q, t, S_{\varepsilon,\pm}(q,t)) \subset \mathbb{R}^2 \times \mathbb{T}^1$ is the stable (for -) and unstable (for +) manifolds for (q_-, p_-) and (q_+, p_+) respectively for the perturbed system. (Here recall that when deriving the Hamilton-Jacobi equation, S is defined by integrating the Lagrangian along orbits of the Euler-Lagrange equation. Here we should integrate along orbits approaching the fixed points $(q_\pm, p_\pm)$ as $t \to \pm\infty$, so the corresponding orbit $(q(t), p(t)) = (q(t), \partial_q S_{\varepsilon,\pm}(q(t), t))$ lies on the stable or unstable manifolds by definition).

We take the expansion

$$S_{\varepsilon,\pm}(q,t) = S_0(q) + \varepsilon S_\pm(q,t) + \cdots$$

and

$$H_\varepsilon(q,p,t) = H_0(q,p) + \varepsilon H_1(q,p,t),$$

to obtain the the equation for $S_\pm$

$$\partial_t S_\pm + \frac{\partial H_0}{\partial p}\Big|_{p(q)} \frac{\partial S_\pm}{\partial q} + H_1(q, p(q), t) = 0.$$

where $p(q) = \partial_q S_0(q)$. Since H_0 is autonomous, the solution $(q_0(t), p_0(t))$ also implies a family of solutions $(q_0(t+\alpha), p_0(t+\alpha)),\ \alpha \in \mathbb{R}$. Using $\frac{\partial H_0}{\partial p} = \dot{q}$ and

$$\partial_t S_\pm + \frac{\partial H_0}{\partial p}\Big|_{p(q)} \frac{\partial S_\pm}{\partial q} = \partial_t S_\pm + \dot{q}\frac{\partial S_\pm}{\partial q} = \frac{d}{dt} S_\pm(q(t+\alpha), t),$$

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we get

$$S_\pm(q_0(t+\alpha),t) = S_\pm(q_0(\alpha),0) - \int_0^t H_1(q_0(t+\alpha),p_0(t+\alpha),t)\,dt.$$

As $t \to \pm\infty$, we have $(q_0(t),p_0(t)) \to 0$ exponentially so $H_1(q_0(t+\alpha),p_0(t+\alpha),t)$ converges to zero exponentially and the integral

$$J(\alpha) = \int_{-\infty}^{\infty} H_1(q_0(t+\alpha),p_0(t+\alpha),t)\,dt$$

converges. Moreover, we have

$$J(\alpha) = S_+(q_0(\alpha),0) - S_-(q_0(\alpha),0) + C,$$

where

$$C = \lim_{t\to\infty} S_+(q_0(t+\alpha),t) - \lim_{t\to-\infty} S_-(q_0(t+\alpha),t)$$

is independent of α. Since $(q,t,\partial_q S_\pm(q,t))$ is the stable and unstable manifolds, we get that to the leading order,

$$J'(\alpha) = (\partial_q S_+(q_0(\alpha),0) - \partial_q S_-(q_0(\alpha),0))\dot{q}_0(\alpha) \neq 0$$

implies that the stable and unstable manifolds $W^s_\varepsilon(q_-,p_-)$ and $W^u_\varepsilon(q_+,p_+)$ do not coincide. This is the Melnikov function

$$\mathcal{M}(\alpha) = \frac{dJ(\alpha)}{d\alpha} = \int_{-\infty}^{\infty} \{H_1,H_0\}(q_0(t+\alpha),p_0(t+\alpha),t)\,dt \neq 0.$$

□

We remark that the Poisson bracket for time dependent Hamiltonians does not involve partial derivatives with respect to t.

When the stable and unstable manifolds $W^s_\varepsilon(q_-,p_-)$ and $W^u_\varepsilon(q_+,p_+)$ intersect transversally, complicated dynamics is created. See the following figure. Suppose $(q_-,p_-) = (q_+,p_+) = 0$ and $W^s_\varepsilon(0)$ intersects $W^u_\varepsilon(0)$ transversally at a point p. Then we have that $\phi^n_{H_\varepsilon}(p) \to 0$ exponentially as $n \to \pm\infty$. Then all the points $\phi^n_{H_\varepsilon}(p)$ will continue to lie in $W^s_\varepsilon(0) \cap W^u_\varepsilon(0)$. Moreover, since the Hamiltonian flow preserves volume, the area bounded by the pieces of $W^s_\varepsilon(0)$ and $W^u_\varepsilon(0)$ between two consecutive intersection points remains constant under iterates. So as $n \to \pm\infty$, the images will be stretched to very thin and stay close to W^u_ε and W^s_ε so new intersections will be created and so on and on.

6. Arnold diffusion

The separatrix splitting can be used to create global instability. For an integrable system $H_0(I)$, the variable I is a constant since $\dot{I} = 0$. Adding a small perturbation $H_\varepsilon(\theta,I) = H_0(I) + \varepsilon H_1(\theta,I)$, KAM theorem implies that orbits on KAM tori with Diophantine frequency stays always on the tori so the corresponding $\{I(t),\ t \in \mathbb{R}\}$ has small oscillation.

However, in statistical physics, we expect the motion is ergodic in the sense that a set of positive volume can visit any region of the phase space. In quantum mechanics, there is always certain probability for particles to visit two given open sets. In general, the problem is to understand the dynamics in the complement of the KAM tori, which is supposed to be very chaotic.

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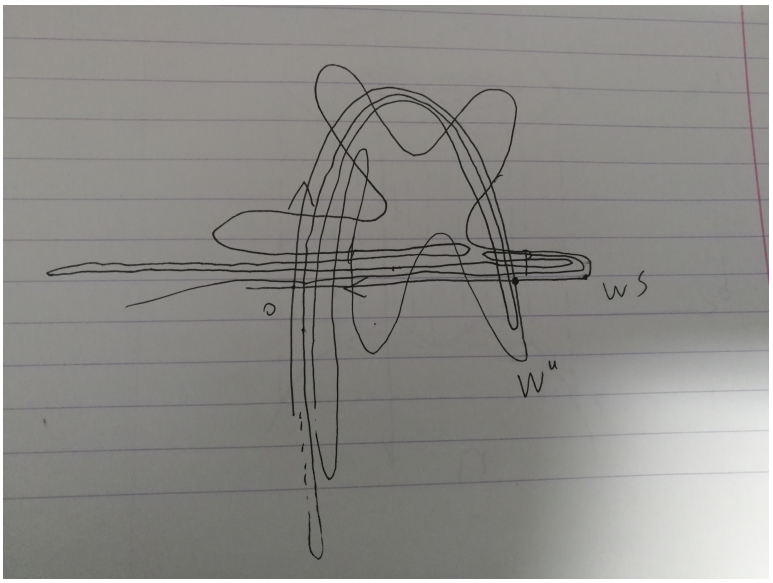


FIGURE 2. Homoclinic intersection created by separatrix splitting

Arnold in [**A1**] constructed the following example

$$H(\theta, I, y, x, t) = \frac{I^2}{2} + \frac{p^2}{2} + (\cos(2\pi q) - 1)(1 + \varepsilon(\cos(2\pi\theta) + \sin(2\pi t))), \tag{6.1}$$

where $(\theta, I; q, p; t) \in T^*\mathbb{T}^1 \times T^*\mathbb{T}^1 \times \mathbb{T}$. It is proved in [**A1**] that

THEOREM 6.1. *In the system* (6.1)*, for any given $A < B$, there is an orbit $\{(\theta(t), I(t), q(t), p(t))\}$ of the system and time $t_1 < t_2$ with $I(t_1) \le A$ and $I(t_2) \ge B$, provided $\varepsilon > 0$ is small enough.*

The theorem can be understood as follows.

(1) when $\varepsilon = 0$, the system decouples into the product system of a rotator $H_0(\theta, I) = \frac{I^2}{2}$ and a pendulum $H_1(q, p) = \dfrac{p^2}{2} + (\cos(2\pi q) - 1)$.
(2) The phase space of H_0 is a cylinder and each I-circle is invariant under the flow. H_1 has a hyperbolic fixed point $(0, 0)$ with stable and unstable manifold merged. So for each I-circle, we have coinciding stable and unstable manifold ($W^s(I)$ and $W^u(I)$) for it.
(3) When the ε-perturbation is added, the rotator and pendulum get coupled together. We can calculate the Melnikov function to show that for each I, we have $W^s_\varepsilon(I)$ intersects $W^u_\varepsilon(I)$ transversally.
(4) The transversality implies that for $|I - I'|$ sufficiently small, we still have $W^s_\varepsilon(I) \cap W^u_\varepsilon(I') \neq \emptyset$ and $W^s_\varepsilon(I') \cap W^u_\varepsilon(I) \neq \emptyset$. An orbit can be found to visit the I- and I-circles.
(5) Given $A < B$, dividing the interval $[A, B]$ into sufficiently small intervals of the form $[I, I']$ such that the transversal intersection of stable and unstable manifolds in the previous item holds for each pair I and I'. Then an orbit can be found to visit the A- and B-circles.

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CHAPTER 8

Homework

1. Homework 1

2. Homework 2

1. Euler method for finding numerical solutions of a ODE.

Consider the IVP $\begin{cases} x' &= f(x,t) \\ x(0) &= x_0 \end{cases}$. We assume f is continuous and bounded.

We first write down the integral equation $x(t) = x_0 + \int_0^t f(x(s), s)\, ds$. Since $x(t)$ in the integrand is not explicitly known, if we pick h sufficiently small, we use $f(x_0, 0)h$ to approximate the integral $\int_0^h f(x(s), s)\, ds$. So we obtain a value $x_1 = x_0 + f(x_0, 0)h$ to approximate $x(h)$. Next, we use $x_2 = x_1 + f(x_1, h)h$ to approximate $x(2h)$.

Inductively, we get a sequence x_k from the inductive formula

$$x_{k+1} = x_k + f(x_k, t_k)h, \quad t_k = kh.$$

We use x_k to approximate $x(kh)$. The general idea is that, at the step k, we start from the point (x_k, t_k) in the (x, t) plane and draw a line with slope $f(x_k, t_k)$ and find the value of x_{k+1} on the line at the time $t_{k+1} = t_k + h$.

This is the easiest numerical method to find a sequence of numbers to approximate the real solution $x(t)$. It is clear from the description that some errors are created and propogated. However, we have the following result: Let $h = 1/n$. As $n \to \infty$, we get

$$|x(1) - x_n| < C/n.$$

The first homework is to prove this statement.

2. Find the interval of existence of the following problems (in case you cannot find the interval exactly, just try your best to give a bound).

(1) $\begin{cases} x' &= t + x^3 \\ x(0) &= 0 \end{cases}$, (2) $\begin{cases} x' &= 2x^2 - t \\ x(1) &= 1 \end{cases}$, (3) $\begin{cases} x' &= t + e^x \\ x(1) &= 0 \end{cases}$

3. Consider the ODE system, $\mathbf{x}' = \mathbf{f}(\mathbf{x}, t)$, where $\mathbf{x}, \mathbf{f} \in \mathbb{R}^n$. More explicitly, if we write $\mathbf{x} = (x_1, \ldots, x_n)$, $\mathbf{f}(\mathbf{x}, t) = (f_1((x_1, \ldots, x_n), t), \ldots, f_n((x_1, \ldots, x_n), t))$. The equation looks like

$$\begin{cases} x_1' &= f_1((x_1, \ldots, x_n), t) \\ &\cdots \\ x_n' &= f_n((x_1, \ldots, x_n), t) \end{cases}$$

Suppose for each initial condition, the solution is known to exist in a time interval. Moreover, suppose for all $\|\mathbf{x}\| \geq b$, we have for some constant c

$$\mathbf{x} \cdot \mathbf{f}(\mathbf{x}, t) \leq k(t)\|\mathbf{x}\|^2 + c, \quad \forall\ t$$

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where $k(t)$ is continuous and bounded. Prove that for any given initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$, the solution exists on the interval $t \in [t_0, \infty)$.

4. Consider the IVP $\begin{cases} x' &= t - x^2 \\ x(t_0) &= x_0 \end{cases}$ where $t_0 \geq 0, x_0 \geq 0$. Prove that the solution $x(t)$ of the IVP satisfies

$$x(t) - \sqrt{t} \to 0, \quad t \to \infty.$$

5. Consider the IVP $\begin{cases} x' &= x + \mu(t + x^2) \\ x(0) &= 1 \end{cases}$. Find $\frac{\partial x}{\partial \mu}|_{\mu=0}$.

6. Consider the IVP $\begin{cases} x' &= x + x^2 + tx^3 \\ x(2) &= x_0 \end{cases}$. Find $\frac{\partial x}{\partial x_0}|_{x_0=0}$.

7. Consider the IVP $\begin{cases} x' &= x^2 + 2\mu t^{-1} \\ x(1) &= -1 \end{cases}$. To see the dependence on μ clearly, we take the power series expansion of x with respect to μ. Suppose $x(t) = v_0(t) + \mu v_1(t) + \mu^2 v_2(t) + \dots$ and substitute it back to the equation we obtain ODEs for $v_0, v_1, v_2, \dots$ respectively. We give initial conditions $v_0(1) = -1, v_1(-1) = 0, v_2(-1) = 0, \dots$. Then work out v_0, v_1, v_2 explicitly.

3. Homework 3

1. Find the general solutions to the linear system

$$\begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ -2 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

2. Find the general solutions to the linear system

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

3. Consider the ODE

$$x'' + p(t)x' + q(t) = 0,$$

where p and q are continuous on the interval (a, b).

(1) Let $\varphi : (a, b) \to \mathbb{R}$ be a nontrivial solution to the ODE. Prove that φ has only nondegenerate zeros (if $\varphi(t_0) = 0$ at some $t_0 \in (a, b)$, then $\varphi'(t_0) \neq 0$). Then prove that on any finite closed intervals $[c, d] \subset (a, b)$, the solution φ has only finitely many zeros.

(2)Let u and v generate all solutions of the ODE via linear combinations. Prove that u and v has no common zeros (there is no $t_0 \in (a, b)$, such that $u(t_0) = v(t_0) = 0$).

4. Find the general solutions to the linear equation

$$x'' + 2x' + x = e^{-t}.$$

5. Work out e^{At} with A being the following matrices

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 3 & -1 \\ 2 & 0 \end{bmatrix}.$$

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4. Homework 4

1. This problem is supposed to give you some extension of the Liouville formula, relating it to the Lie theory. Denote by $\mathrm{SL}(n,\mathbb{R})$ the set of $n\times n$ real matrices with determinant 1 and by $\mathfrak{sl}(n,\mathbb{R})$ the set of $n\times n$ real matrices with trace 0. If a matrix $A\in\mathfrak{sl}(n,\mathbb{R})$, then Liouville formula implies that $e^A\in\mathrm{SL}(n,\mathbb{R})$.

A vector space V with a binary operator $[\cdot,\cdot]: V\times V\to V$ satisfying the following three conditions is called a Lie algebra: for all $a,b,c\in V$, we have

(a) linearity, $[a,b+\lambda c]=[a,b]+\lambda[a,c]$, for all $\lambda\in\mathbb{R}$;

(b) anti-symmetry, $[a,b]=-[b,a]$;

(c) Jacobi identity, $[[a,b],c]+[[b,c],a]+[[c,a],b]=0$.

(1) Prove that $\mathfrak{sl}(n,\mathbb{R})$ endowed with the operation $[A,B]=AB-BA$ is a Lie algebra.

(2) Do the symmetric matrices form a Lie algebra with the same operation? What about anti-symmetric matrices?

(3) Prove that the map $\mathfrak{sl}(n,\mathbb{R})\to\mathrm{SL}(n,\mathbb{R})$ via $A\mapsto e^A:=\sum_{k=0}^{\infty}\frac{A^k}{k!}$ is a diffeomorphism from a small neighborhood of zero to a small neighborhood of identity. (Hint, try the implicit function theorem).

(4) $\mathfrak{sl}(2,\mathbb{R})$ is a vector space of three dimensions. We pick a basis $\{u,l,g\}$, where $u=\begin{bmatrix}0&1\\0&0\end{bmatrix}$, $l=\begin{bmatrix}0&0\\1&0\end{bmatrix}$, and $g=\begin{bmatrix}1/2&0\\0&-1/2\end{bmatrix}$. We next introduce a linear operator $L_g:\mathfrak{sl}(2,\mathbb{R})\to\mathfrak{sl}(2,\mathbb{R})$ via $v\mapsto[g,v]$. The problem is to solve the ODE $X'=L_gX$ on $\mathfrak{sl}(2,\mathbb{R})$.

2. Consider the equation $X'=AX+f(t)$ where $A\in\mathbb{R}^{2\times2}$ and $f:\mathbb{R}\to\mathbb{R}^2$ is periodic with period $2\pi/\omega$, i.e. $f(t+\frac{2\pi}{\omega})=f(t)$. Find (as good as possible) sufficient conditions on the eigenvalues of A to guarantee that all solutions of the equation is periodic. (Hint, variation of constant formula)

If you do not understand Fourier series, you can simply consider the case $f(t)=(\sin(\omega t),\cos(\omega t))$.

3. Prove that the ratio of any two linearly independent solutions of the equation $x''+p(t)x'+q(t)x=0$, where p,q are continuous, has no local maximum.

4. Consider the equation $x''+bx'+cx=f(t)$ where f is continuous and $|f(t)|<1$ for all $t\in\mathbb{R}$. Suppose the homogeneous part has two solutions $e^{\lambda_1 t}$ and $e^{\lambda_2 t}$ with $\lambda_1<\lambda_2<0$. Prove that the equation has a solution that is bounded for all $t\in\mathbb{R}$, and all other solutions converges to it as $t\to\infty$. (Hint, use the formula (6.60) in the book that is the formula of variation of constant for second order equations, then take the integral lower bound to be ∞ or $-\infty$ to make the integral converge).

5. Homework 5

1. Draw the planar phase portrait of the equation $X'=AX$ where A is the following matrices

$$\begin{bmatrix}1&-2\\4&3\end{bmatrix},\quad\begin{bmatrix}2&0\\1&1\end{bmatrix},\quad\begin{bmatrix}2&-1\\1&-1\end{bmatrix}.$$

Please make sure the key features of the phase portraits in the following checklist are included.

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(1) If A is diagonalizable in $\mathbb{R}^{2\times 2}$, make sure that the special solutions along the eigenvectors are drawn, the arrows are correct and the tangencies at 0 are along the correct directions.
(2) If A has complex eigenvalues $\alpha \pm i\beta$, make sure the information provided by the signs of α and β are included.

2. Study the fixed points of the following equation

$$\begin{cases} x' &= y^2 - 4x^2 \\ y' &= 4y - 8 \end{cases}.$$

You are supposed to find the fixed points, linearize the equation at each fixed point and draw the phase portrait for each linearized equation.

3. The following equation is written in polar coordinates (r, θ). Please draw the corresponding phase portrait in the (x, y) plane.

$$\begin{cases} r' &= r(r-1)(r-2) \\ \theta' &= 1 \end{cases}.$$

4. Study the stability of the zero solution by constructing a Lyapunov function.

$$\begin{cases} x' &= y - x + xy \\ y' &= x - y - x^2 - y^3 \end{cases}.$$

6. Homework 6

1. Prove the statement:

THEOREM 6.1. *Suppose in the equation $X' = AX + f(X)$, we have*

(1) *all eigenvalues of A has negative real part;*
(2) *$f(t, X) = o(\|X\|)$ uniformly for all $t \in \mathbb{R}$ as $\|X\| \to 0$.*

Then the zero solution is asymptotically stable.

(Hint: use the DuHamel Principle, then Gronwall inequality $y(t) \le c + b\int_0^t y(s)ds \Longrightarrow y(t) \le ce^{bt}$).

2. We have learned an example of Hopf bifurcation in class. Try to formulate a general statement about Hopf bifurcation. Suppose we are given an ODE $X' = AX + F(X, \varepsilon)$ on the plane depending on the parameter ε. Try to make assumptions on A and F to guarantee that the system has the behavior of Hopf bifurcation when we change the parameter ε.

Please also give a sketch of proof. You do not have to give detailed proof. Your statement may contain errors but should at least sound plausible and be supported by the sketched proof.

3. Consider the map $E_2 : \mathbb{T}^1 \to \mathbb{T}^1$ via $x \mapsto 2x \bmod 1$. State and prove the structural stability for this map.

4. Prove the mixing property of the map $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} : \mathbb{T}^2 \to \mathbb{T}^2$, i.e. for all $f, g : \mathbb{T}^2 \to \mathbb{R}$ continuous, we have

$$\int_{\mathbb{T}^2} f(A^n x) g(x) dx \to \int_{\mathbb{T}^2} f(x)\, dx \int_{\mathbb{T}^2} g(x)\, dx, \quad \text{as } n \to \infty.$$

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Hint: use Fourier series. This is a strong form of instability. It shows that though the map A is deterministic, after many iterations, it has similar behavior as independent random variables in probability (two events A, B are independent if $P(A \cap B) = P(A) \cap P(B)$).

7. Homework 7

1. Prove the fundamental theorem of algebra (Every polynomial $z^n + a_1 z^{n-1} + \dots a_{n-1} z + a_n = 0$ admits a zero on the complex plane $\mathbb{C}$) following the instructions.

The proof needs the following result.

PROPOSITION 7.1. *The rotation degree of a simple closed curve γ is invariant under a continuous deformation of the vector field, provided during the deformation process, no zero appears on γ.*

The result is intuitively clear, since the degree is an integer, but its proof is quite involved. A rigorous proof needs differential topology. Let us assume it.

Step 1, We identify $\mathbb{C}$ with $\mathbb{R}^2$. Let $z = x + iy$, $i^2 = -1$. Then the vector field $V(z) = z^n$ has a zero when $z = 0$. Prove that the index of zero is n. (For example, when $n = 2$, $z^2 = (x + iy)^2 = (x^2 - y^2) + 2xyi$. On $\mathbb{R}^2$, the vector field is $V(x, y) = (x^2 - y^2, 2xy)$. To solve this problem, you do not have to do the calculation in the x-y coordinates. It is enough to use the definitions of rotation degree and index.)

Step 2, Construct a simple closed curve γ, and for the vector field $V(z) = z^n + a_1 z^{n-1} + \dots a_{n-1} z + a_n$, construct a continuous deformation to deform it to z^n such that during the deformation process there is no zero appearing on γ.

Step 3, conclude the proof by combining Step 1 and Step 2.

2. Prove the following version of Brouwer fixed point theorem.

THEOREM 7.2. *Let f be a continuous map from $D := \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$ to D. Then f has a fixed point.*

Hint, construct a vector field $V(X) = f(X) - X$, $X = (x, y)$. Prove that if there is no fixed point on the boundary of D, then the rotation degree of the unit circle is nonvanishing.

3. Let $x(t), t > 0$, be a solution to the equation $x'' + q(t)x = 0$ with $q(t) > 0$ for $t > 0$. Suppose $x(t)$ has zeros $0 < t_1 < t_2 < \dots < t_n < \dots$.

(1) Suppose q is continuous and increasing. Prove that $t_{n+1} - t_n < t_n - t_{n-1}$ (the distance between neighboring zeros decrease).
(2) Suppose in addition that $q(t) \to C$ as $t \to \infty$, prove that $\lim_{n\to\infty}(t_n - t_{n-1}) = \frac{\pi}{\sqrt{C}}$.

4. For what $a \in \mathbb{R}$, the boundary value problem

$$x'' + ax = 1, \quad x(0) = 0,\ x(1) = 0$$

has no solution?

8. Homework 8

1. Prove that the operator $A:\ \ell^2(\mathbb{Z}) \to \ell^2(\mathbb{Z})$ via $(a_n) \mapsto (\frac{1}{|n|+1} a_n)$ is compact.
2. Work out the Green's function of the following BVP. $\begin{cases} x'' + x & = 0, \\ x(0) = x(1) & = 0 \end{cases}$

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3. Suppose $0 \le f(t) \le m$. Try to estimate from below and above the solution $x(t)$ that is bounded as $t \to 0$ and $t \to \infty$ of the equation

$$t^2x'' + 2tx' - 2x = f(t).$$

Hint: work out the Green's function.

In practice, there are some Sturm-Liouville boundary value problems whose eigenfunctions are widely used and are called special functions. In the following exercises, try to work on some of them.

4. Verify that the Legendre polynomials

$$P_n(t) := \frac{1}{2^n n!}\frac{d^n}{dt^n}(t^2-1)^n$$

are eigenfunctions of the linear operator

$$Lx = -\frac{d}{dt}((1-t^2)\frac{d}{dt}x)$$

with eigenvalue $\lambda_n = n(n+1)$.

5. Prove that the Tchebyshev polynomials $T_n(t) = \cos(n\theta)$ where $\cos\theta = t$ and $\theta \in [0, \pi]$ are orthogonal with respect to the inner product

$$\langle f, g\rangle = \int_{-1}^{1} \frac{f(t)\overline{g(t)}}{\sqrt{1-t^2}}\, dt.$$

6. Define the Hermite polynomials H_n by

$$H_n(t) = (-1)^n e^{t^2}\frac{d^n}{dt^n}(e^{-t^2}).$$

(a) Prove that $\varphi_n(t) = e^{-t^2/2}H_n(t)$ are orthogonal in $L^2(\mathbb{R})$.

(b) Prove that the n-th function $\varphi_n(t)$ is an eigenfunction of the operator

$$H = -\frac{d^2}{dt^2} + t^2$$

With eigenvalue $\lambda_n = 2n + 1$.

Hint: this is the harmonic oscillator in quantum mechanics. We give a hint having ideas from representation theory. Let

$$A = \frac{d}{dt} + t, \quad A^* = -\frac{d}{dt} + t.$$

Verify that $A\varphi_n = 2n\varphi_{n-1}$ (annihilation operator) and $A^*\varphi_n = \varphi_{n+1}$ (creation operator) and $H = AA^* - 1$ (Hamiltonian). To verify the last equality, remember that H is an operator and should be applied to functions.

9. Homework 9

1. Let $f : U(\subset \mathbb{R}^n) \to \mathbb{R}$ be analytic, where U is open. Suppose $x_* \in U$ is an isolated critical point of f. Prove that the flow near x_* is nonrecurrent.

For any ε, there exists δ, such that if an orbit $x(t)$ satisfies

$$|x(0) - x_*| < \delta, \quad |x(T) - x_*| < \delta$$

for some $T > 0$, then we have

$$|x(t) - x_*| < \varepsilon, \quad \forall\ t \in [0, T].$$

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Remark: This means that it cannot happen that an orbit starting near x_*, moves far away and then come back to a neighborhood of x_*. Whenever it leaves a neighborhood of x_*, it never comes back again.

note that x_* may be a degenerate critical point, i.e. $D^2 f(x_*)$ may be degenerate. So this is a case where one should consider Lojasiewicz.

2. For some fixed $g \in C([0,1];\mathbb{R})$, define the linear operator T from $C([0,1];\mathbb{R})$ to $C([0,1];\mathbb{R})$ by

$$(Tf)(x) = \int_0^x f(t)g(t)\,dt.$$

Prove that this operator is compact.

3. A bounded linear operator is of finite rank if its image is finite-dimensional. Note that a finite-rank operator is compact, since all balls are pre-compact in a finite-dimensional space.

Prove: A compact operator $T : H \to H$ with H a Hilbert space is a uniform operator norm limit of finite rank operators. In other words, there exists a sequence $T_n : H \to H$, each of which is of finite rank and $\|T - T_n\| \to 0$ as $n \to \infty$.

4. A C^1 vector field over $\mathbb{R}^n \setminus \{0\}$ is called a central field if

$$V(T(p)) = T(V(p)) \qquad \forall\ T \in O(n)$$

where $O(n)$ is the orthogonal group. Prove that a central field is always a gradient field, and analyze the solutions of the negative gradient flow.

10. Homework 10

1. In this problem, we try to understand the Kepler problem. The Hamiltonian is written as $H : \mathbb{R}^4 \to \mathbb{R}$

$$H(x_1, x_2, p_1, p_2) = \frac{1}{2m}(p_1^2 + p_2^2) - \frac{k}{\sqrt{x_1^2 + x_2^2}}.$$

This means that a particle of mass m is attracted by the force field centered at the origin. The number $k > 0$ absorbs the gravitational constant and masses.

In the first step, we perform a polar coordinate transform

$$(x_1, x_2; y_1, y_2) \mapsto (r, \theta; R, \Theta)$$

with $x_1 = r\cos\theta,\ x_2 = r\sin\theta, \quad y_1 = R\cos\theta - \frac{\Theta}{r}\sin\theta, \quad y_1 = R\sin\theta + \frac{\Theta}{r}\cos\theta.$

Prove that this transform is symplectic.

After the polar coordinate transform, you are supposed to obtain a Hamiltonian of the form

$$H = \frac{1}{2m}(R^2 + \frac{\Theta^2}{r^2}) - \frac{k}{r}.$$

Note that the Hamiltonian does not depend on θ, so we get $\dot{\Theta} = 0$ and Θ is a constant. Actually Θ has the physical meaning of the angular momentum. So it is essentially a system with one degrees of freedom in the variables (r, R). Draw its phase portrait for a given $\Theta > 0$.

In this problem, we only show that the bounded solutions are periodic, but we do not pursue that the bounded solutions are all elliptic.

2.

(1) Work out the Legendre transform of $f(x) = e^x$, $f : \mathbb{R} \to \mathbb{R}$.

(2) Work out the Legendre transform of $f : \mathbb{R} \to \mathbb{R}$ where $f(x) = a(x) - cx$ where a is a convex function and c is a constant.

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(3) Let $f : \mathbb{R}^n \to \mathbb{R}$ be the quadratic form $f(x_1, \ldots, x_n) = \sum_{i,j=1}^n a_{ij} x_i x_j$ where a_{ij} is positive definite. Work out its Legendre transform $g(y_1, \ldots, y_n) = \sum_{i,j=1}^n b_{ij} y_i y_j$. Prove that the matrix $(b_i j)$ is the inverse of (a_{ij}). Prove also that the values of the both functions coincide at the corresponding points

$$f(x(y)) = g(y), \quad g(y(x)) = f(x).$$

3. Work out the Euler-Lagrange equation for the variational principle.

(1) Let D be an open bounded and simply-connected domain in $\mathbb{R}^2$, consider the functional $\int_D |\nabla u|^2 dx$ defined on the set of functions $\{u \in C^1(D; \mathbb{R}) \mid u|_{\partial D} = 0\}$. You are supposed to get $\Delta u = 0$.

(2) Consider the functional $\int_{\mathbb{R}^2} |\nabla u|^2 e^{-\frac{|x|^2}{2}} dx$ on the set of functions $\{u \in C^1(\mathbb{R}^2; \mathbb{R}) \mid \lim_{|x|\to\infty} u(x) = 0\}$. You are supposed to get $\Delta u - x \cdot \nabla u = 0$. This is the so-called Uhlenbeck-Ornstein operator.

4. (1)Verify the Jacobi identity

$$\{\{H_1, H_2\}, H_3\} + \{\{H_2, H_3\}, H_1\} + \{\{H_3, H_1\}, H_2\} = 0$$

for the Poisson bracket

$$\{H_1, H_2\} := \sum_{i=1}^n \frac{\partial H_1}{\partial q_i} \frac{\partial H_2}{\partial p_i} - \frac{\partial H_1}{\partial p_i} \frac{\partial H_2}{\partial q_i}$$

This implies that C^∞ functions on $\mathbb{R}^{2n}$ with the Poisson bracket form a Lie algebra.

(2) Prove Poisson theorem: if $\{F_1, H\} = 0$ and $\{F_2, H\} = 0$, then $\{\{F_1, F_2\}, H\} = 0$.

(3) Prove that Poisson bracket is invariant under sympletic transform. Suppose the map $(q, p) \to (Q, P)$ via $Q = Q(q, p)$ and $P = P(q, p)$ is symplectic, then the Poisson bracket computed using the coordinates (q, p) is the same as that computed using the coordinates (Q, P).

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